CSE 317

Data Communication Level – 3 (Spring Term)

Maj Syed Ashraf Shahed, Inf

Instr Class B, CSE Dept

MIST



Data Communications and Networking Fourth Edition

Forouzan

Chapter 2

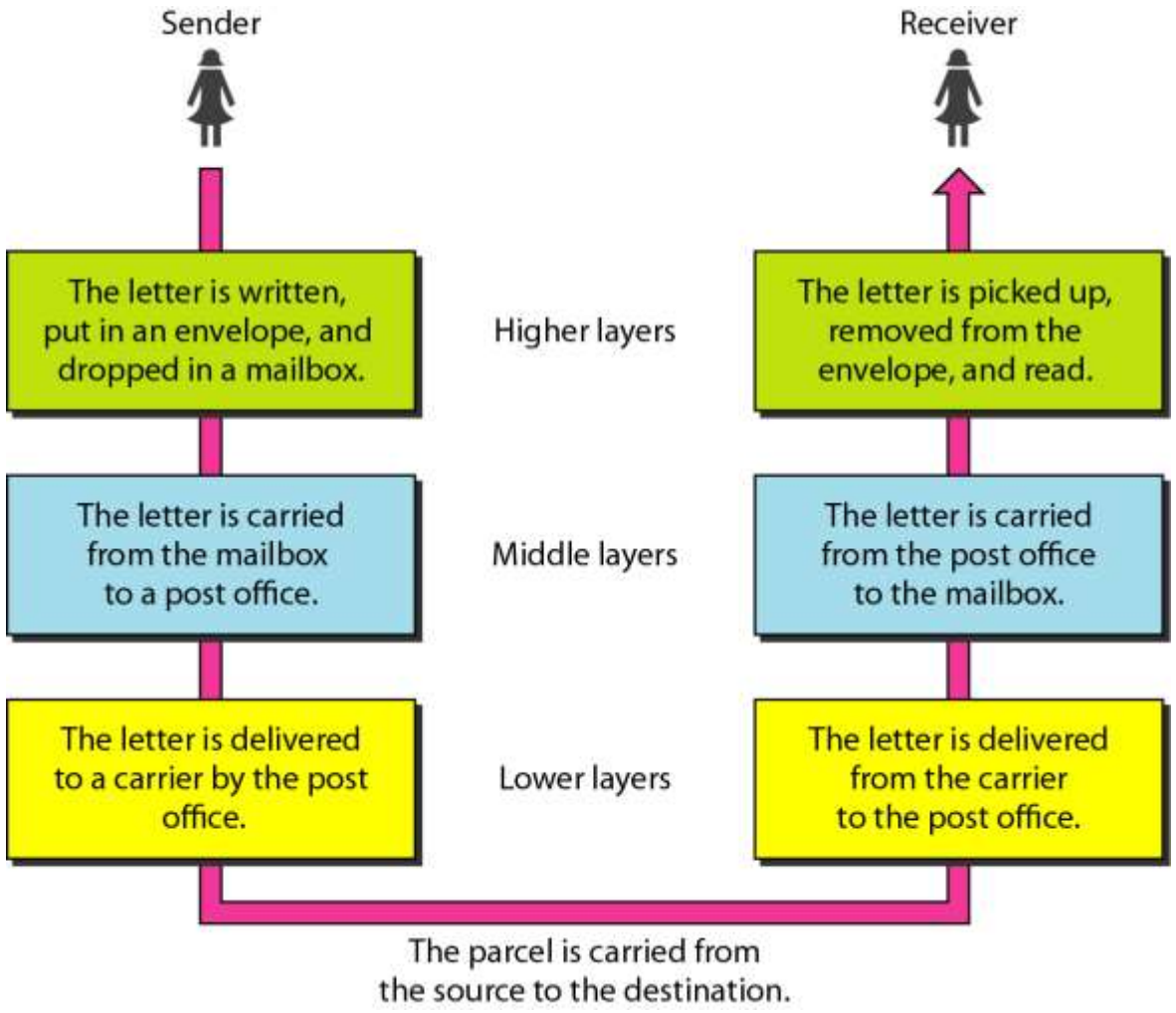
Network Models

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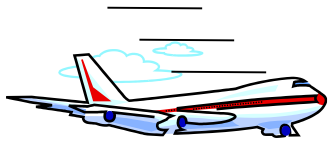
2-1 LAYERED TASKS

We use the concept of layers in our daily life. As an example, let us consider two friends who communicate through postal mail. The process of sending a letter to a friend would be complex if there were no services available from the post office.

Figure: Tasks involved in sending a letter



Example: organization of air travel



end-to-end transfer of person ~~plus baggage~~

ticket (purchase)

baggage (check)

gates (load)

runway takeoff

airplane routing

ticket (complain)

baggage (claim)

gates (unload)

runway landing

airplane routing

airplane routing

How would you *define/discuss* the *system* of airline travel?

- a series of steps, involving many services

Example: organization of air travel



layers: each layer implements a service

- via its own internal-layer actions
- relying on services provided by layer below

2-2 THE OSI MODEL

Established in 1947, the International Standards Organization (ISO) is a multinational body dedicated to worldwide agreement on international standards. An ISO standard that covers all aspects of network communications is the Open Systems Interconnection (OSI) model. It was first introduced in the late 1970s.

Note

International Standards Organization (ISO) is the organization.

Open Systems Interconnection (OSI) is the model.

Figure: *Seven layers of the OSI model*

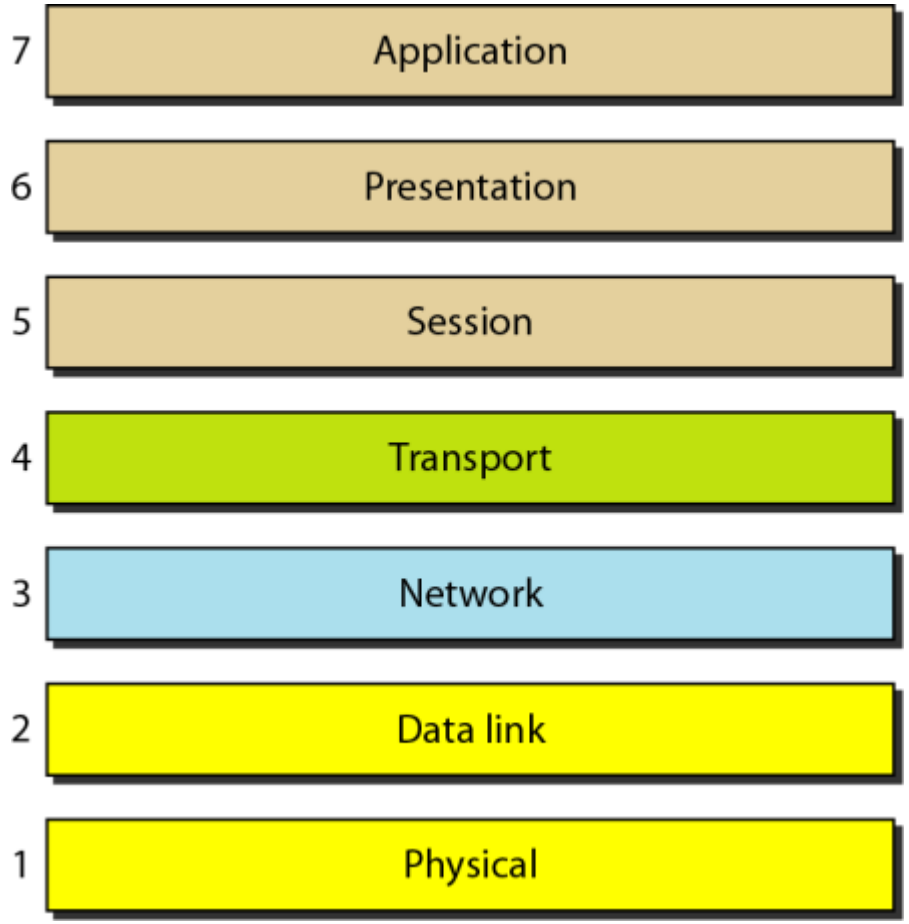


Figure: *The interaction between layers in the OSI model*

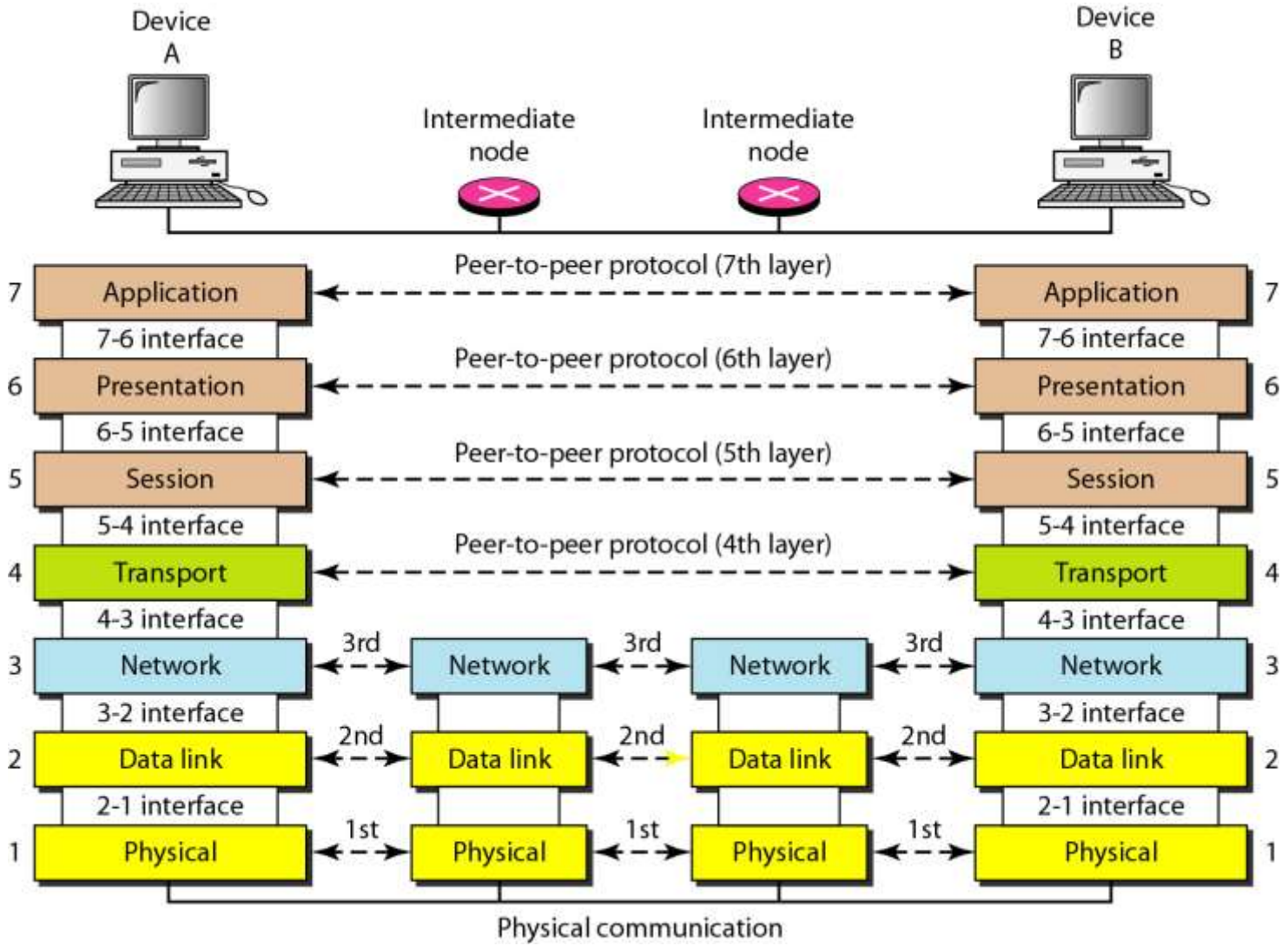


Figure: *An exchange using the OSI model*

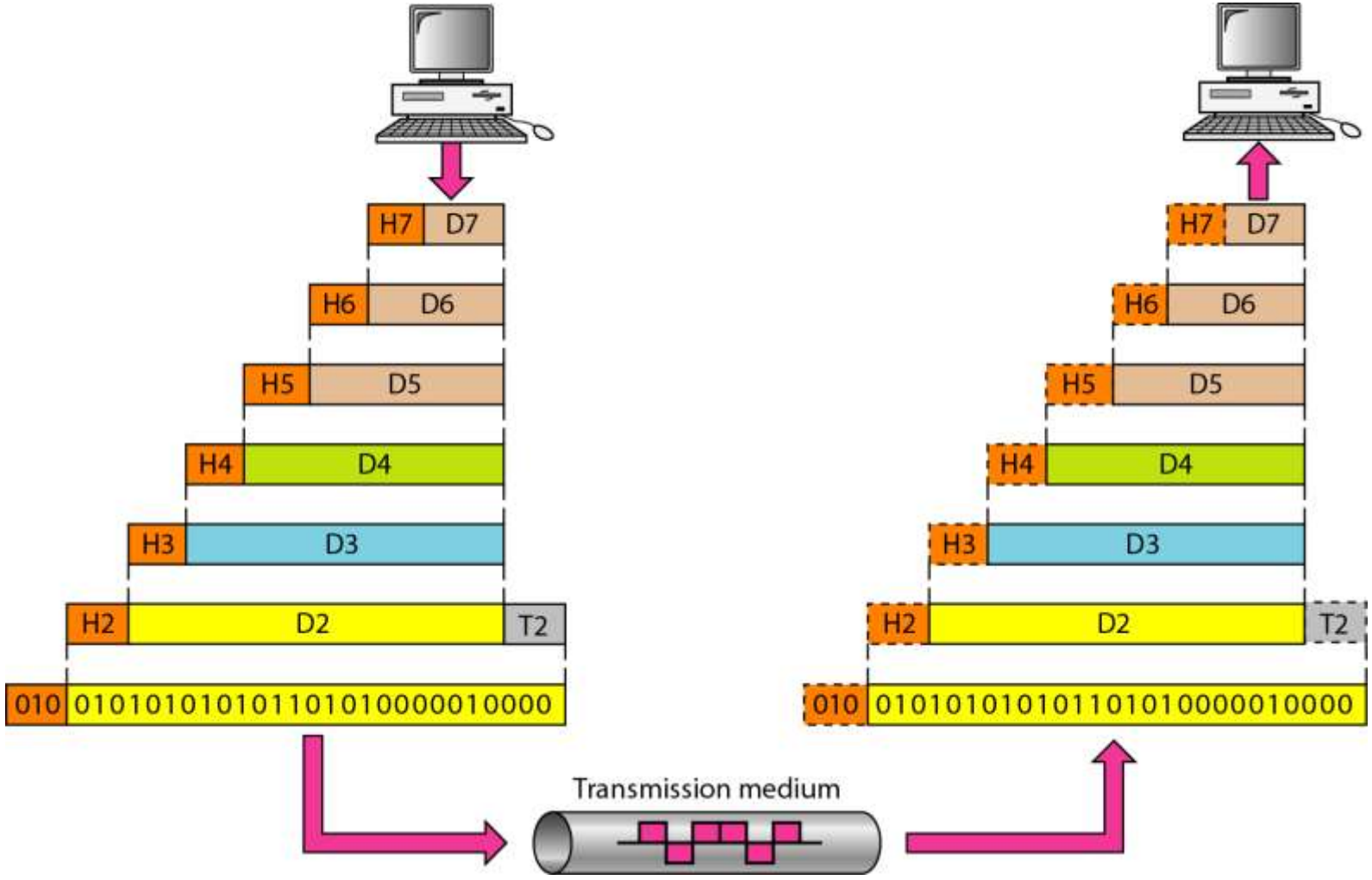
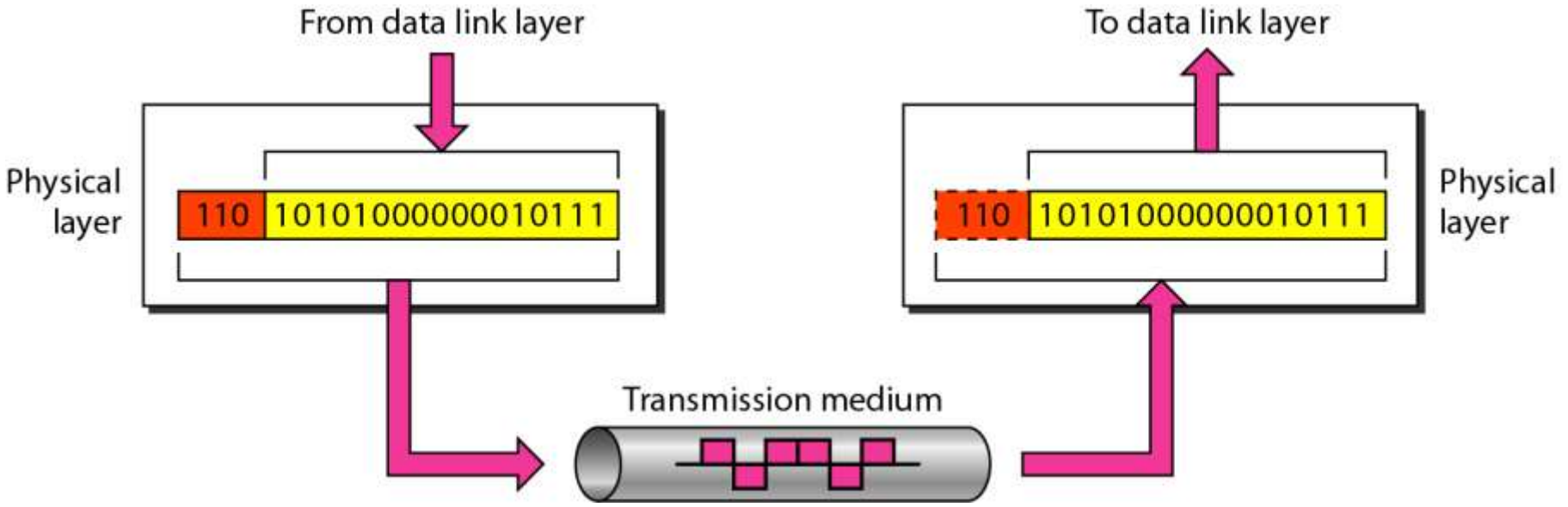


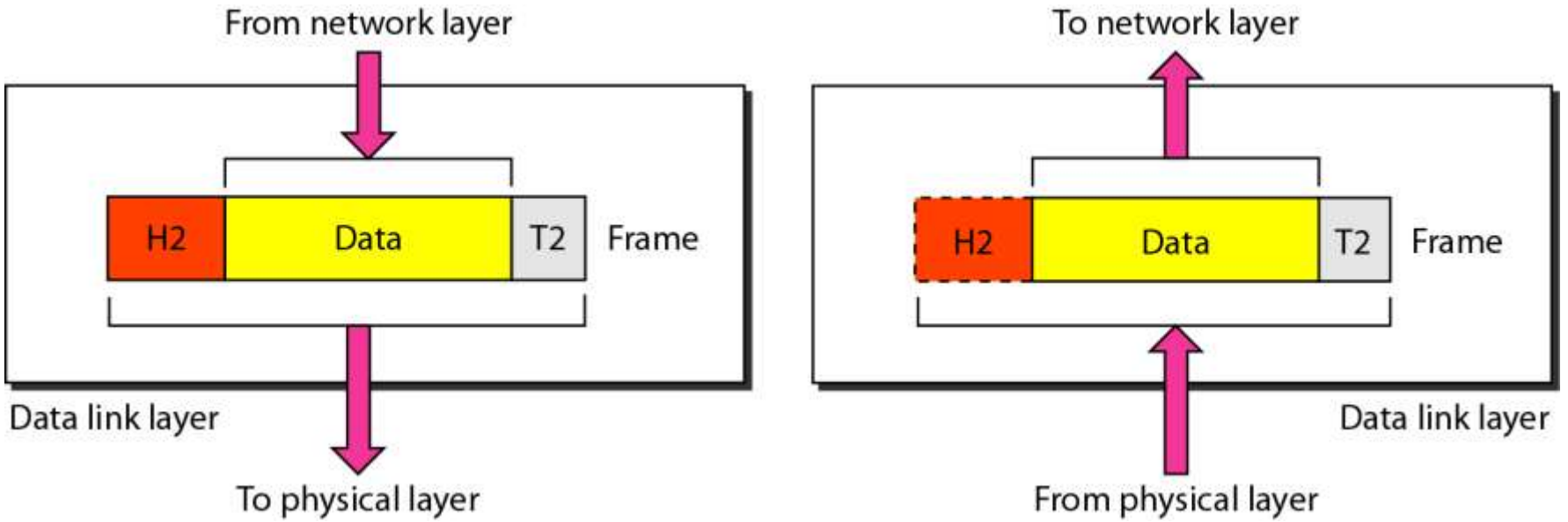
Figure: *Physical Layer*



Note

The physical layer is responsible for movements of individual bits from one hop (node) to the next.

Figure: *Data Link Layer*





Note

The data link layer is responsible for moving frames from one hop (node) to the next.

Figure: *Hop-to-hop delivery*

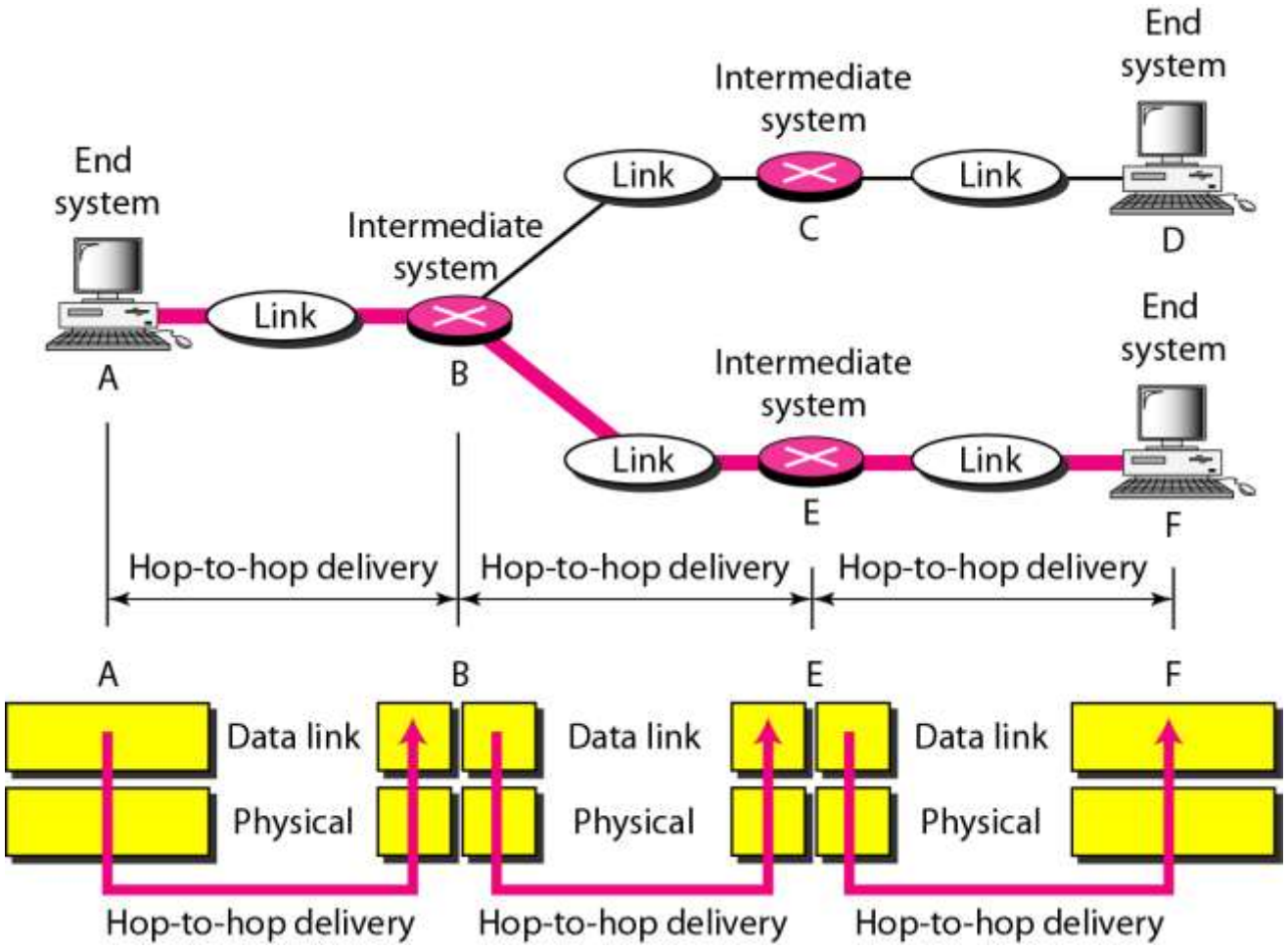
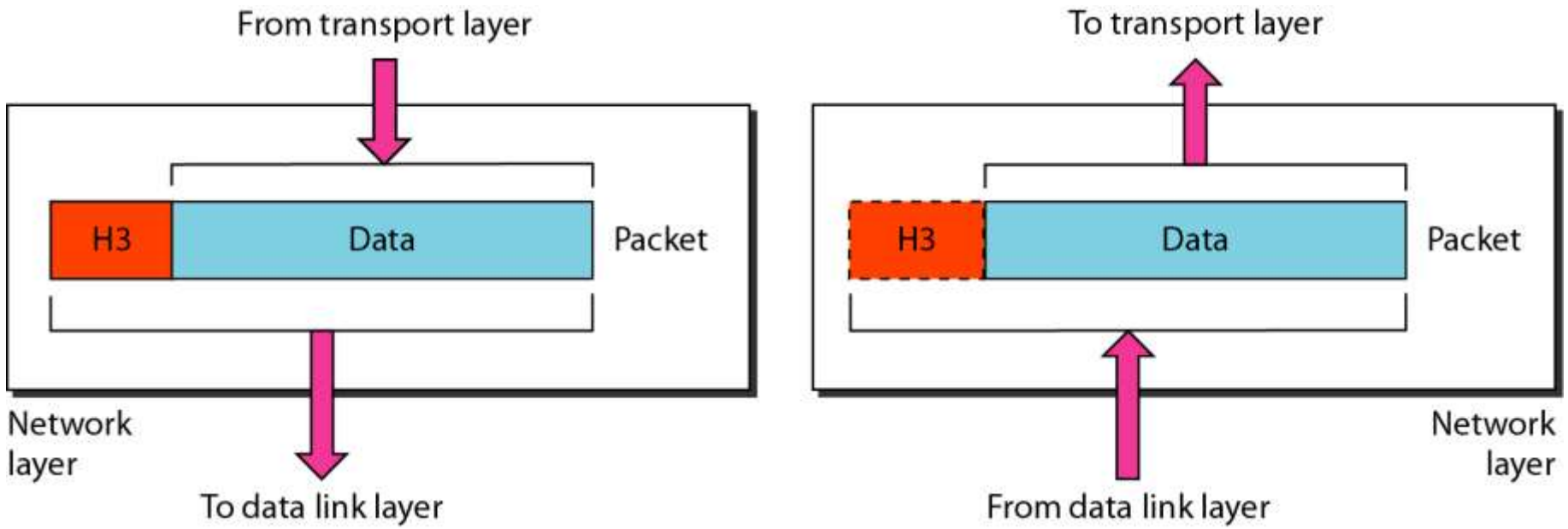


Figure: *Network Layer*



Note

The network layer is responsible for the delivery of individual packets (Datagram) from the source host to the destination host.

Figure: *Source-to-destination delivery*

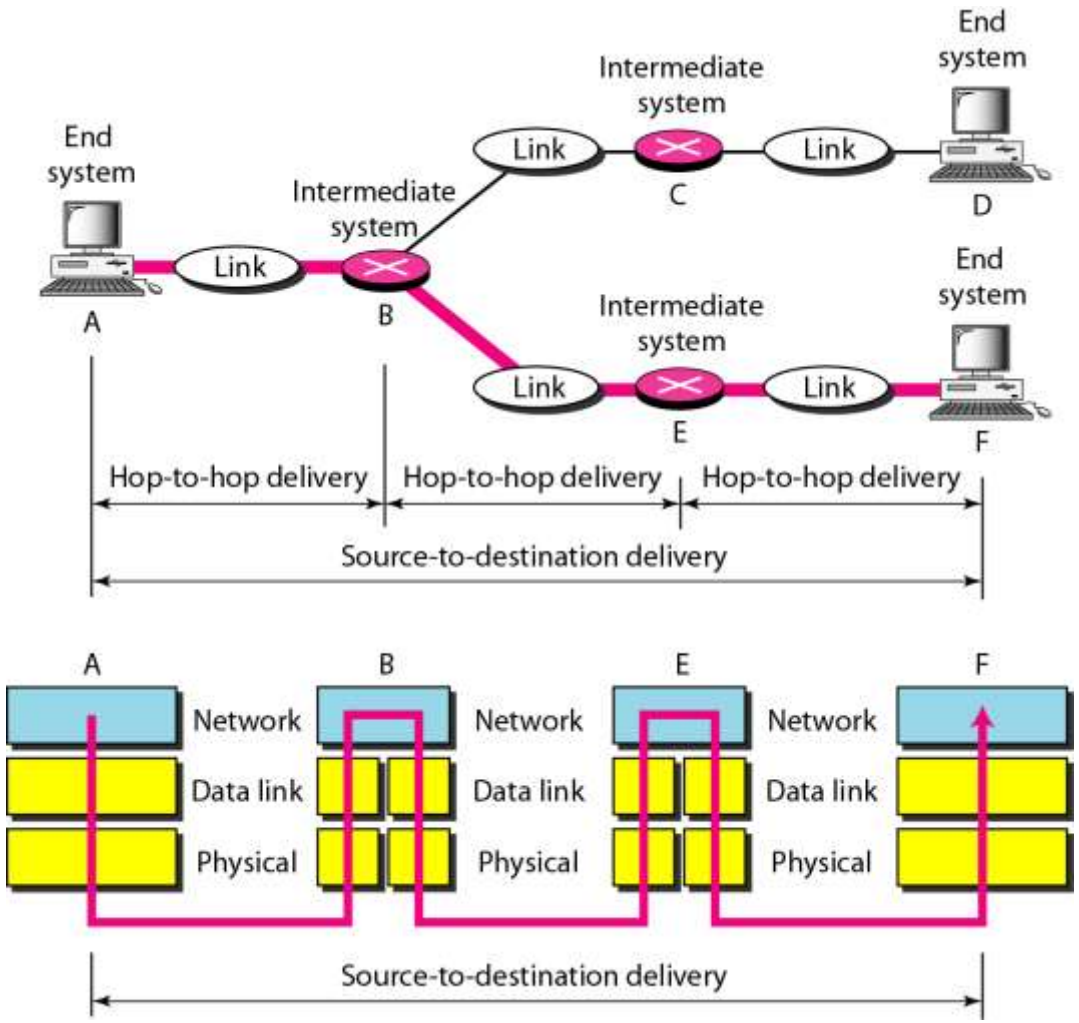
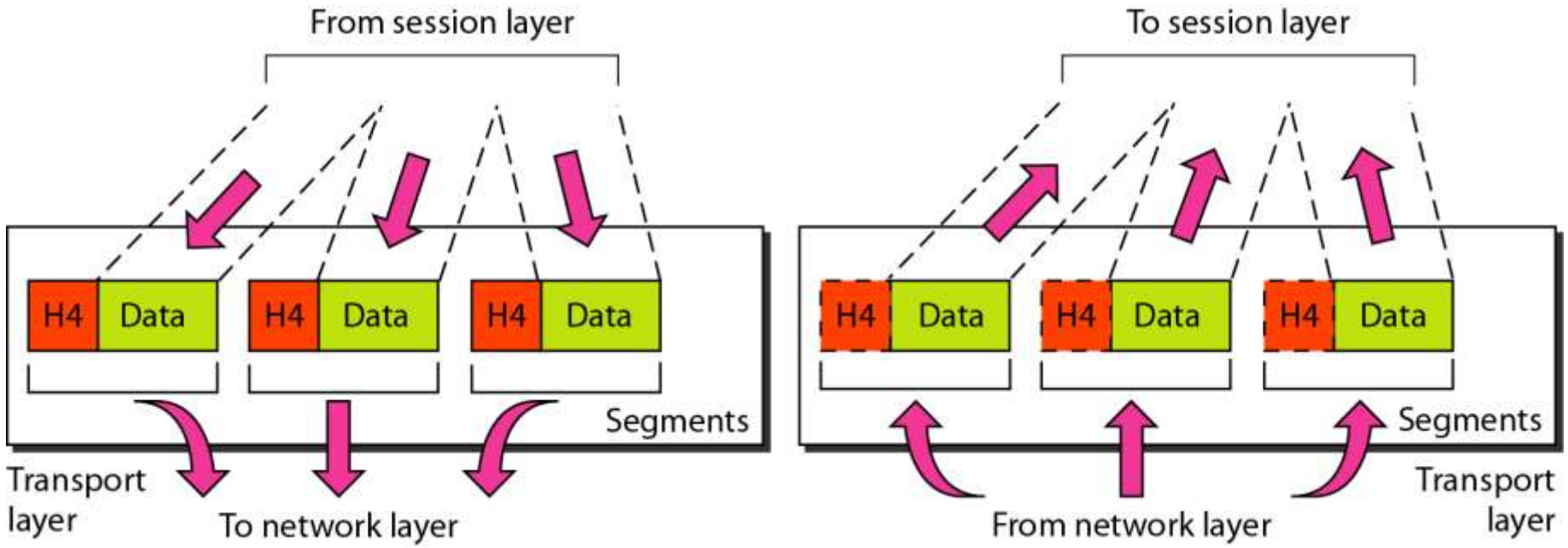


Figure: *Transport layer*



Note

The transport layer is responsible for the delivery

of a message (Segment) from one process to another.

Figure: *Reliable process-to-process delivery of a message*

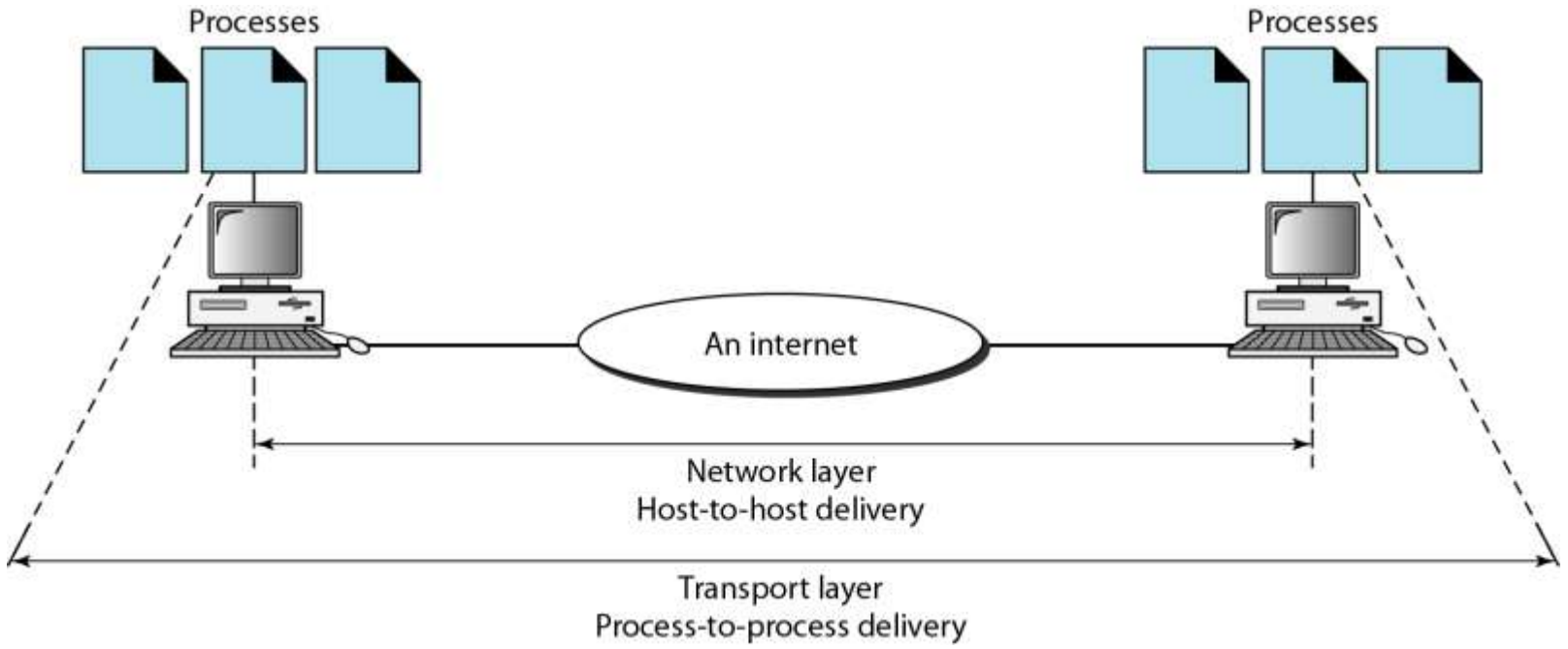
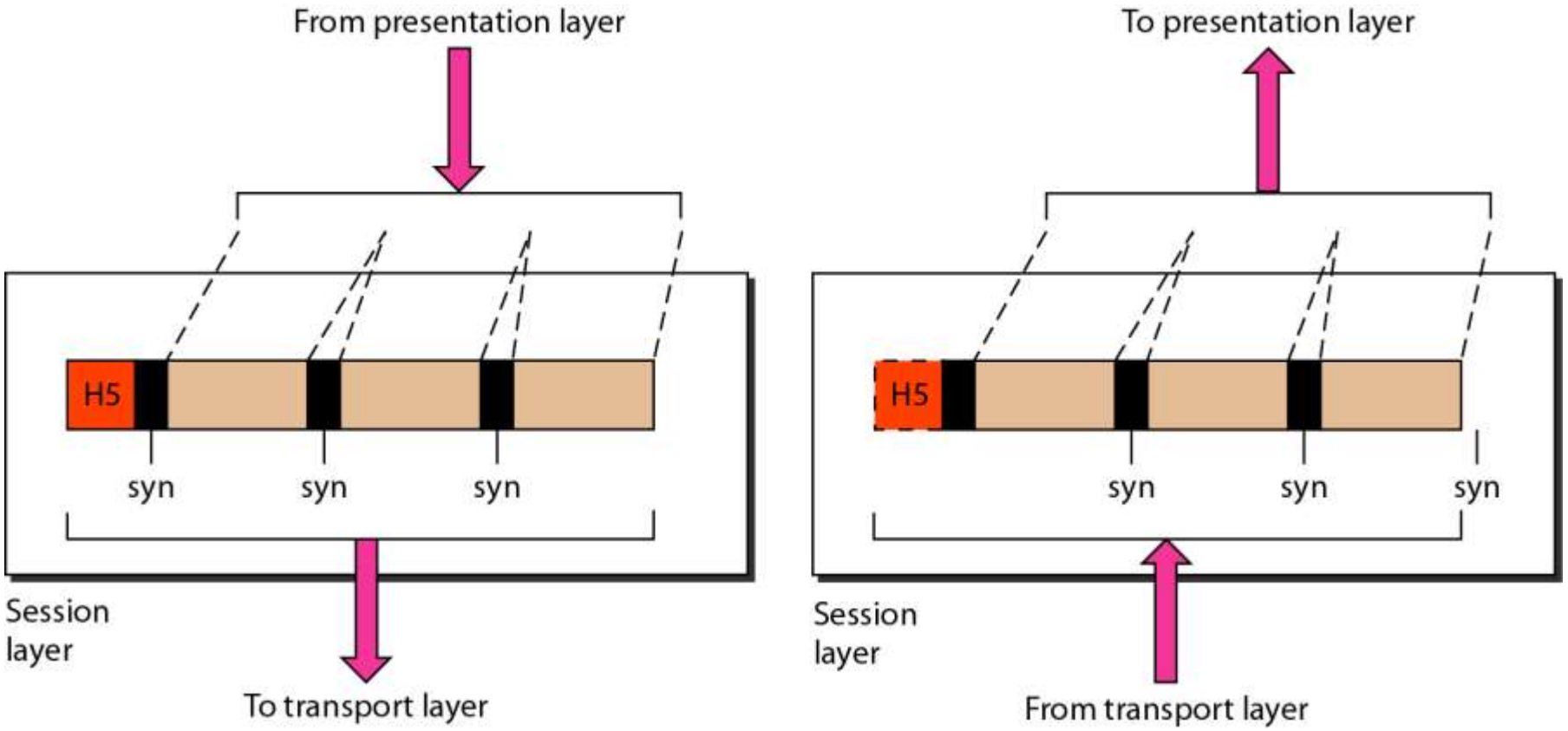


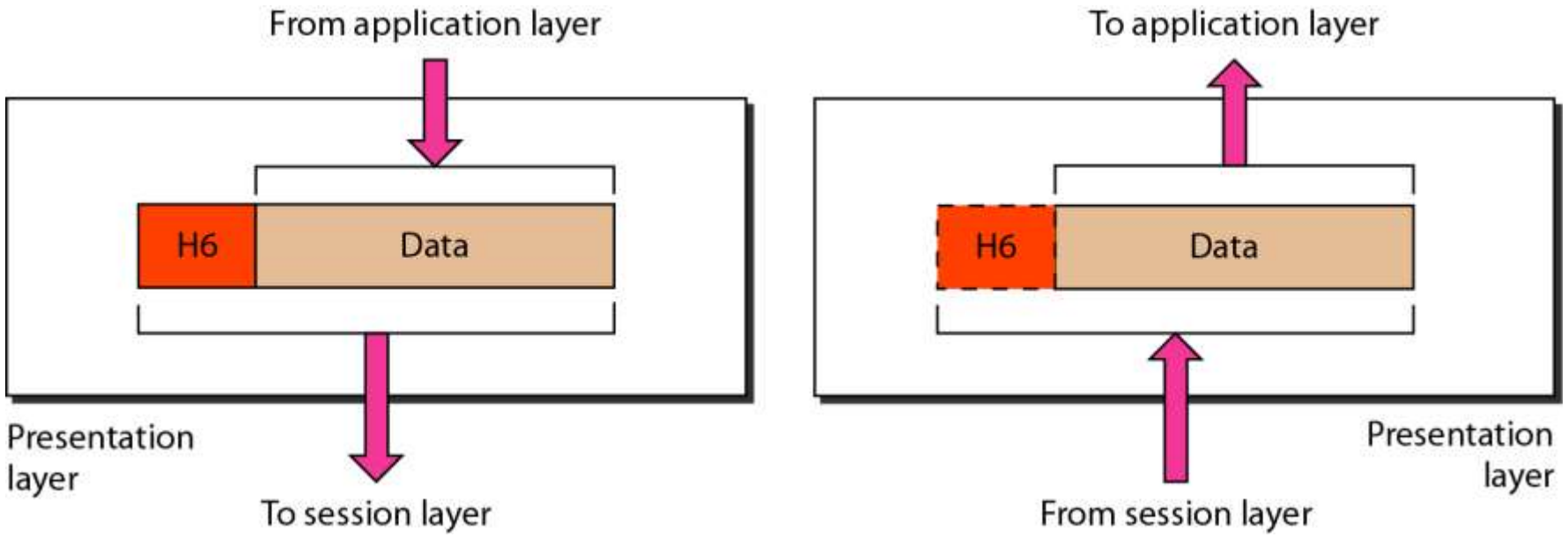
Figure: *Session Layer*





The session layer is responsible for dialog control and synchronization.

Figure: *Presentation Layer*

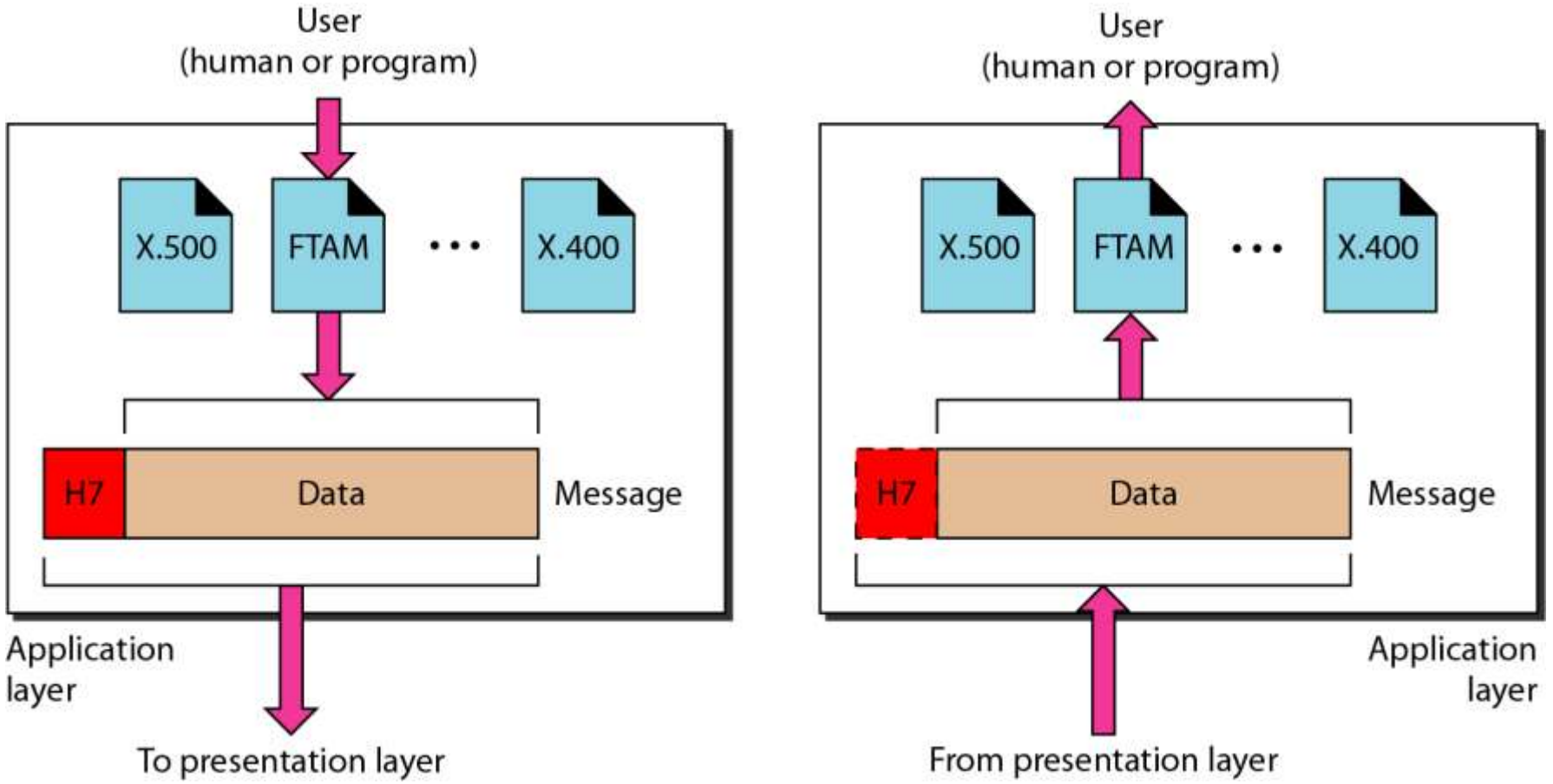




Note

The presentation layer is responsible for translation, compression, and encryption.

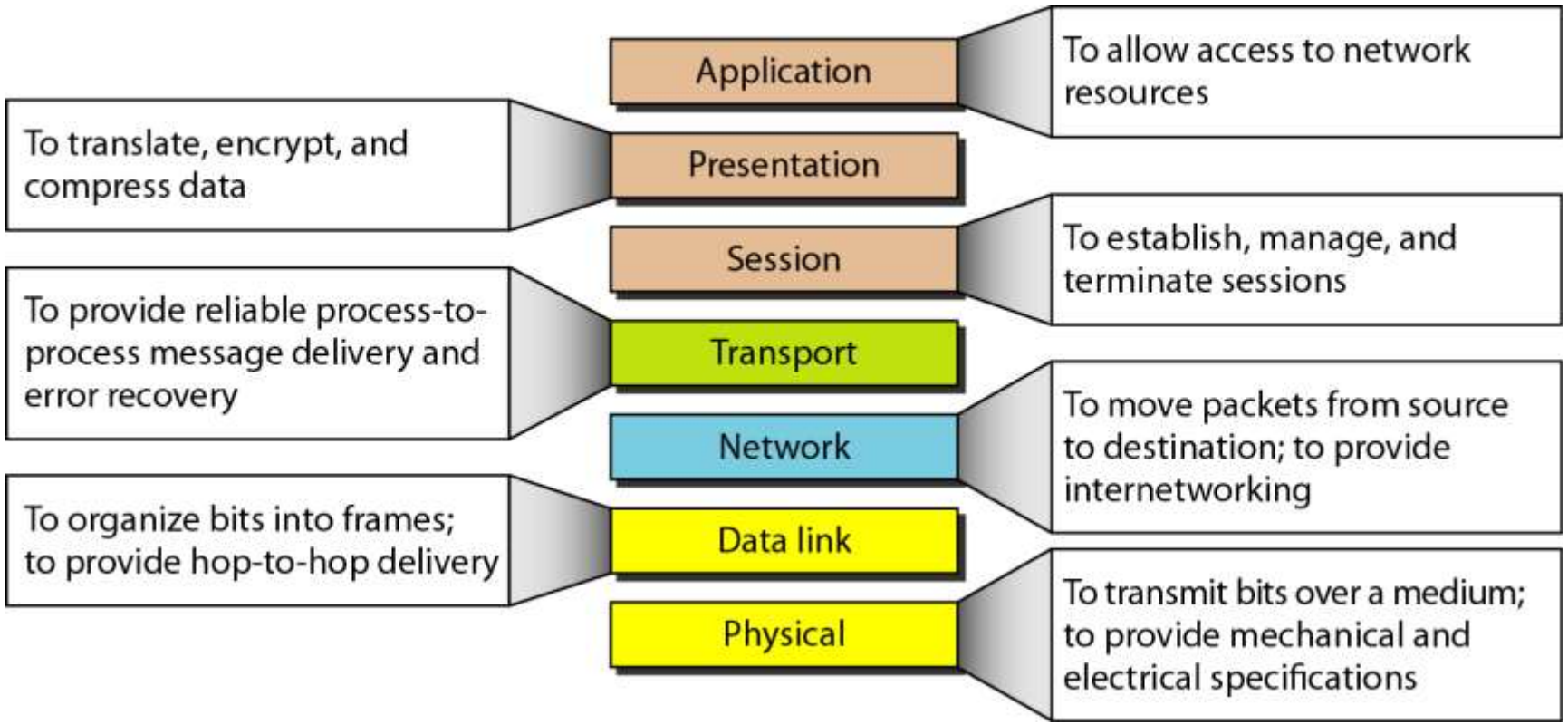
Figure: *Application Layer*



Note

The application layer is responsible for providing services to the user.

Figure 2.15 *Summary of Layers*



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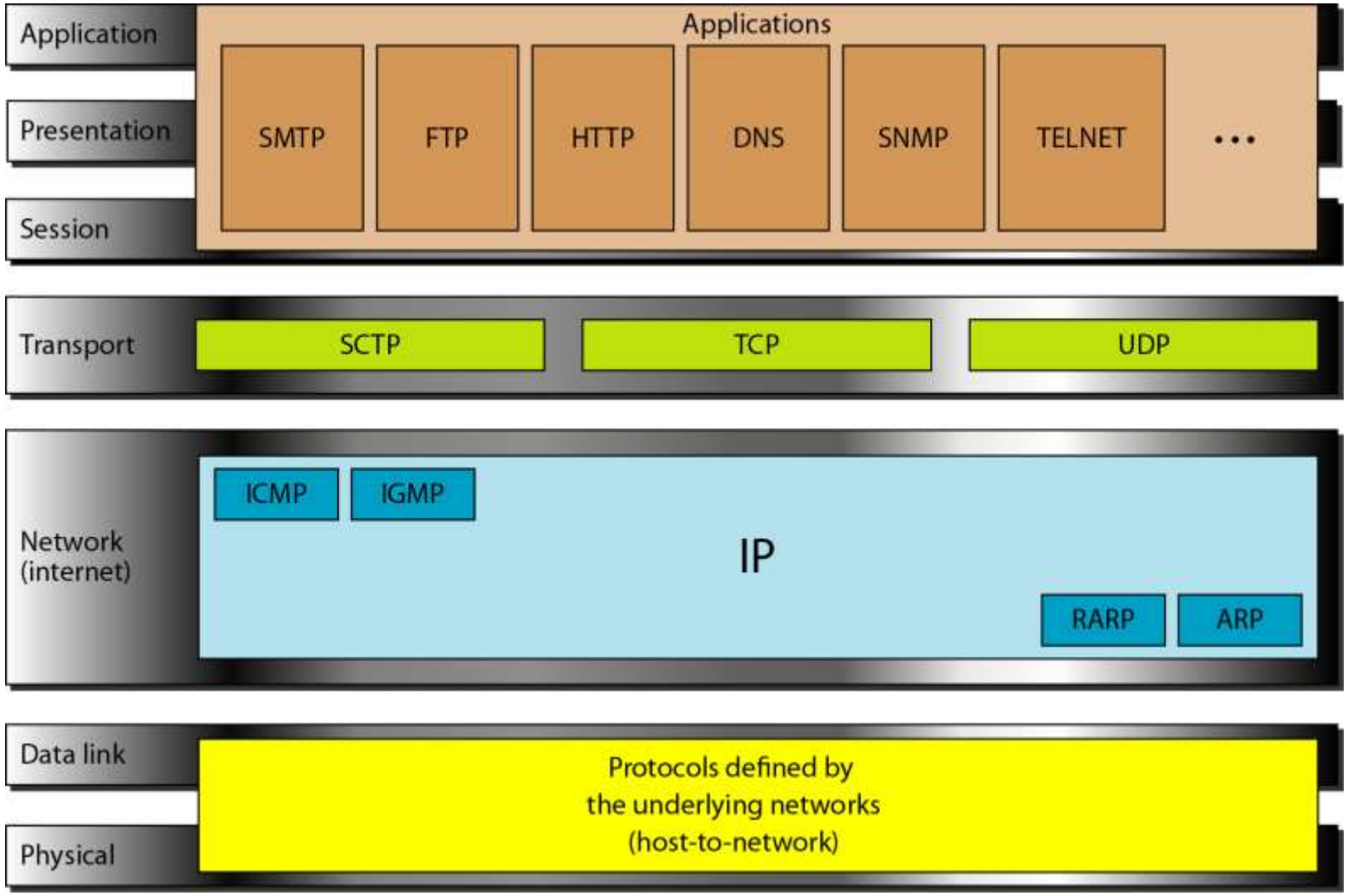
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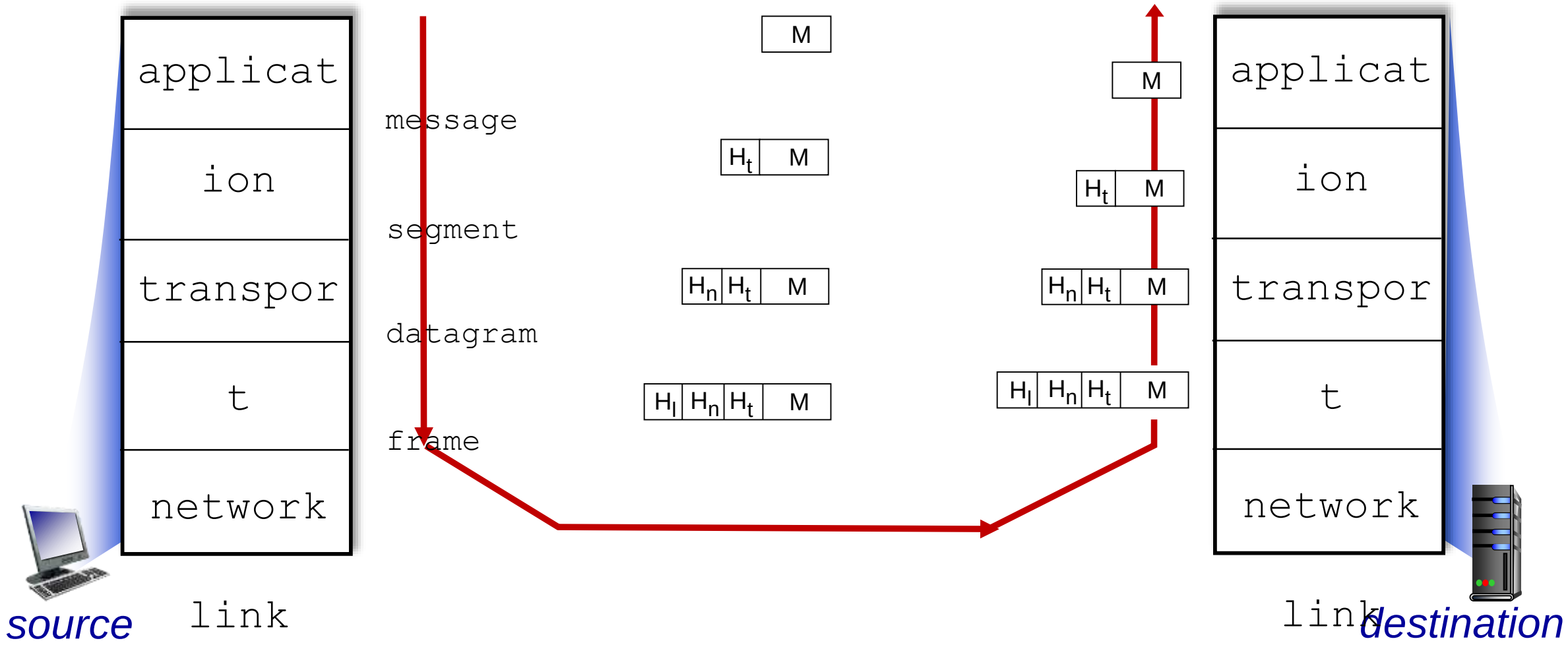
MIST

Lec 2

Figure: *TCP/IP and OSI Model*

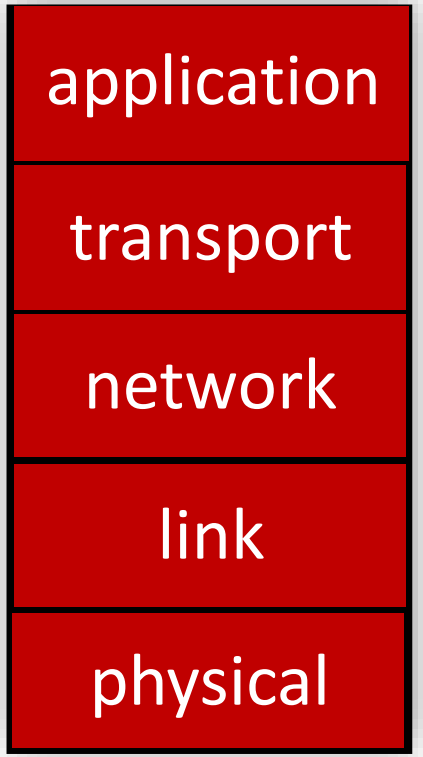


Services, Layering and Encapsulation

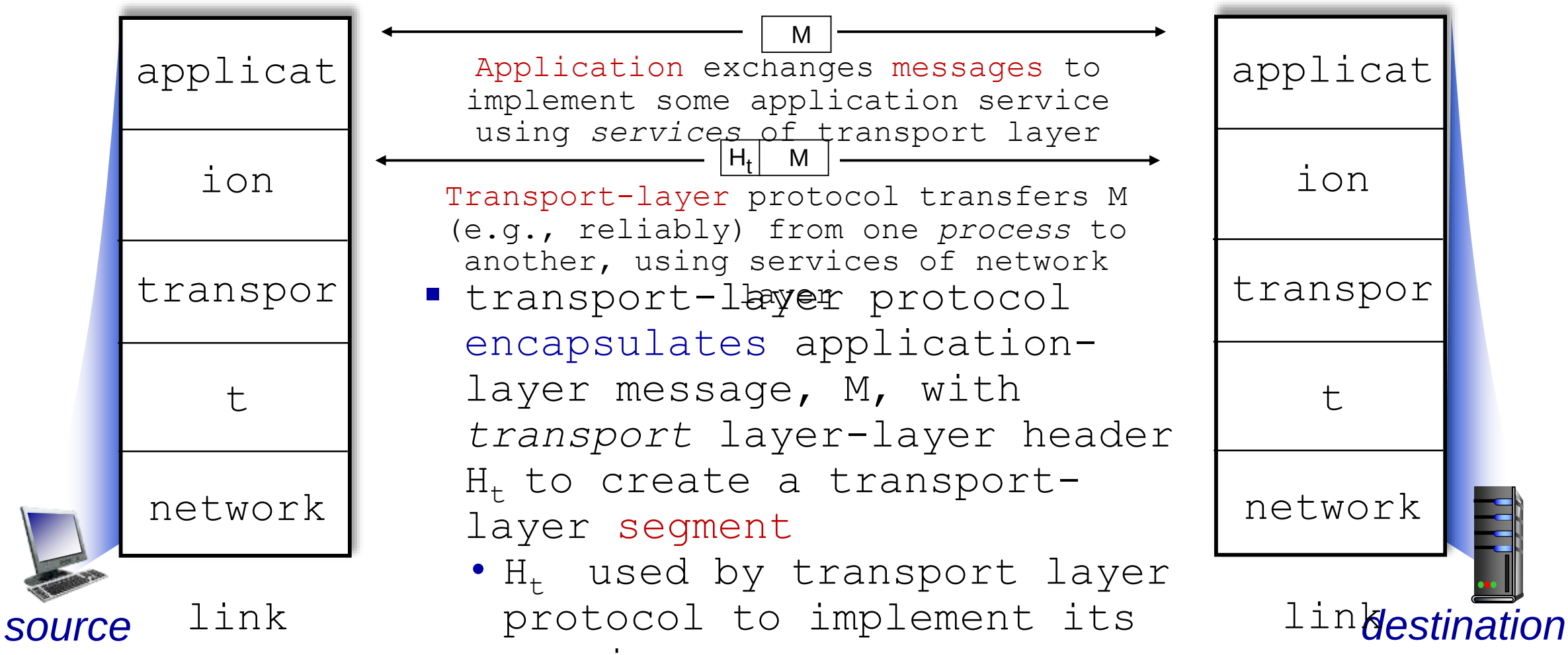


Layered Internet protocol stack

- *application:* supporting network applications
  - HTTP, IMAP, SMTP, DNS
- *transport:* process-process data transfer
  - TCP, UDP
- *network:* routing of datagrams from source to destination
  - IP, routing protocols
- *link:* data transfer between neighboring network elements
  - Ethernet, 802.11 (WiFi), PPP
- *physical:* bits “on the wire”

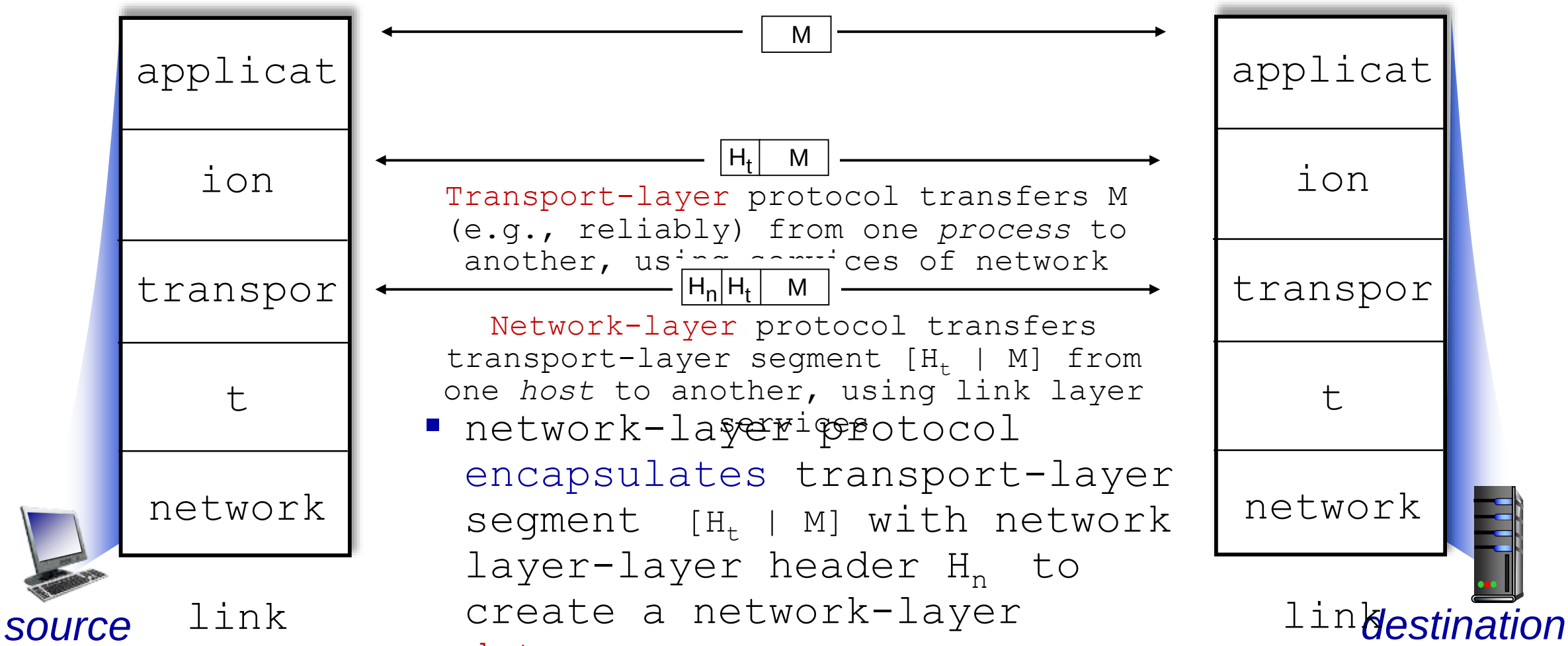


Services, Layering and Encapsulation



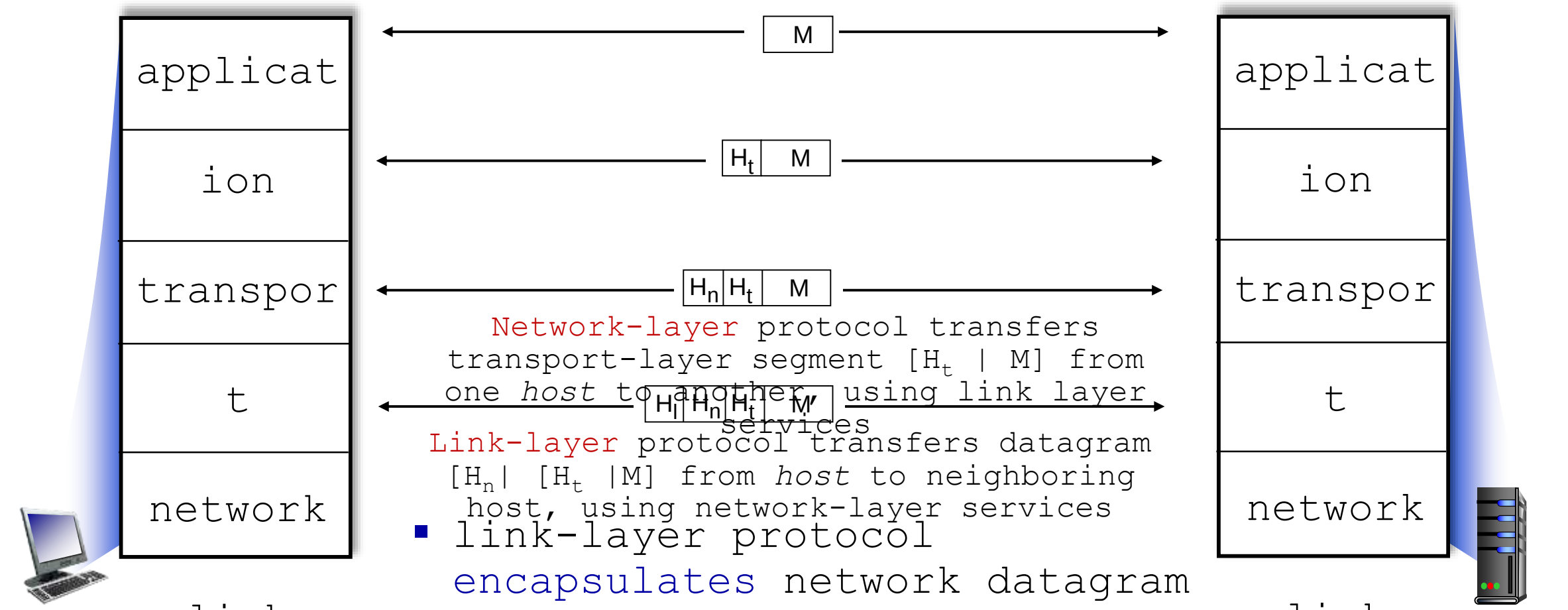
- transport-layer protocol encapsulates application-layer message, M, with *transport* layer-layer header H_t to create a transport-layer segment
  - H_t used by transport layer protocol to implement its service

Services, Layering and Encapsulation



- network-layer protocol encapsulates transport-layer segment [H_t | M] with network layer-layer header H_n to create a network-layer datagram
  - H_n used by network layer protocol to implement its

Services, Layering and Encapsulation

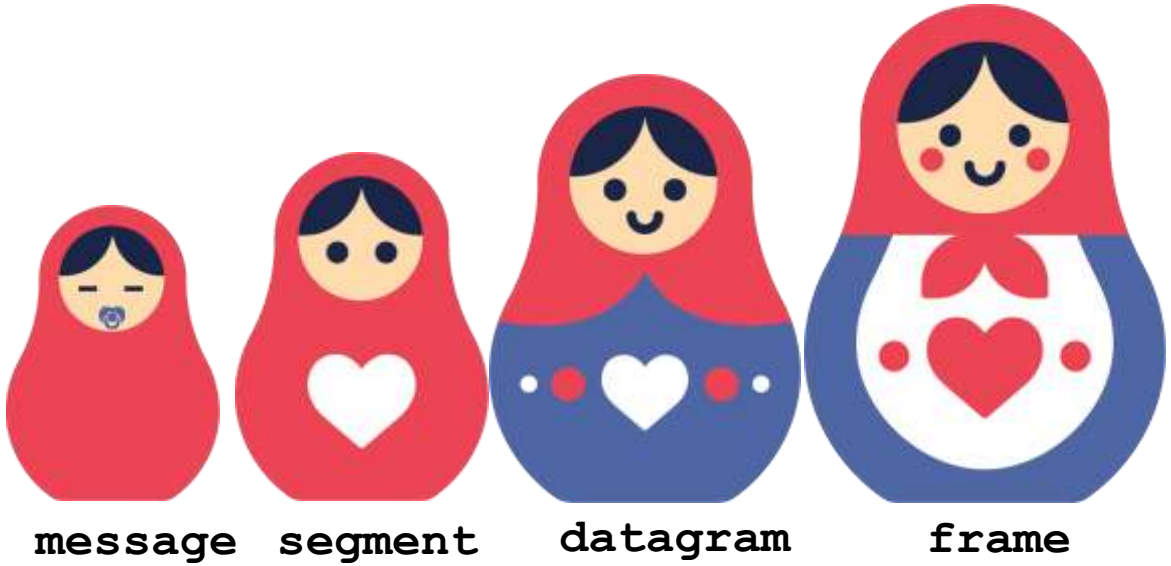


- link-layer protocol encapsulates network datagram [H_n| [H_t |M], with link-layer header H_l to create a link-layer frame

Encapsulation

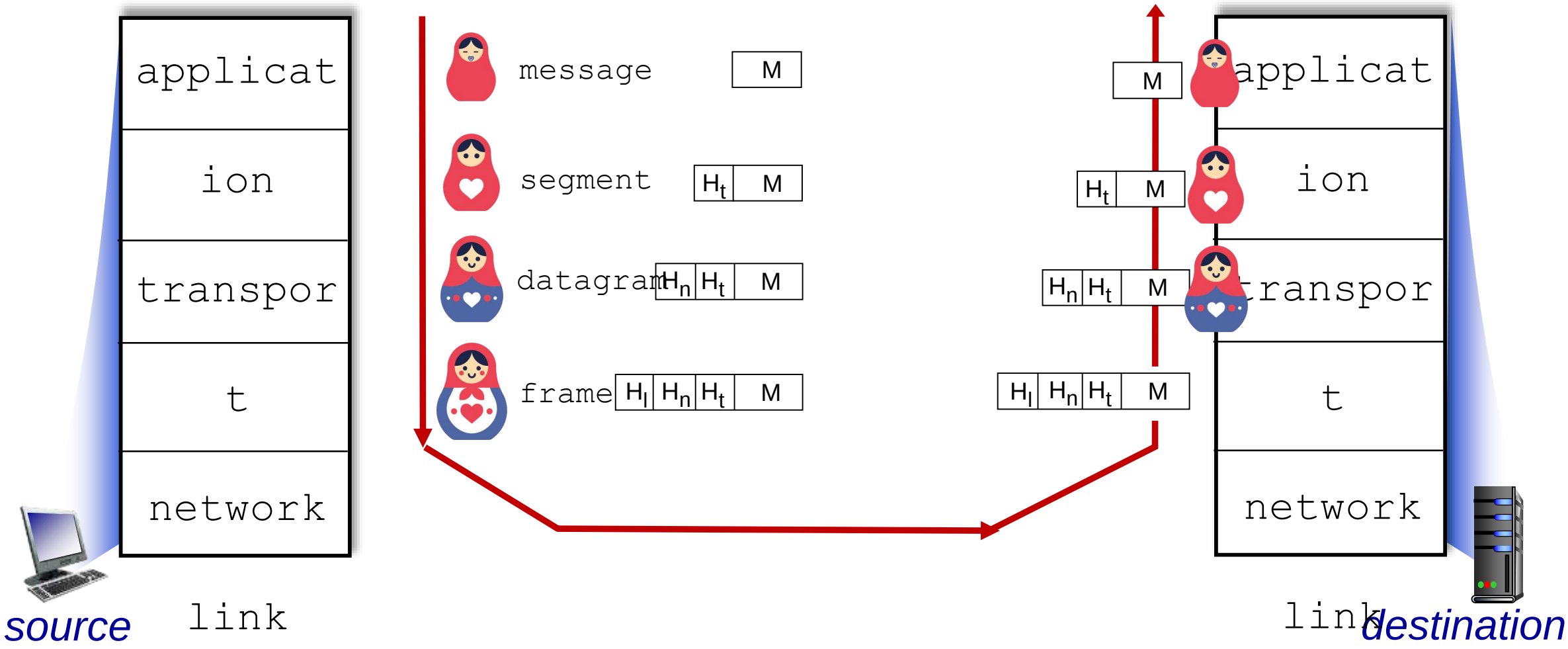
Matryoshka dolls (stacking dolls)



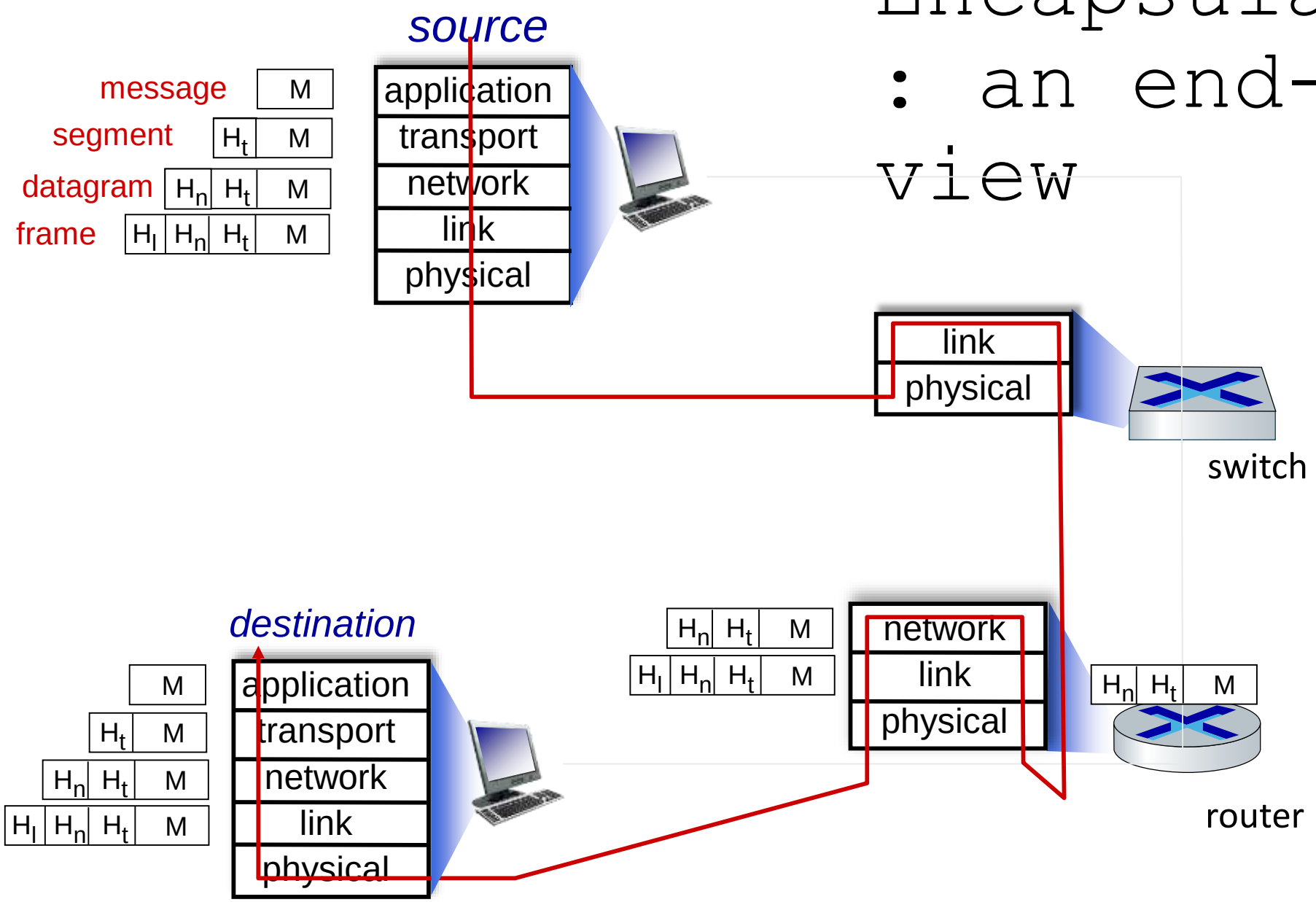


Credit: https://dribbble.com/shots/7182188-Babushka-Boi

Services, Layering and Encapsulation



Encapsulation : an end-end view



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Lec 3

2-5 ADDRESSING

Four levels of addresses are used in an internet employing the TCP/IP protocols: physical, logical, port, and specific.

Topics discussed in this section:

Physical Addresses
Logical Addresses
Port Addresses
Specific Addresses

Figure: *Addresses in TCP/IP*

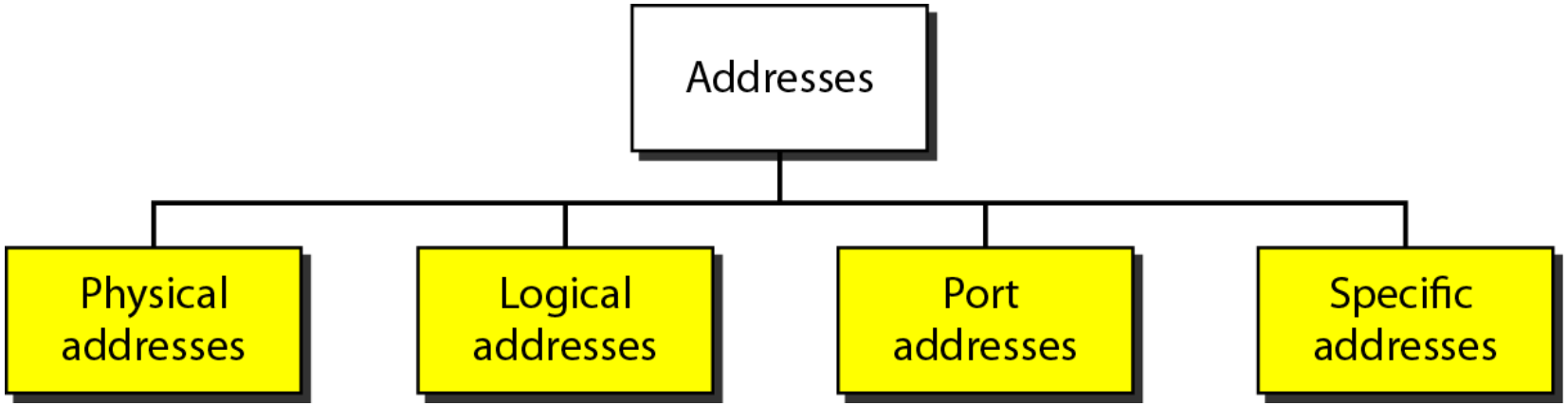


Figure: *Relationship of layers and addresses in TCP/IP*

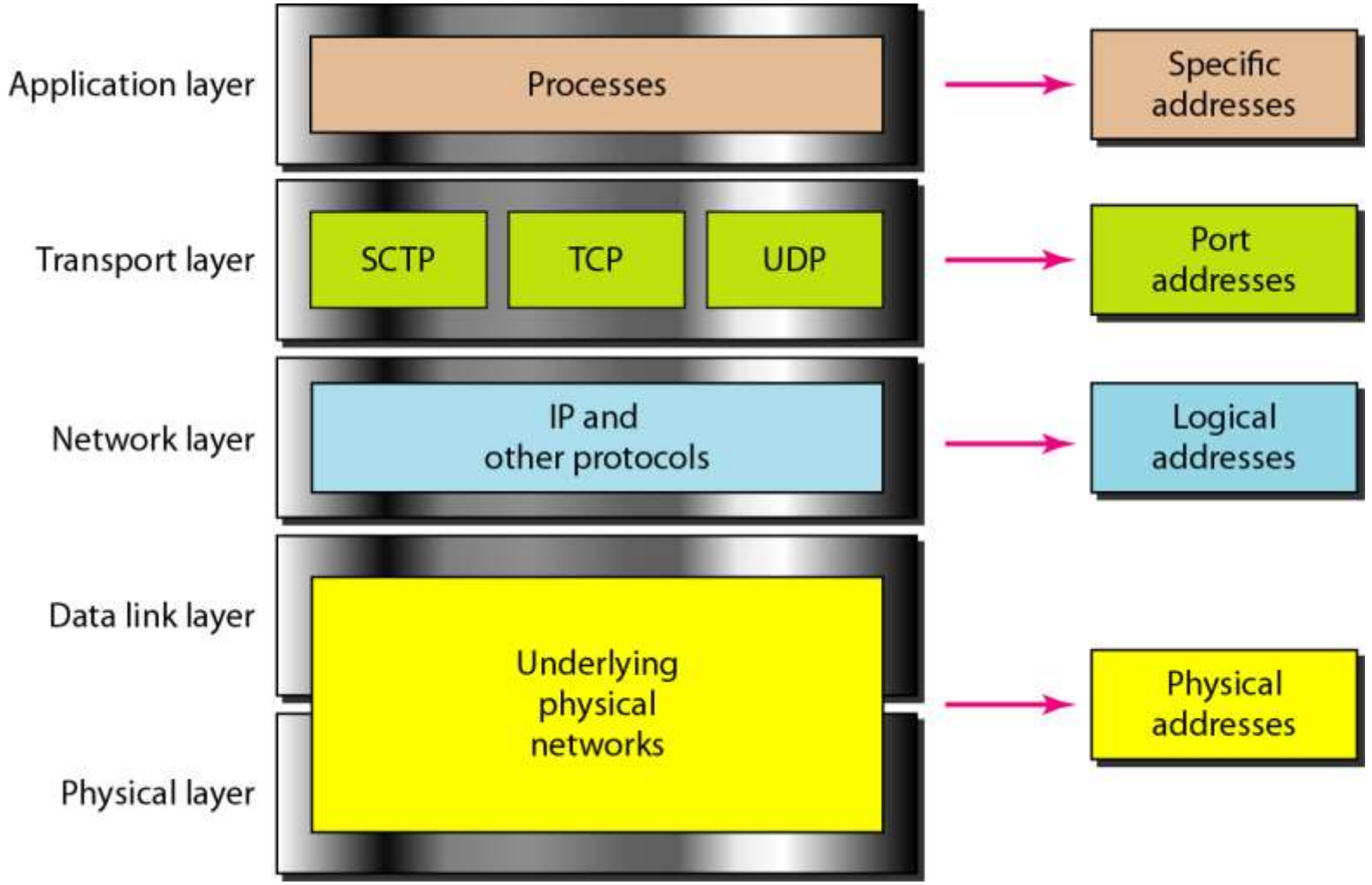
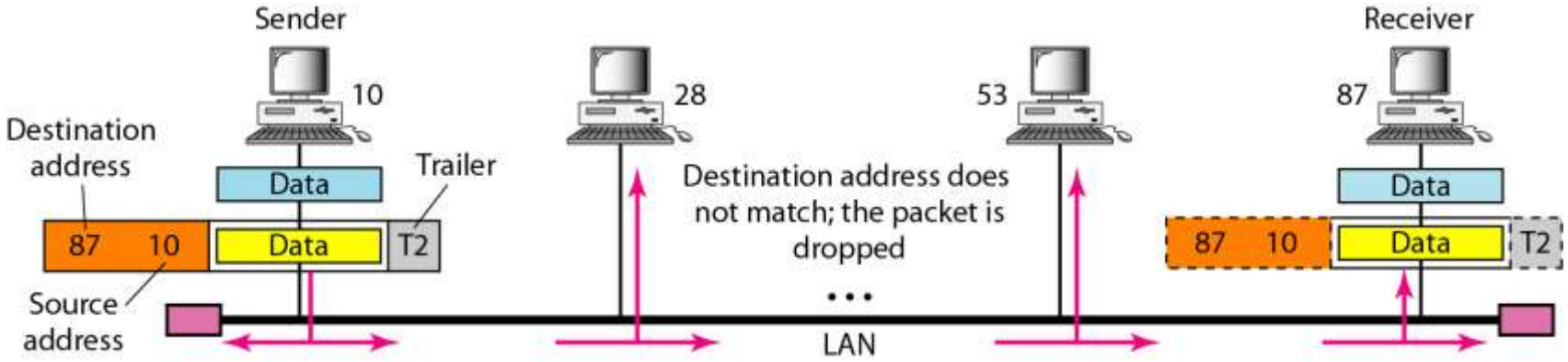


Figure: *Physical Addresses*



In Figure a node with physical address 10 sends a frame to a node with physical address 87. The two nodes are connected by a link (bus topology LAN). As the figure shows, the computer with physical address 10 is the sender, and the computer with physical address 87 is the receiver.



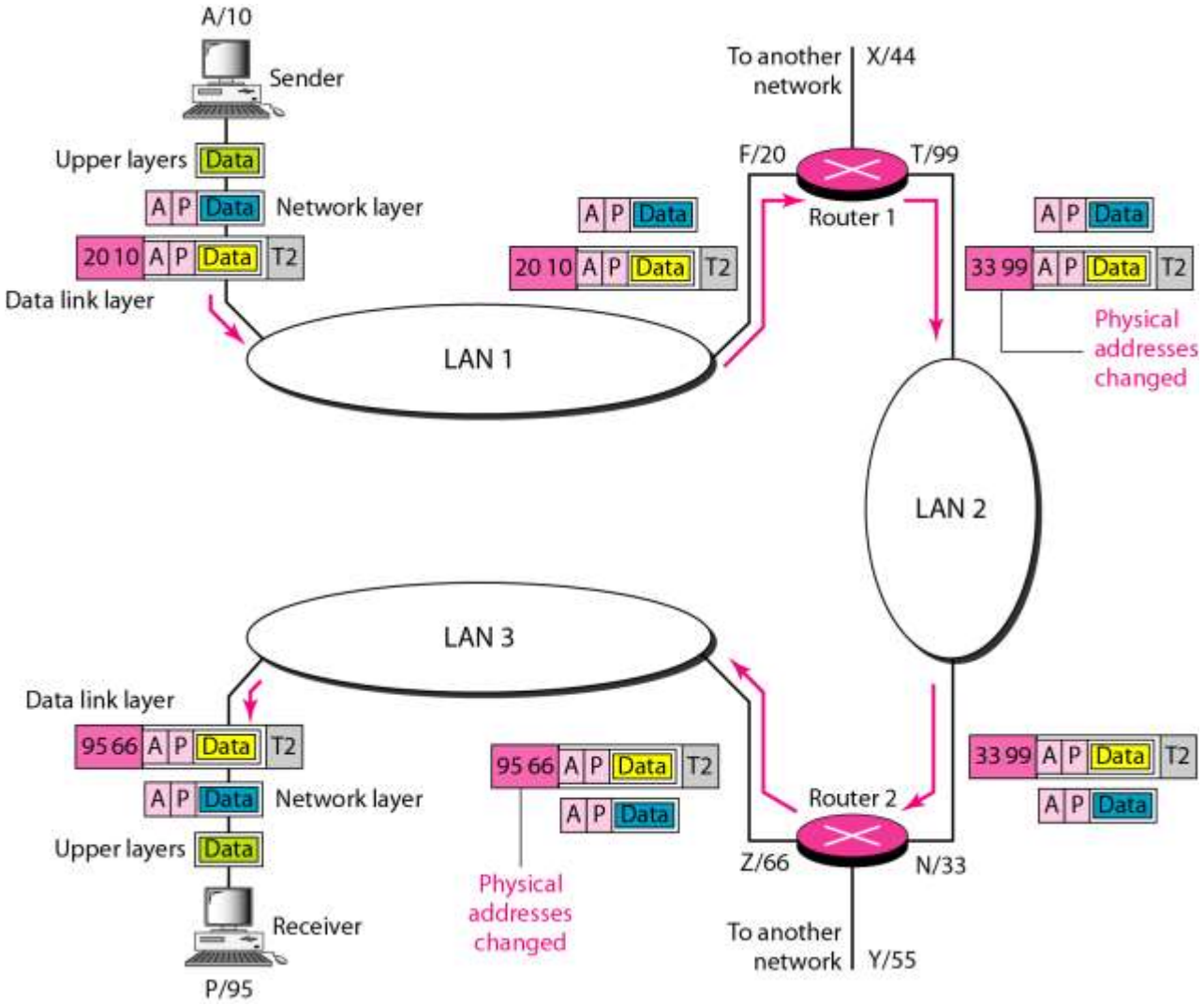
Example 2.2

As we will see in Chapter 13, most local-area networks use a 48-bit (6-byte) physical address written as 12 hexadecimal digits; every byte (2 hexadecimal digits) is separated by a

07:01:02:01:2C:4B

A 6-byte (12 hexadecimal digits) physical address.

Figure: Logical Address (*IP addresses – 192.168.1.10*)

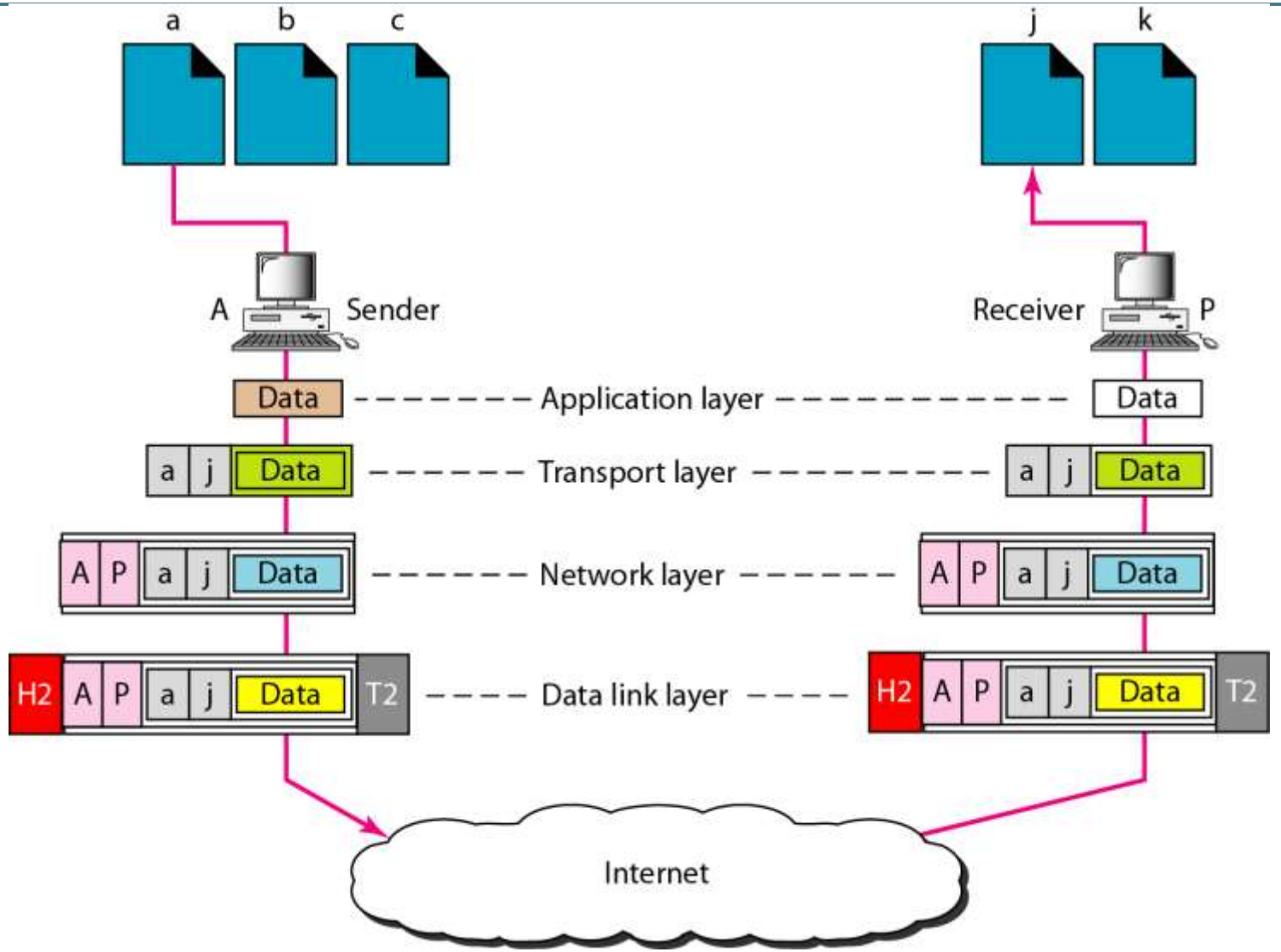




Example 2.3

Figure shows a part of an internet with two routers connecting three LANs. Each device (computer or router) has a pair of addresses (logical and physical) for each connection. In this case, each computer is connected to only one link and therefore has only one pair of addresses. Each router, however, is connected to three networks (only two are shown in the figure). So each router has three pairs of addresses, one for each connection.

Figure: *Port Addresses (HTTP-80, HTTPS-443, FTP-21)*





Example 2.4

Figure shows two computers communicating via the Internet. The sending computer is running three processes at this time with port addresses a, b, and c. The receiving computer is running two processes at this time with port addresses j and k. Process a in the sending computer needs to communicate with process j in the receiving computer. Note that although physical addresses change from hop to hop, logical and port addresses remain the same from the source to



Example 2.5

As we will see in Chapter 23, a port address is a 16-bit address represented by one decimal number as shown.

0 to 65535

A 16-bit port address represented as one single number.

Note

The physical addresses change from hop to hop,

but the logical and port addresses usually remain the same.



Specific Addresses

Human-readable address used at the application layer.
Identifies a particular resource, service, mailbox, or webpage.
Examples:
Email: john@example.com

Specific Address: www.google.com
Logical Address: DNS translates it to an IP, e.g., 142.250.190.78
Port Address: Browser uses port 443 (HTTPS)
Physical Address: Your computer sends frames using MAC addresses on the local network.
Memory Tip:
MAC → IP → Port → Resource
(Physical → Logical → Port → Specific)

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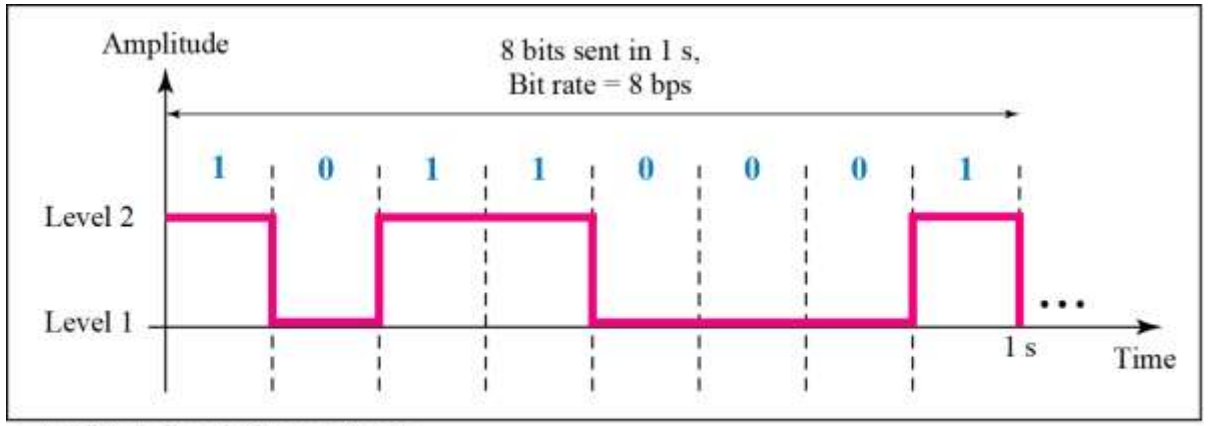
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Lec 4 (Digital Signal)

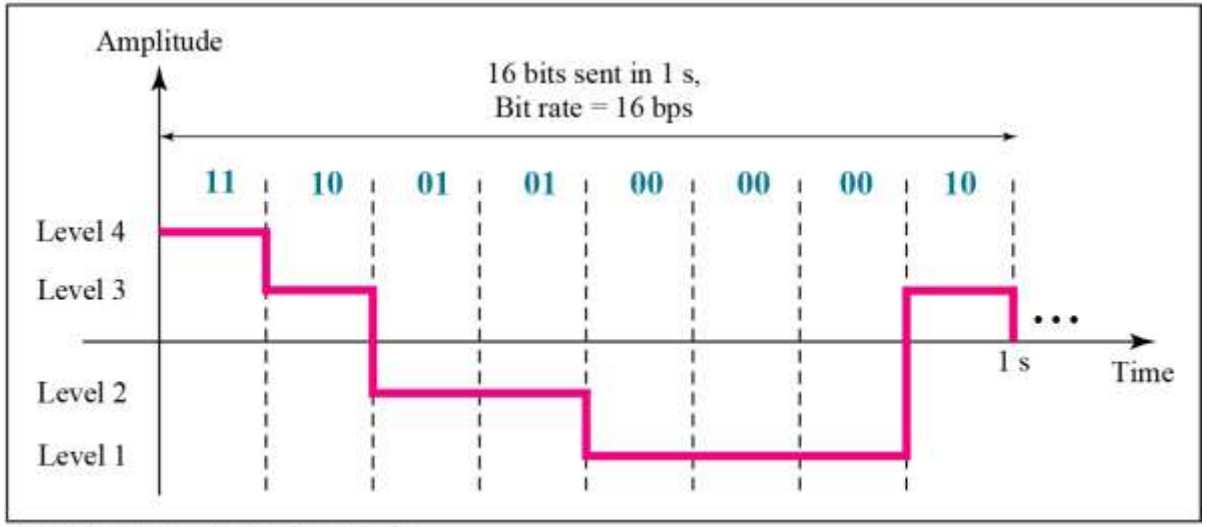
3-3 DIGITAL SIGNALS

In addition to being represented by an analog signal, information can also be represented by a digital signal. For example, a 1 can be encoded as a positive voltage and a 0 as zero voltage. A digital signal can have more than two levels. In this case, we can send more than 1 bit for each level.

Figure 3.16 Two digital signals: one with two signal levels and with four signal levels



a. A digital signal with two levels



b. A digital signal with four levels



Bit Rate

Note:

- Bit Rate = $\text{Log}_2 L$
- The number of bits sent per level needs to be an integer as

Example 3.16

A digital signal has eight levels. How many bits are needed per level?

We calculate the number of bits from the formula

$$\text{Number of bits per level} = \log_2 8 = 3$$

Each signal level is represented by 3 bits.

Example 3.17

A digital signal has nine levels. How many bits are needed per level?

We calculate the number of bits by using the formula.

Each signal level is represented by 3.17 bits.

However, this answer is not realistic. The number of bits sent per level needs to be an integer as well as a power of 2.

For this example, 4 bits can represent one level.



Example 3.18

Assume we need to download text documents at the rate of 100 pages per second. What is the required bit rate of the channel?

Solution

A page is an average of 24 lines with 80 characters in each line. If we assume that one character requires 8 bits, the bit rate is

$$100 \times 24 \times 80 \times 8 = 1{,}536{,}000 \text{ bps} = 1.536 \text{ Mbps}$$

Example 3.19

A digitized voice channel is made by digitizing a 4-kHz bandwidth analog voice signal. We need to sample the signal at twice the highest frequency (two samples per hertz). We assume that each sample requires 8 bits. What is the required bit rate?

Solution

The bit rate can be calculated as:

$$\text{Bit Rate} = \text{Bandwidth} \times \text{No of Sample} \times \text{Sample Size}$$
$$= 4 \text{ kHz} \times 2 \times 8 \text{ bit}$$
$$= 64 \text{ k bit Hz}$$
$$= 64 \text{ kbps}$$

Example 3.20

What is the bit rate for high-definition TV (HDTV)?

Solution

HDTV uses digital signals to broadcast high quality video signals. The HDTV screen is normally a ratio of 16 : 9. There are 1920 by 1080 pixels per screen, and the screen is renewed 30 times per second. Twenty-four bits represents one color pixel.

$$1920 \times 1080 \times 30 \times 24 = 1{,}492{,}992{,}000 \text{ or } 1.5 \text{ Gbps}$$

The TV stations reduce this rate to 20 to 40 Mbps through compression.



Bit Length

Note:

- Bit Length = Propagation Speed x Bit Duration = Propagation Speed / Bandwidth
- Bit Duration = 1 / Bandwidth



What is the length of a bit in a channel with a propagation speed 2×10^8 m/s, if the channel Bandwidth is:
a. 1 Mbps. b. 10 Mbps.

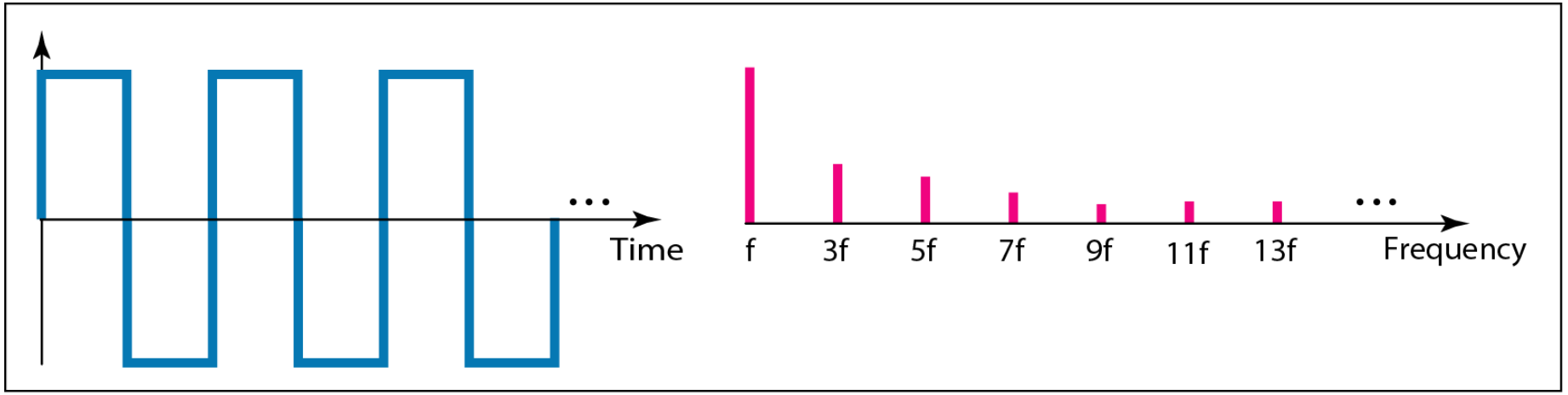
Solution

a. Bit Length = Propagation Speed / Bandwidth

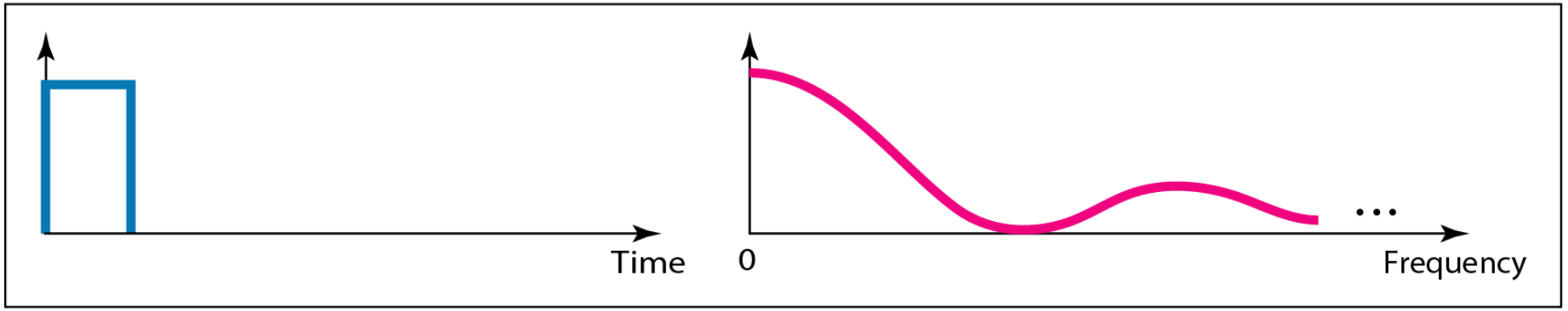
$= 2 \times 10^8$ m/s / 1 Mbps = 200 m

b. Bit Length = 20 m

Figure 3.17 The time and frequency domains of periodic and nonpe digital signals



a. Time and frequency domains of periodic digital signal



b. Time and frequency domains of nonperiodic digital signal

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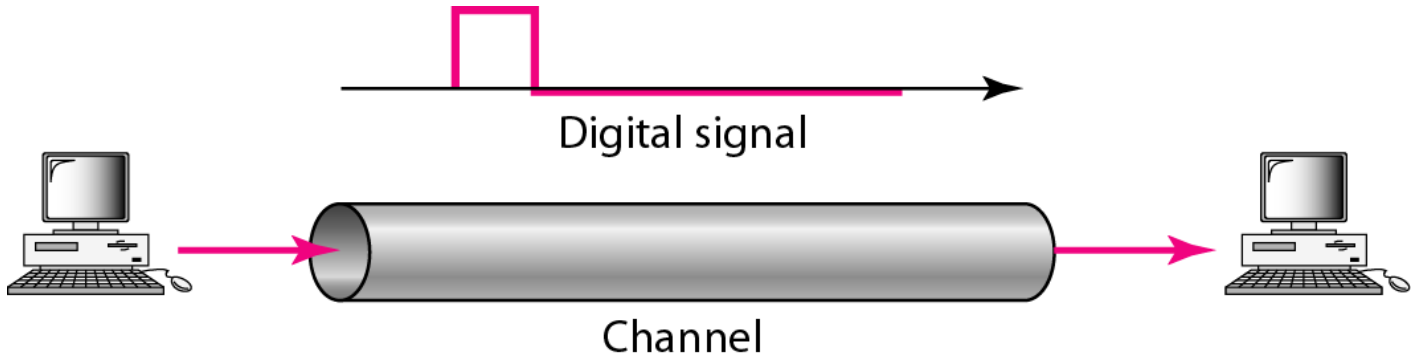
Lec 5 (Digital Signal Transmission)



Transmission of Digital Signals

1. Baseband Transmission
2. Broadband Transmission

Figure 3.18 Baseband transmission



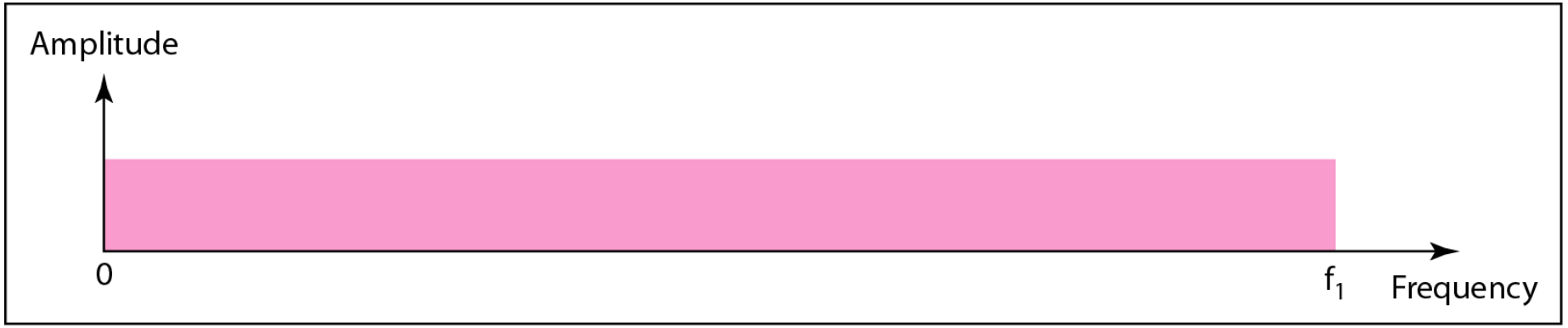
Note: Dedicated low-pass channel



Note

A digital signal is a composite analog signal with an infinite bandwidth.

Figure 3.19 Bandwidths of two low-pass channels

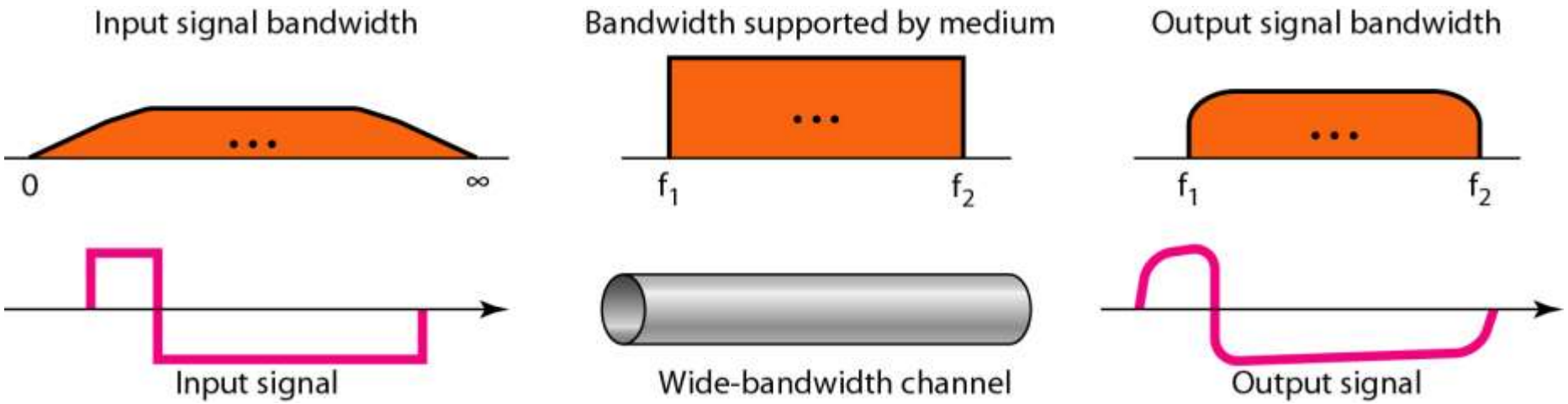


a. Low-pass channel, wide bandwidth



b. Low-pass channel, narrow bandwidth

Figure 3.20 Baseband transmission using a dedicated medium (Low-Pass Channel with Wide Bandwidth)



Note

Baseband transmission of a digital signal that preserves the shape of the digital signal is possible only if we have a low-pass channel with an infinite or very wide bandwidth.



Example 3.21

An example of a dedicated channel where the entire bandwidth of the medium is used as one single channel is a LAN. Almost every wired LAN today uses a dedicated channel for two stations communicating with each other. In a bus topology LAN with multipoint connections, only two stations can communicate with each other at each moment in time (timesharing); the other stations need to refrain from sending data. In a star topology LAN, the entire channel between each station and the hub is used for communication between these two entities. We study LANs in Chapter 14.

Figure 3.21 Rough approximation of a digital signal using the fi
for worst case (Low-Pass Channel with Lim

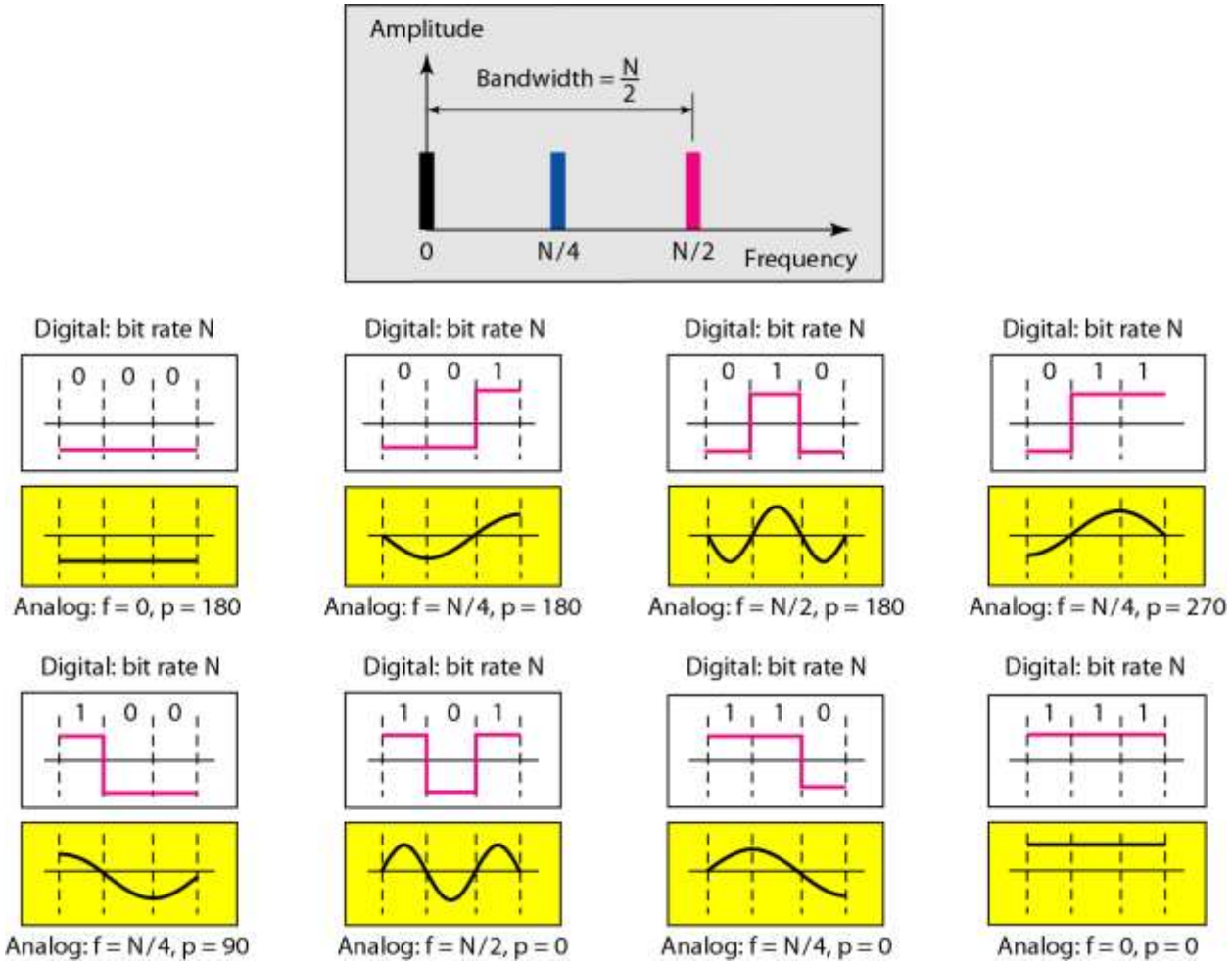
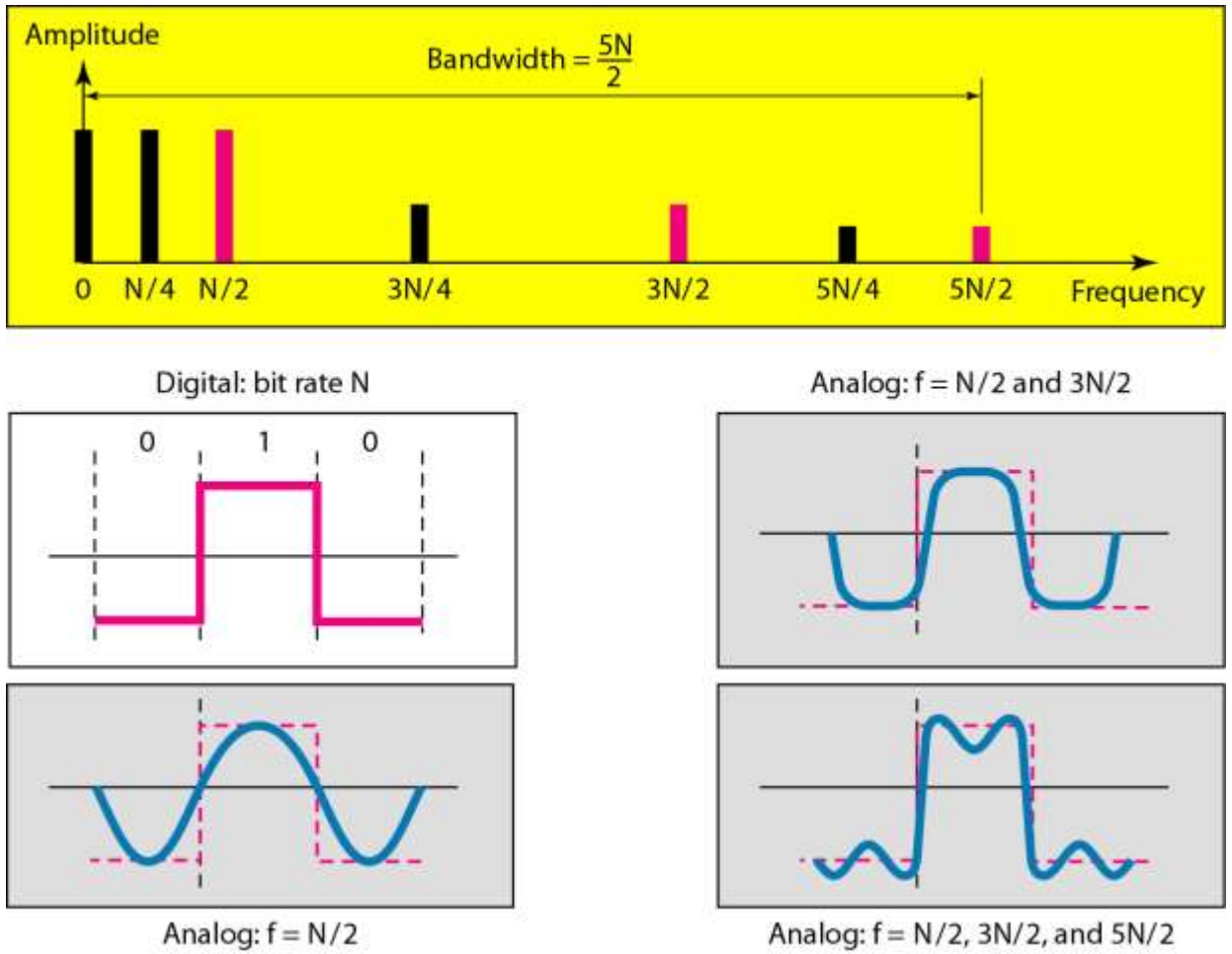


Figure 3.22 Simulating a digital signal with first three harmoni



Note

In baseband transmission, the required bandwidth is proportional to the bit rate; if we need to send bits faster, we need more bandwidth.

Table 3.2 Bandwidth requirements

Bit Rate	*Harmonic 1*	*Harmonics 1, 3*	*Harmonics 1, 3, 5*
$n = 1$ kbps	$B = 500$ Hz	$B = 1.5$ kHz	$B = 2.5$ kHz
$n = 10$ kbps	$B = 5$ kHz	$B = 15$ kHz	$B = 25$ kHz
$n = 100$ kbps	$B = 50$ kHz	$B = 150$ kHz	$B = 250$ kHz

Note:
B = N / 2

Example 3.22

What is the required bandwidth of a low-pass channel if we need to send 1 Mbps by using baseband transmission?

Solution

The answer depends on the accuracy desired.

a. The minimum bandwidth, is B = bit rate /2, or 500 kHz.

b. A better solution is to use the first and the third harmonics with B = 3 × 500 kHz = 1.5 MHz.

c. Still a better solution is to use the first, third, and fifth harmonics with B = 5 × 500 kHz = 2.5 MHz.

Example 3.22

We have a low-pass channel with bandwidth 100 kHz. What is the maximum bit rate of this channel?

Solution

The maximum bit rate can be achieved if we use the first harmonic. The bit rate is 2 times the available bandwidth, or 200 kbps.

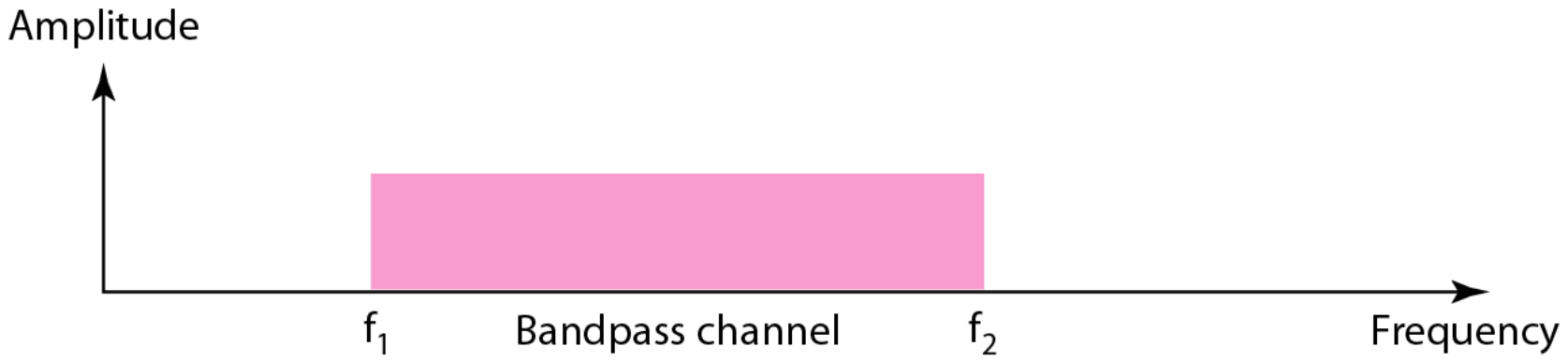
Note:

B = N / 2

N = 2B

Figure 3.23 Bandwidth of a bandpass channel

Broadband Transmission (Using Modulation)

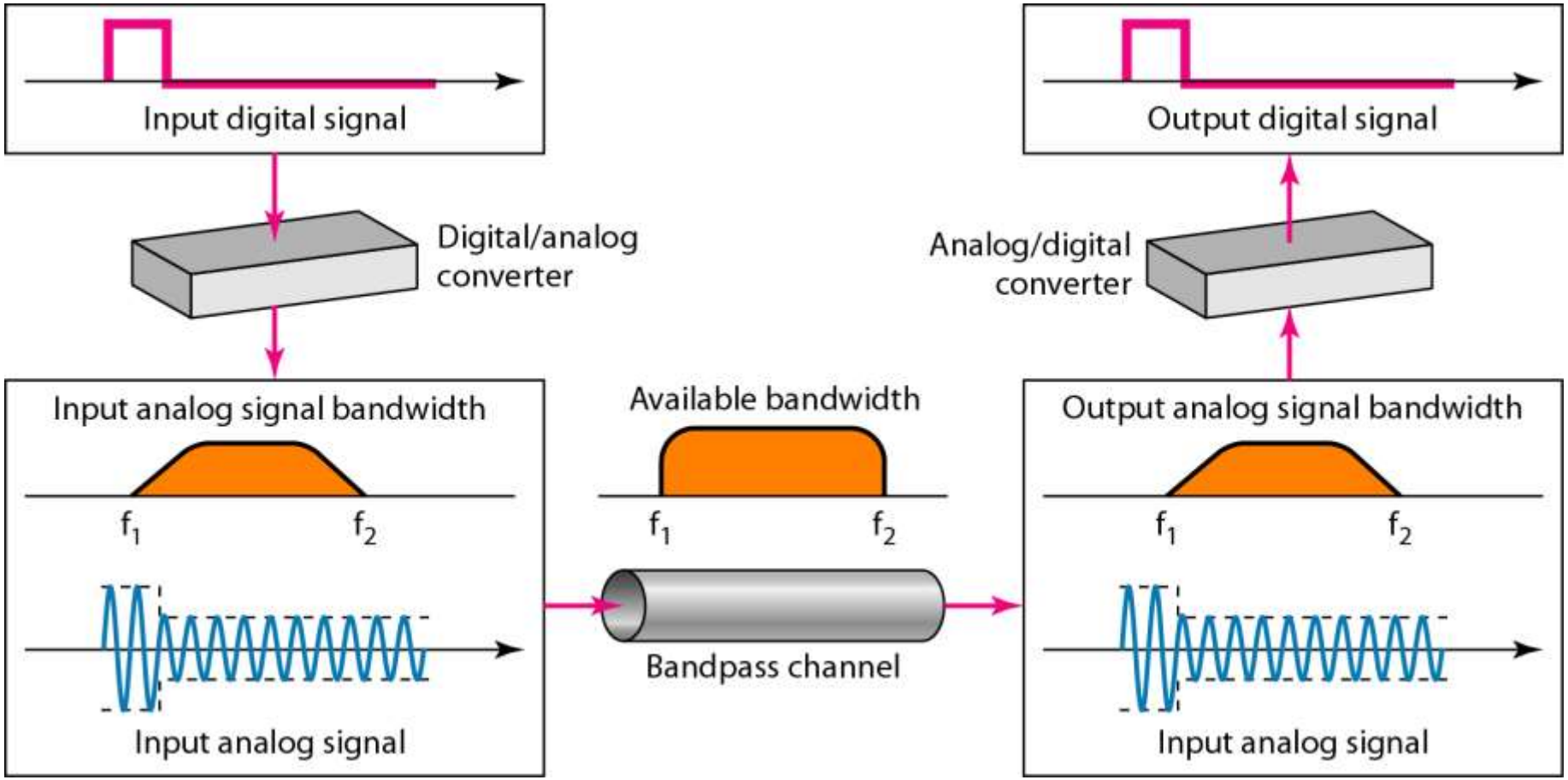


Note: Bandpass Channel is a channel with a bandwidth does not start from Ze

Note

If the available channel is a bandpass channel, we cannot send the digital signal directly to the channel;
we need to convert the digital signal to an analog signal before transmission.

Figure 3.24 Modulation of a digital signal for transmission on a channel





Example 3.24

An example of broadband transmission using modulation is the sending of computer data through a telephone subscriber line, the line connecting a resident to the central telephone office. These lines are designed to carry voice with a limited bandwidth. The channel is considered a bandpass channel. We convert the digital signal from the computer to an analog signal, and send the analog signal. We can install two converters to change the digital signal to analog and vice versa at the receiving end. The converter, in this case, is called a modem which we discuss in detail in Chapter 5.



Example 3.25

A second example is the digital cellular telephone. For better reception, digital cellular phones convert the analog voice signal to a digital signal (see Chapter 16). Although the bandwidth allocated to a company providing digital cellular phone service is very wide, we still cannot send the digital signal without conversion. The reason is that we only have a bandpass channel available between caller and callee. We need to convert the digitized voice to a composite analog signal before sending.

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Lec 6 (Transmission Impairment)

3-4 TRANSMISSION IMPAIRMENT

Signals travel through transmission media, which are not perfect. The imperfection causes signal impairment. This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium. What is sent is not what is received. Three causes of impairment are attenuation, distortion, and noise.

Topics discussed in this section:

Attenuation
Distortion
Noise

Figure 3.25 Causes of impairment

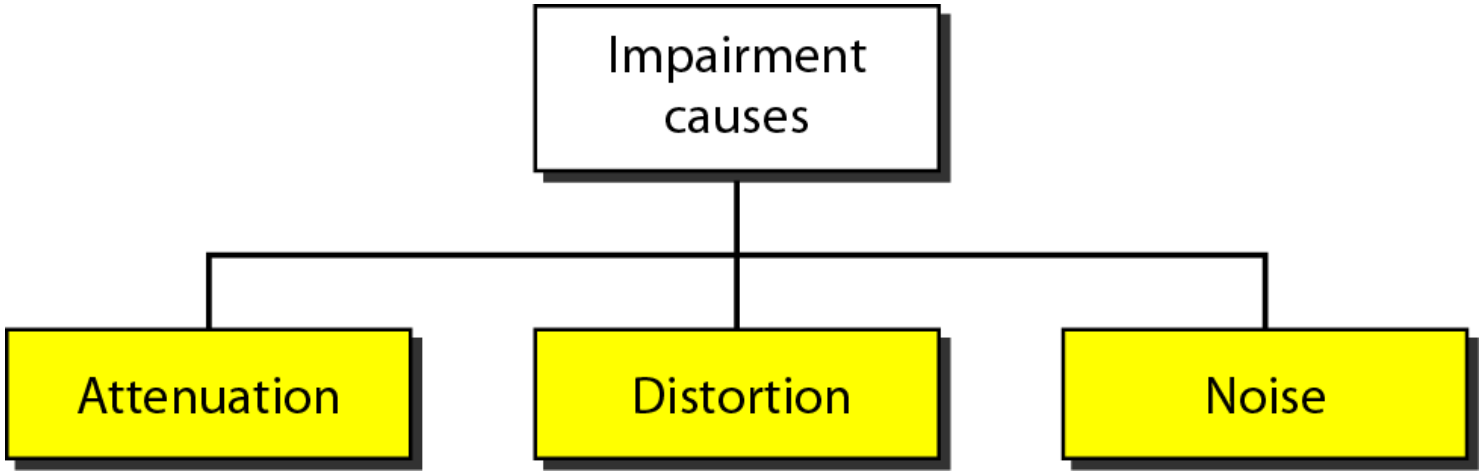
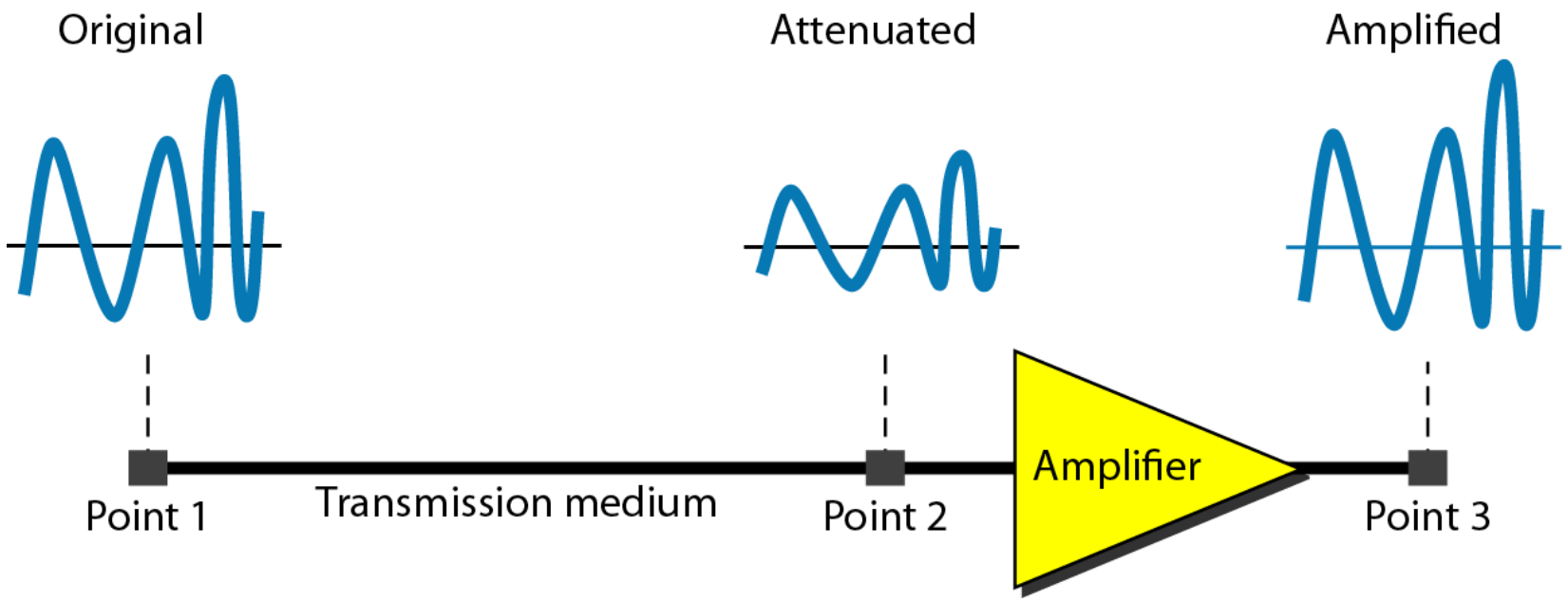


Figure 3.26 Attenuation



Example 3.26

Suppose a signal travels through a transmission medium and its power is reduced to one-half. This means that P2 is (1/2)P1. In this case, the attenuation (loss of power) can be calculated as

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{0.5 P_1}{P_1} = 10 \log_{10} 0.5 = 10(-0.3) = -3 \text{ dB}$$

A loss of 3 dB (−3 dB) is equivalent to losing one-half the power.

Note:
dB = decibel

$$dB = 10 \log_{10} \left(\frac{P_2}{P_1} \right)$$

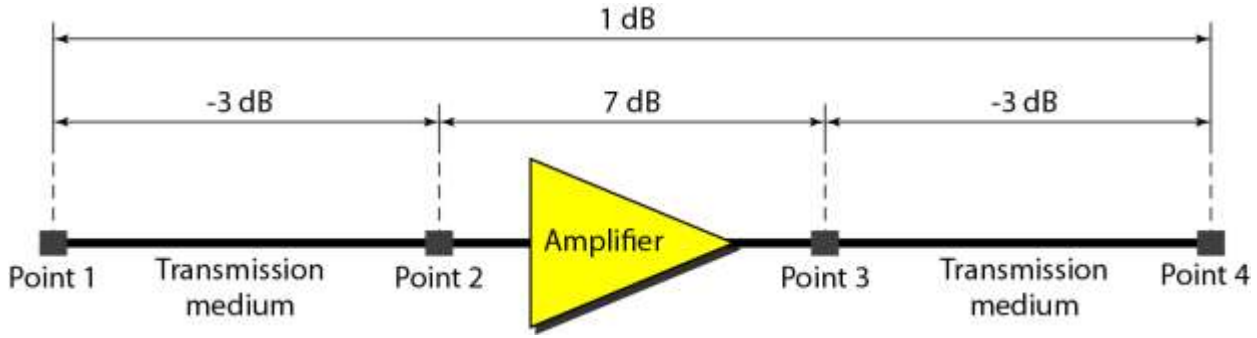
Example 3.27

A signal travels through an amplifier, and its power is increased 10 times. This means that $P_2 = 10P_1$. In this case, the amplification (gain of power) can be calculated as

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{10P_1}{P_1}$$

$$= 10 \log_{10} 10 = 10(1) = 10 \text{ dB}$$

Figure 3.27 Decibels for Example 3.28



One reason that engineers use the decibel to measure the changes in the strength of a signal is that decibel numbers can be added (or subtracted) when we are measuring several points (cascading) instead of just two. In Figure 3.27 a signal travels from point 1 to point 4. In this case, the decibel value can be calculated as

$$\text{dB} = -3 + 7 - 3 = +1$$

Example 3.29

Sometimes the decibel is used to measure signal power in milliwatts. In this case, it is referred to as dB_m and is calculated as $dB_m = 10 \log_{10} P_m$, where P_m is the power in milliwatts. Calculate the power of a signal with $dB_m = -30$.

Solution

We can calculate the power in the signal as

$$dB_m = 10 \log_{10} P_m = -30$$

$$\log_{10} P_m = -3 \qquad P_m = 10^{-3} \text{ mW}$$

Note:
Pm = Power in milliwatts

Example 3.30

The loss in a cable is usually defined in decibels per kilometer (dB/km). If the signal at the beginning of a cable with −0.3 dB/km has a power of 2 mW, what is the power of the signal at 5 km?

Solution

The loss in the cable in decibels is 5 × (−0.3) = −1.5 dB. We can calculate the power as

$$\text{dB} = 10 \log_{10} \frac{P_2}{P_1} = -1.5$$

$$\frac{P_2}{P_1} = 10^{-0.15} = 0.71$$

$$P_2 = 0.71 P_1 = 0.7 \times 2 = 1.4 \text{ mW}$$

Figure 3.28 Distortion

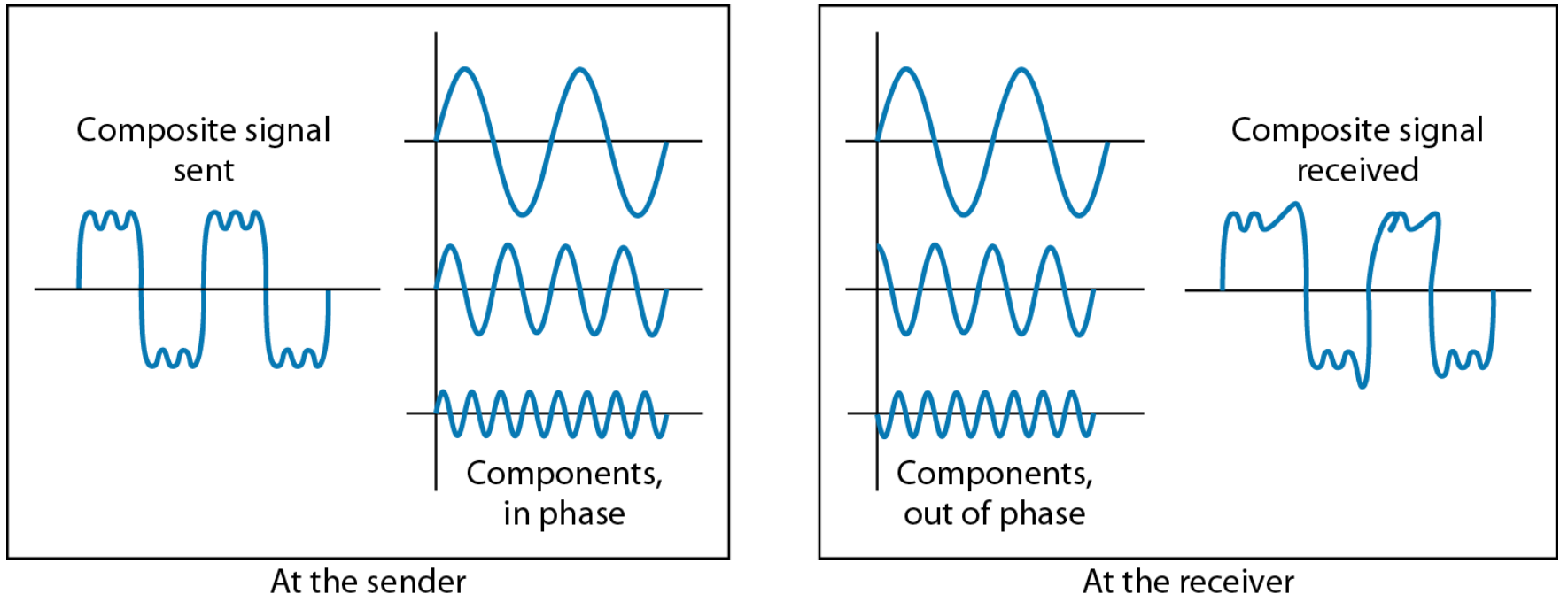
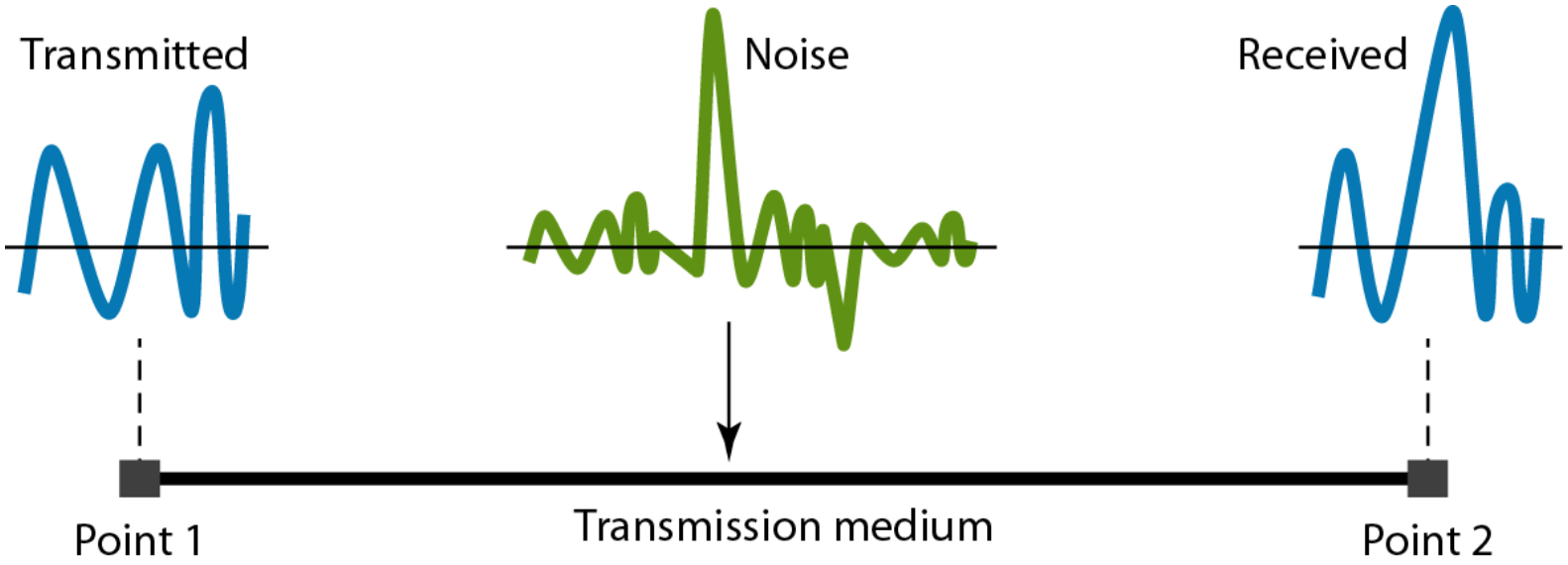


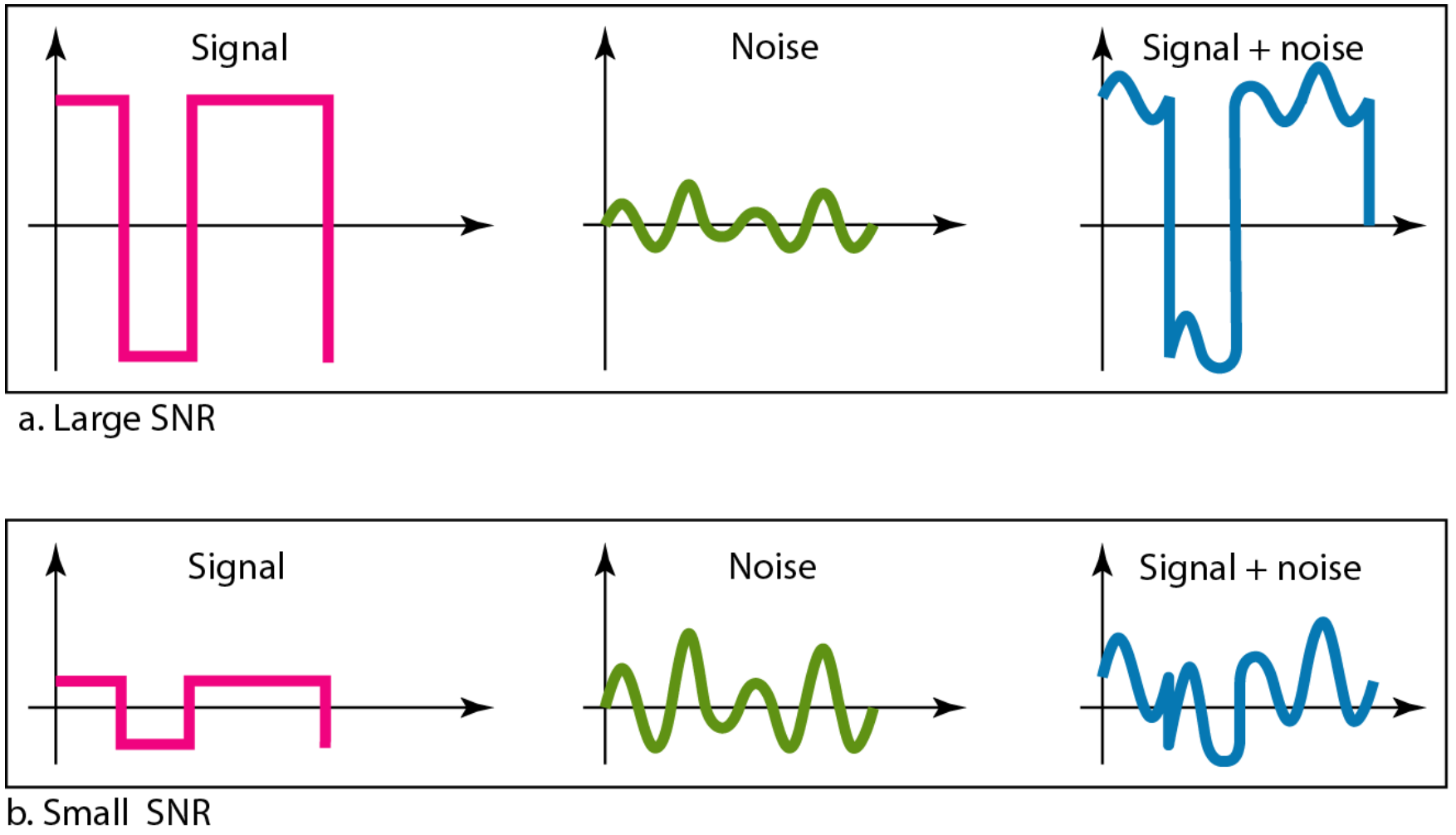
Figure 3.29 Noise



Note:

1. Thermal Noise – Random motion of Electron.
2. Induced Noise – Equipment.
3. Crosstalk – Crossover wires.
4. Impulse Noise – Spike.

Figure 3.30 Two cases of SNR: a high SNR and a low SNR (Signal t



Example 3.31

The power of a signal is 10 mW and the power of the noise is 1 μW; what are the values of SNR and SNR_{dB} ?

Solution
The values of SNR and SNRdB can be calculated as follows:

$$SNR = \frac{10{,}000\ \mu W}{1\ mW} = 10{,}000$$

$$SNR_{dB} = 10 \log_{10} 10{,}000 = 10 \log_{10} 10^4 = 40$$

Note:
Signal to Noise Ratio = SNR
= Average Signal Power / Average Noise Power

Example 3.32

The values of SNR and SNRdB for a noiseless channel are

$$\text{SNR} = \frac{\text{signal power}}{0} = \infty$$

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \infty = \infty$$

We can never achieve this ratio in real life; it is an ideal.

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Lec 7 (Digital-to-digital Conversion)



Data Communications and Networking Fourth Edition

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Chapter 4

Digital Transmission

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4-1 DIGITAL-TO-DIGITAL CONVERSION

In this section, we see how we can represent digital data by using digital signals. The conversion involves three techniques: line coding, block coding, and scrambling. Line coding is always needed; block coding and scrambling may or may not be needed.

Topics discussed in this section:

- Line Coding
- Line Coding Schemes
- Block Coding
- Scrambling

Figure 4.1 *Line coding and decoding*

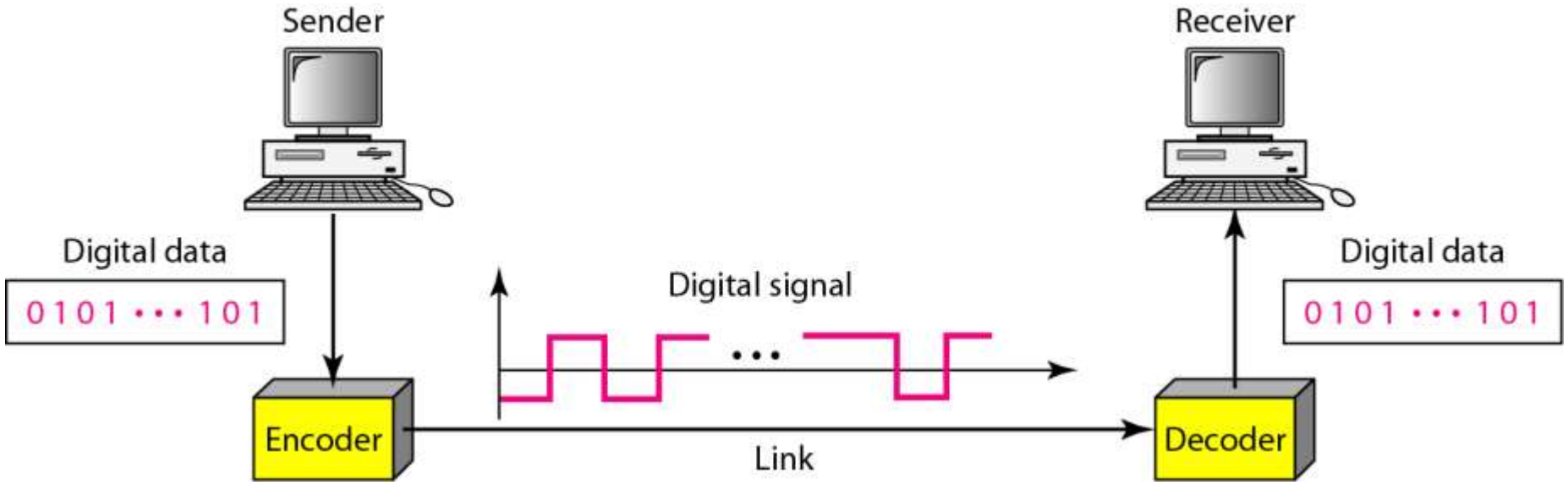
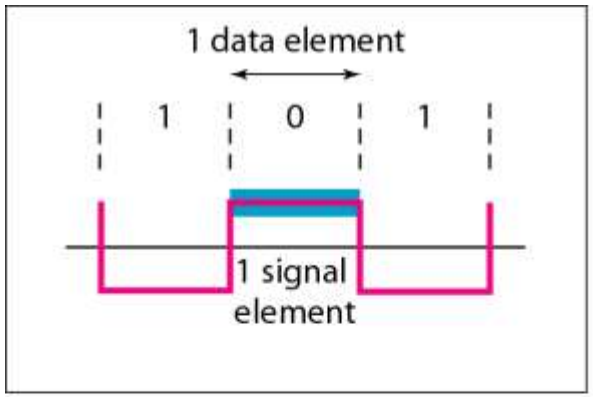
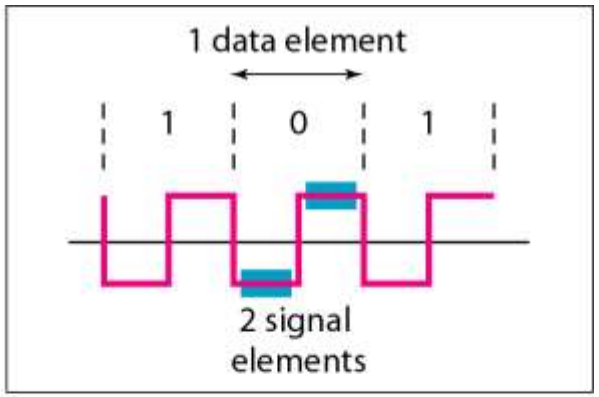


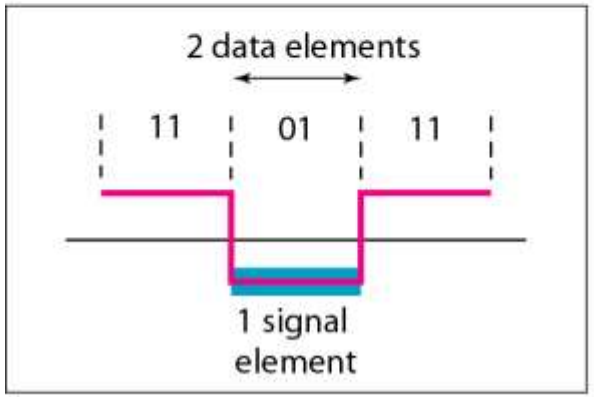
Figure 4.2 *Signal element versus data element*



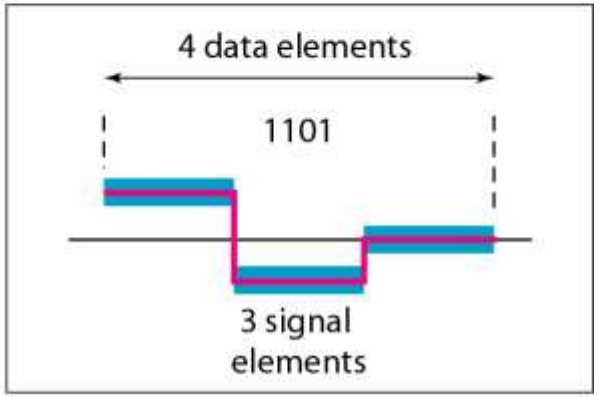
a. One data element per one signal element (r = 1)



b. One data element per two signal elements $\left(r = \frac{1}{2}\right)$



c. Two data elements per one signal element (r = 2)



d. Four data elements per three signal elements $\left(r = \frac{4}{3}\right)$

Figure 4.4 *Line coding schemes*

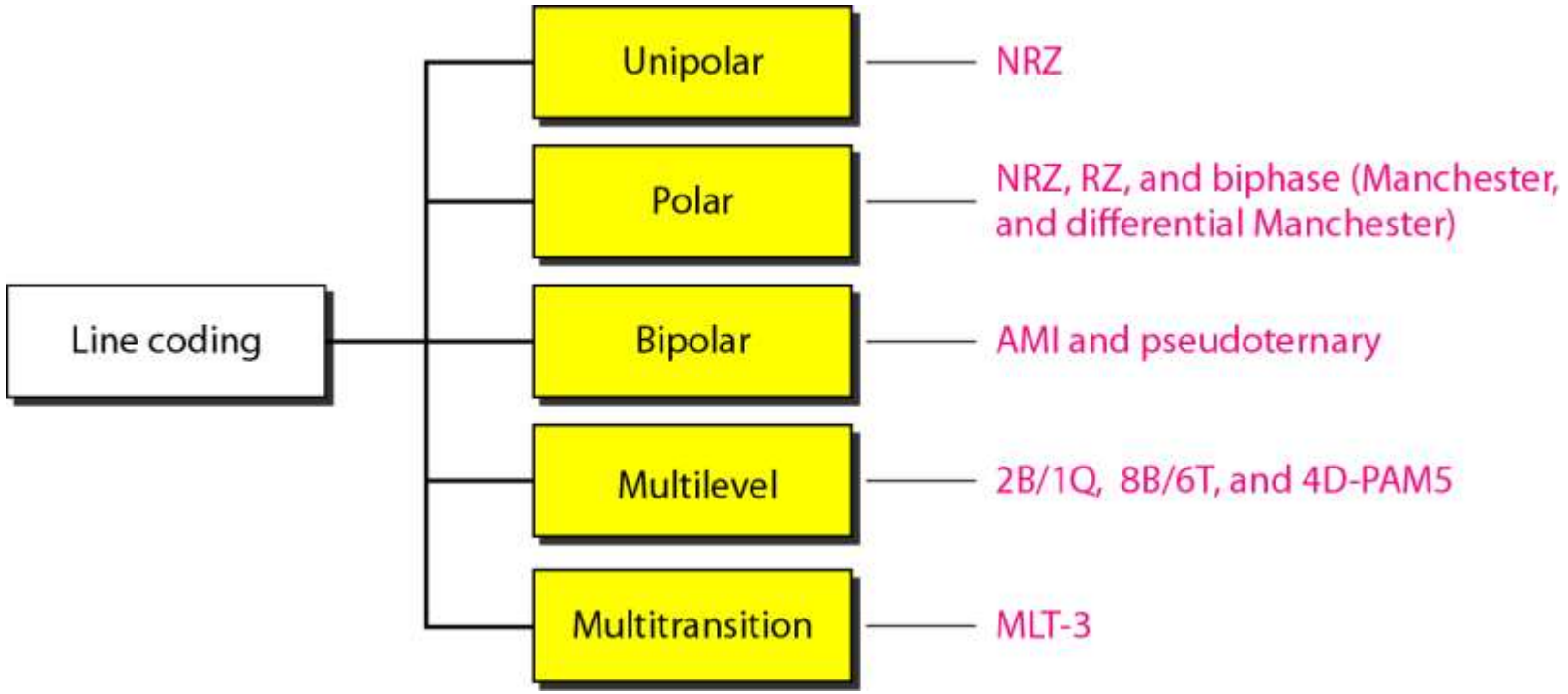
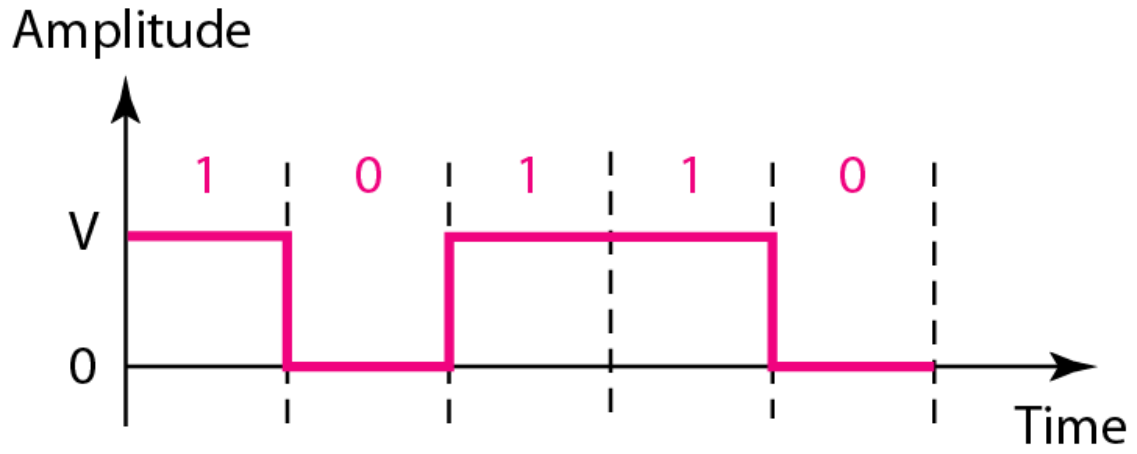


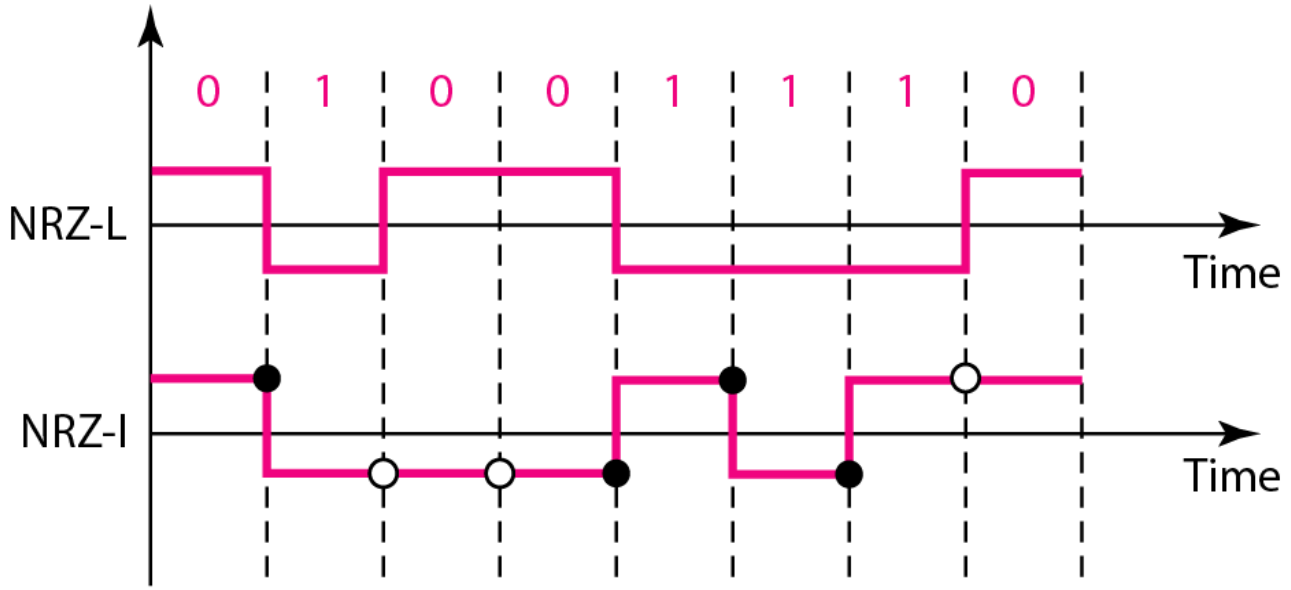
Figure 4.5 *Unipolar NRZ scheme*



Rules:
1 = Positive Voltage
0 = Zero Voltage

Note:
NZR = Non Return to Zero

Figure 4.6 *Polar NRZ-L and NRZ-I schemes*

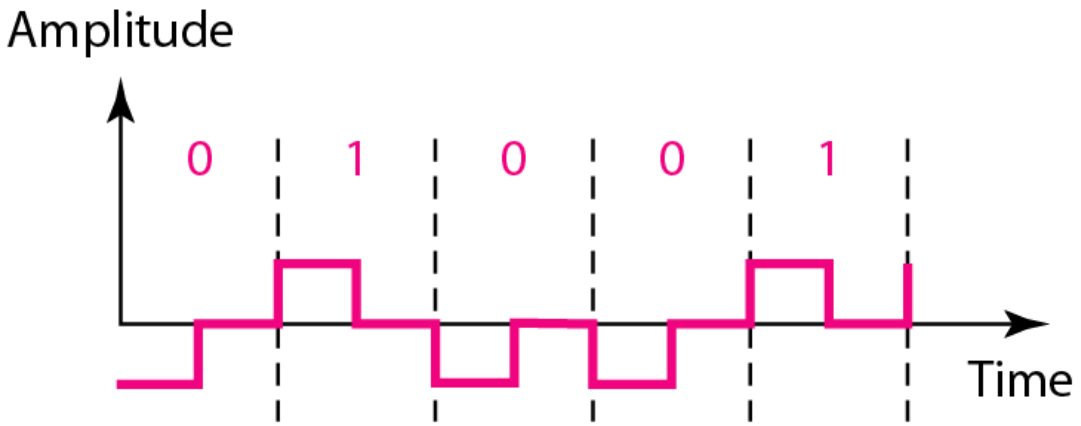


Rules for NRZ-L:
1 = Negative Voltage
0 = Positive Voltage

Rules for NZR-I:
Next bit 1 = Inversion
Next bit 0 = No Inversion

Note:
1. NZR = Non Return to Zero.
2. Next bit of all Inverted Line is 1 and rest all are 0.

Figure 4.7 *Polar RZ scheme*



Rules:

0 = Negative to Zero

1 = Positive to Zero

Figure 4.8 *Polar biphase: Manchester and differential Manchester schem*

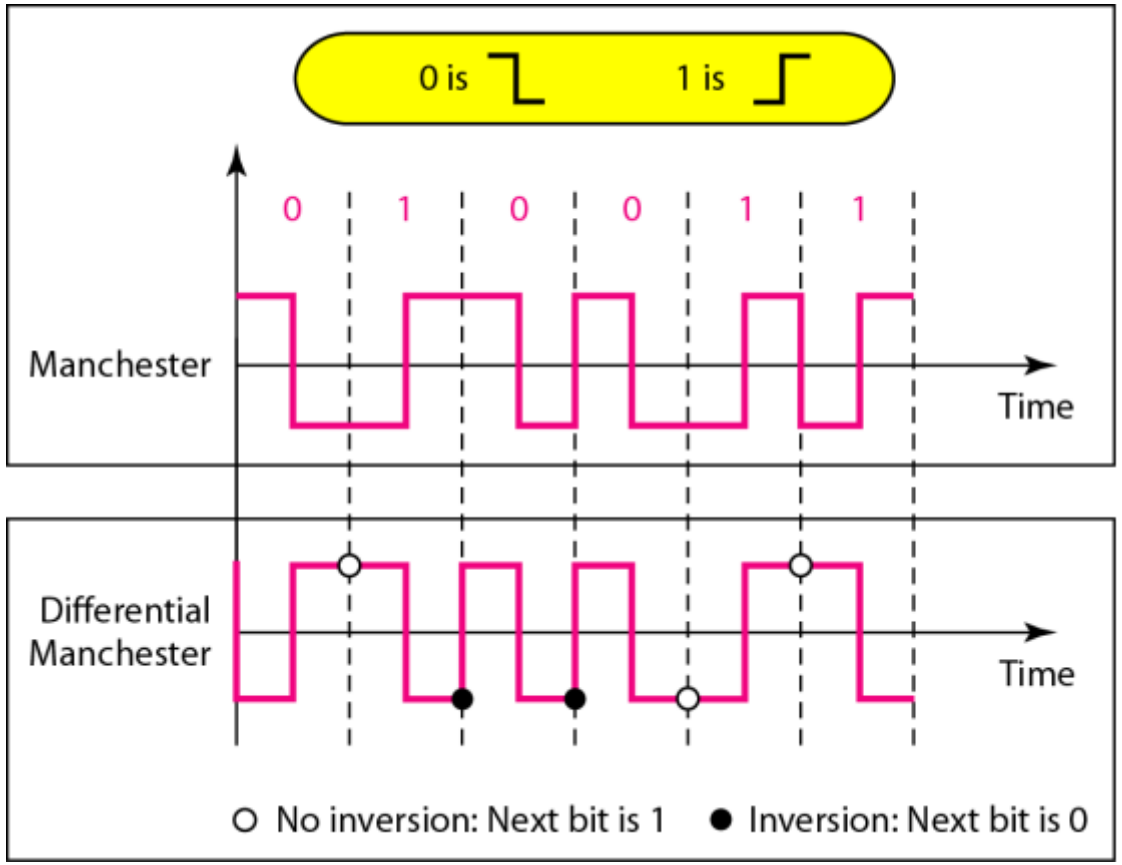
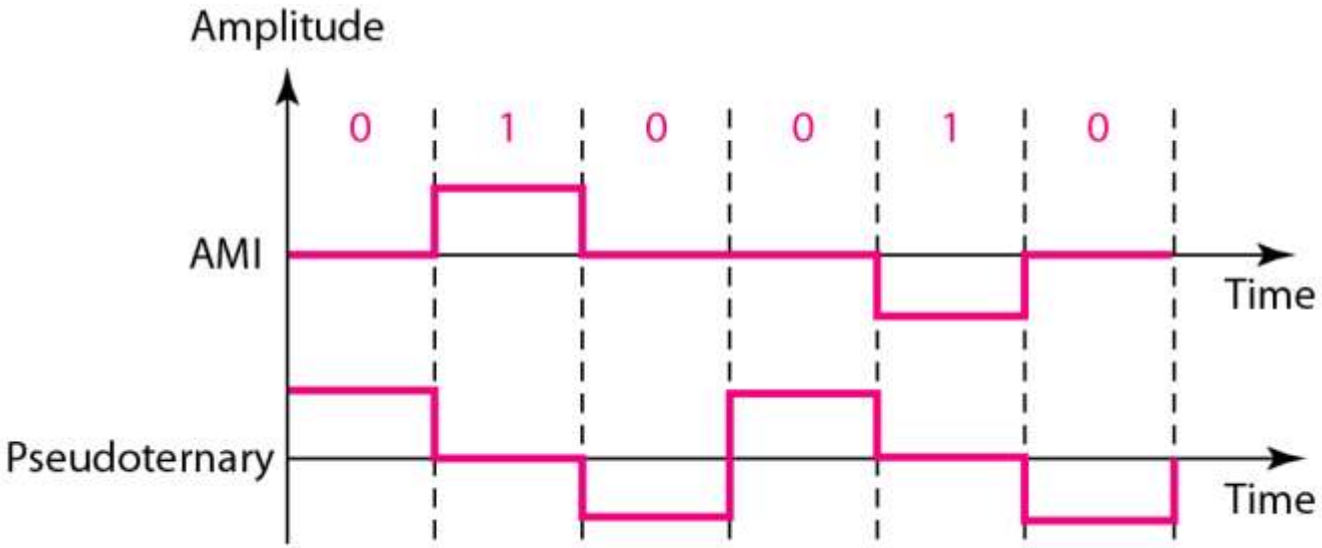


Figure 4.9 *Bipolar schemes: AMI (Alternate Mark Inversion) and Pseudoternary*



Rules:
0 = Zero Voltage
1 = Alternate

Rules:
1 = Zero Voltage
0 = Alternate

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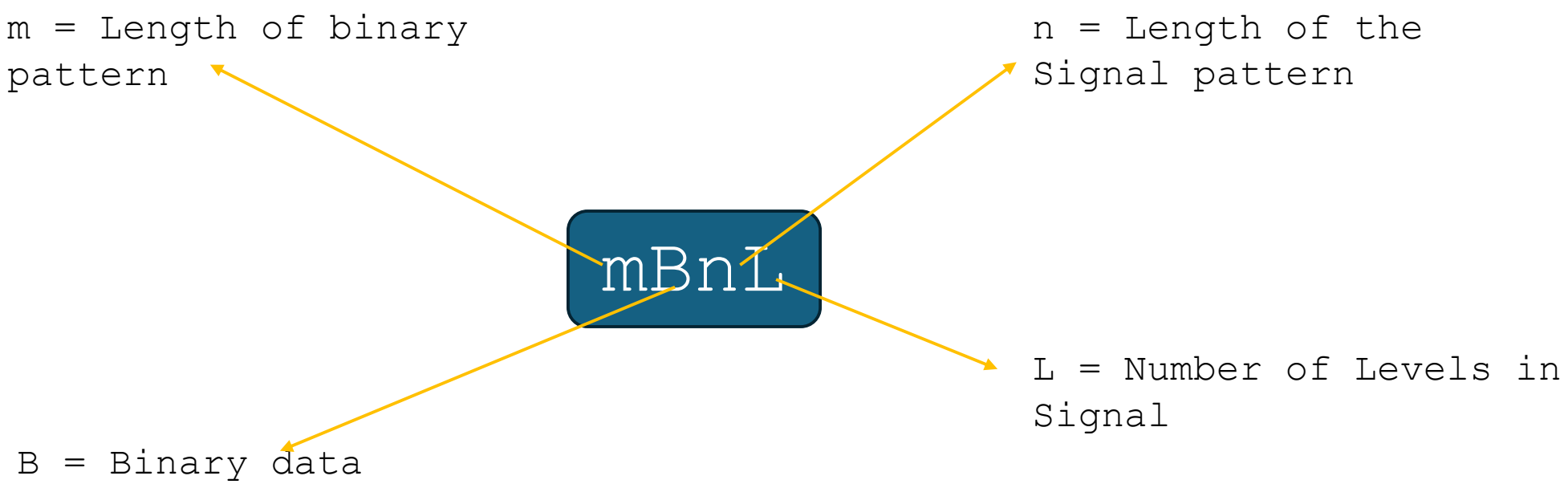
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Lec 8 (Digital-to-digital Conversion)

Multilevel Encoding: mBnL



Number of Data Pattern $= n^m$

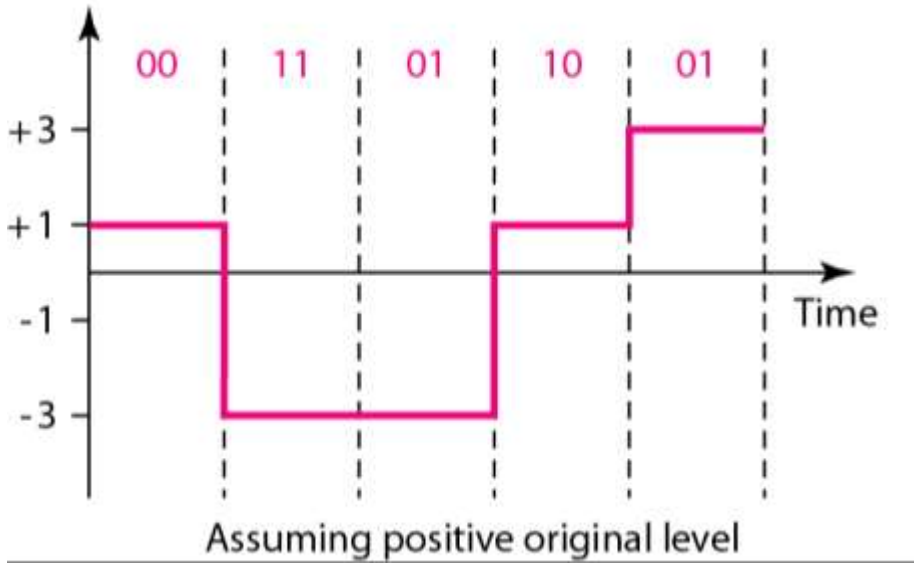
Number of Signal Pattern = L

Example: 2B1Q, 8B6T etc

Figure 4.10 *Multilevel: 2B1Q scheme*

Next bits	Previous level: positive — Next level	Previous level: negative — Next level
00	+1	-1
01	+3	-3
10	-1	+1
11	-3	+3

Transition table



Assuming positive original level

Number of Data Pattern $= 2^m = 4$

Number of Signal Pattern $= L^n = 4$

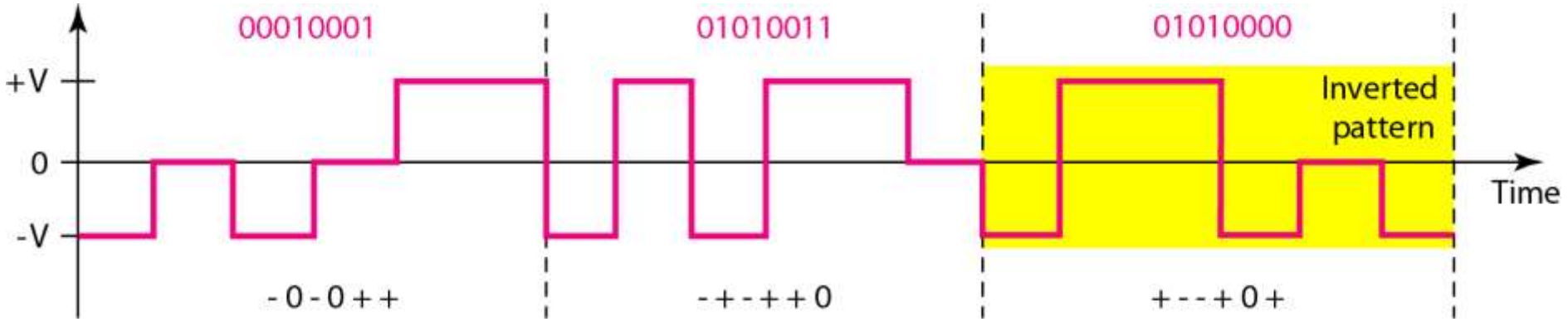
Figure 4.11 *Multilevel: 8B6T scheme*

Mapping Table 8B6T (Partial)

Data	Code	Data	Code
00	- + 0 0 - +	2A	+ 0 + - 0 -
11	- 0 - 0 + +	50	+ - - + 0 +
0A	0 - + + - 0	53	- + - + + 0

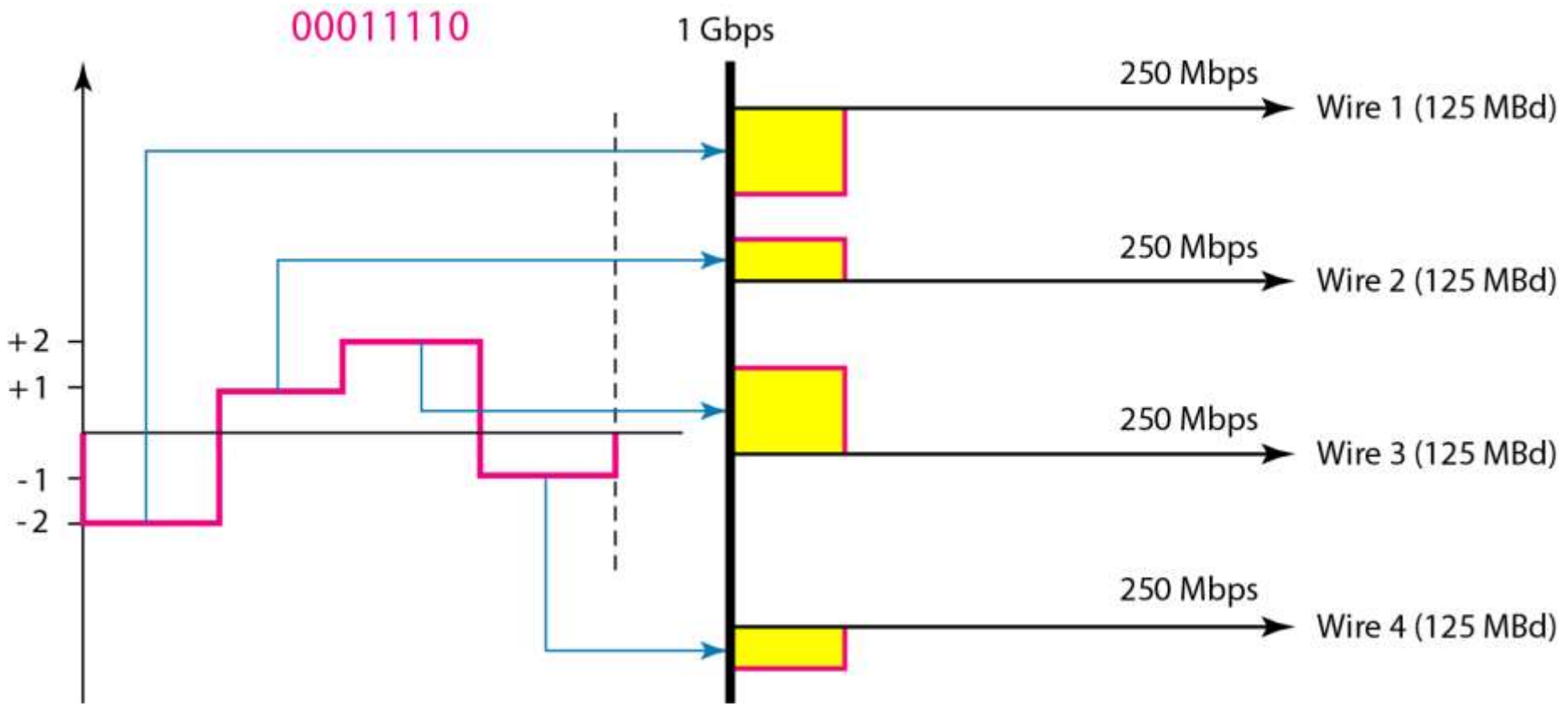
11 = 0001 0001
53 = 0101 0011
50 = 0101 0000
(Hexadecimal Numbers)

DC Balance
- + 0 0 - + : Weight 0
- + - + + 0 : Weight +1



Number of Data Pattern = 2^m = 256
Number of Signal Pattern = L^n = 729
Unused = 729 – 256 = 473 (Redundant and used for Synchronization, Error detection, DC balance)

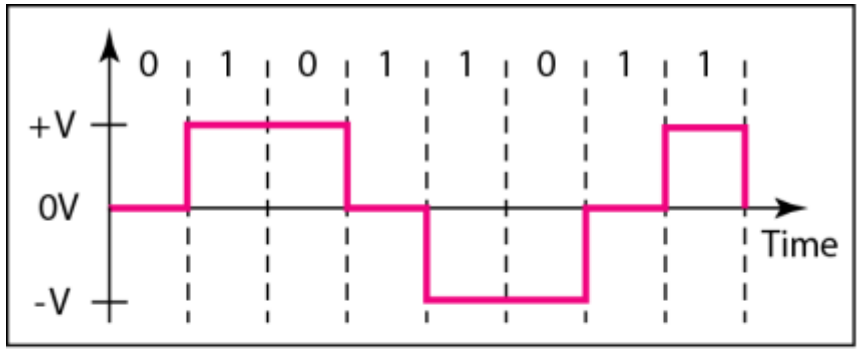
Figure 4.12 *Multilevel: 4D-PAM5 scheme (Pulse Amplitude Modu*



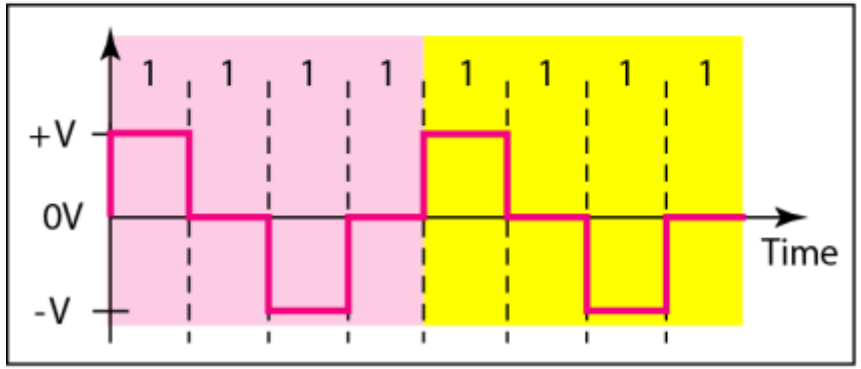
Note: 8B4Q, 0 Level works for Error detection

Figure 4.13 *Multitransition: MLT-3 scheme (Multiline Transi*

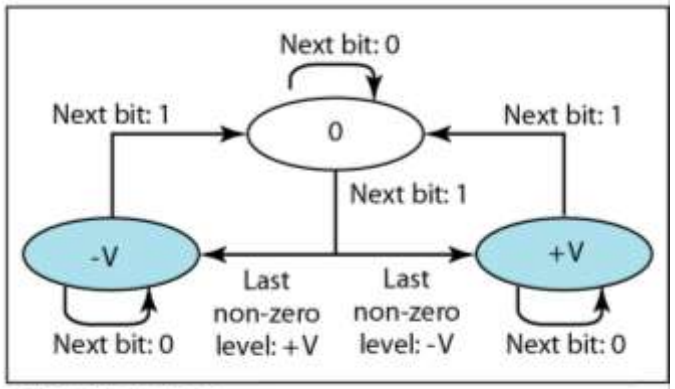
Let us assume that, the last level was at 0 voltage and the last non-zero puls



a. Typical case



b. Worse case



c. Transition states

Rules:

Next Bit 0 = No transition

Next Bit 1 = 0 (Current level not 0)

Next Bit 1 = Opposite of last non-zero (Current level 0)

Table 4.1 *Summary of line coding schemes*

Category	*Scheme*	*Bandwidth (average)*	*Characteristics*
Unipolar	NRZ	$B = N/2$	Costly, no self-synchronization if long 0s or 1s, DC
Unipolar	NRZ-L	$B = N/2$	No self-synchronization if long 0s or 1s, DC
	NRZ-I	$B = N/2$	No self-synchronization for long 0s, DC
	Biphase	$B = N$	Self-synchronization, no DC, high bandwidth
Bipolar	AMI	$B = N/2$	No self-synchronization for long 0s, DC
Multilevel	2B1Q	$B = N/4$	No self-synchronization for long same double bits
	8B6T	$B = 3N/4$	Self-synchronization, no DC
	4D-PAM5	$B = N/8$	Self-synchronization, no DC
Multiline	MLT-3	$B = N/3$	No self-synchronization for long 0s

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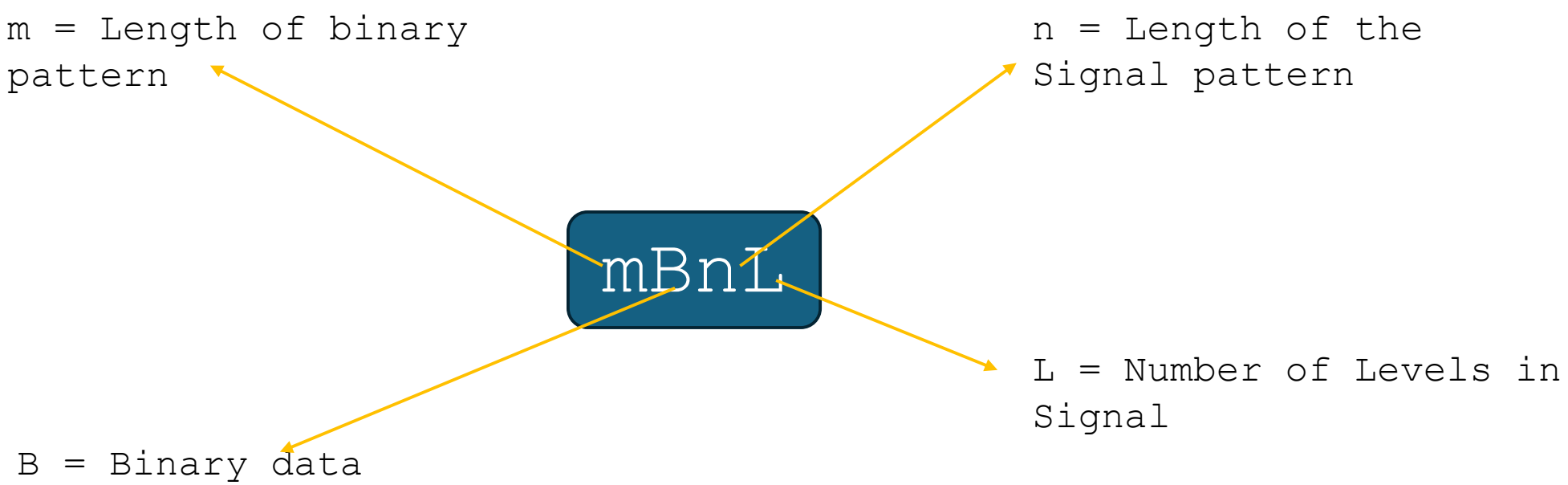
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Lec 8 (Digital-to-digital Conversion)

Multilevel Encoding: mBnL



Number of Data Pattern $= n^m$

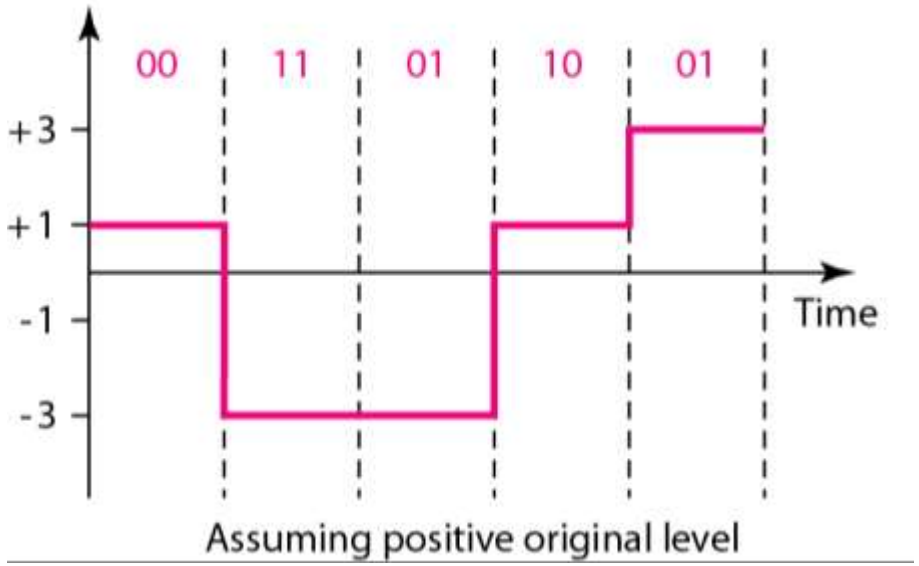
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Example: 2B1Q, 8B6T etc

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11	-3	+3

Transition table



Assuming positive original level

Number of Data Pattern $= 2^m = 4$

Number of Signal Pattern $= L^n = 4$

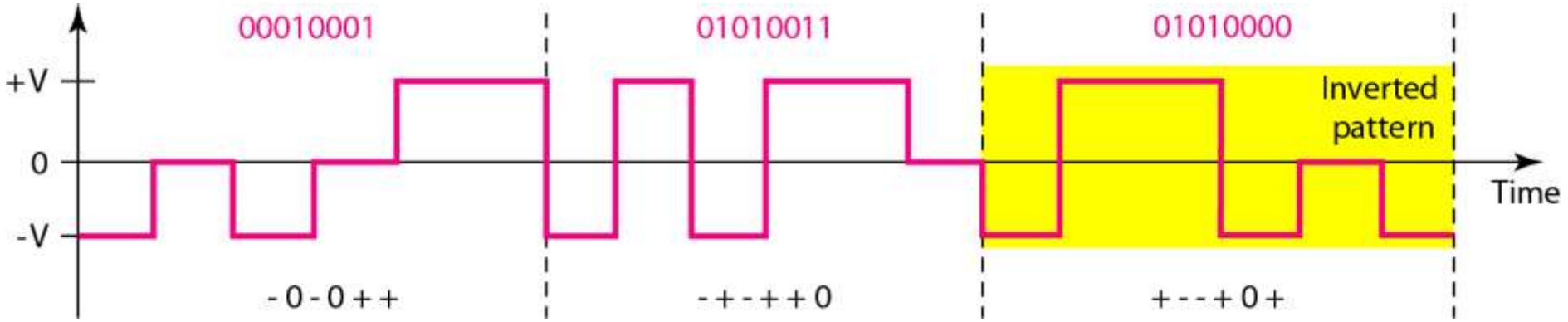
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Mapping Table 8B6T (Partial)

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0A	0 - + + - 0	53	- + - + + 0

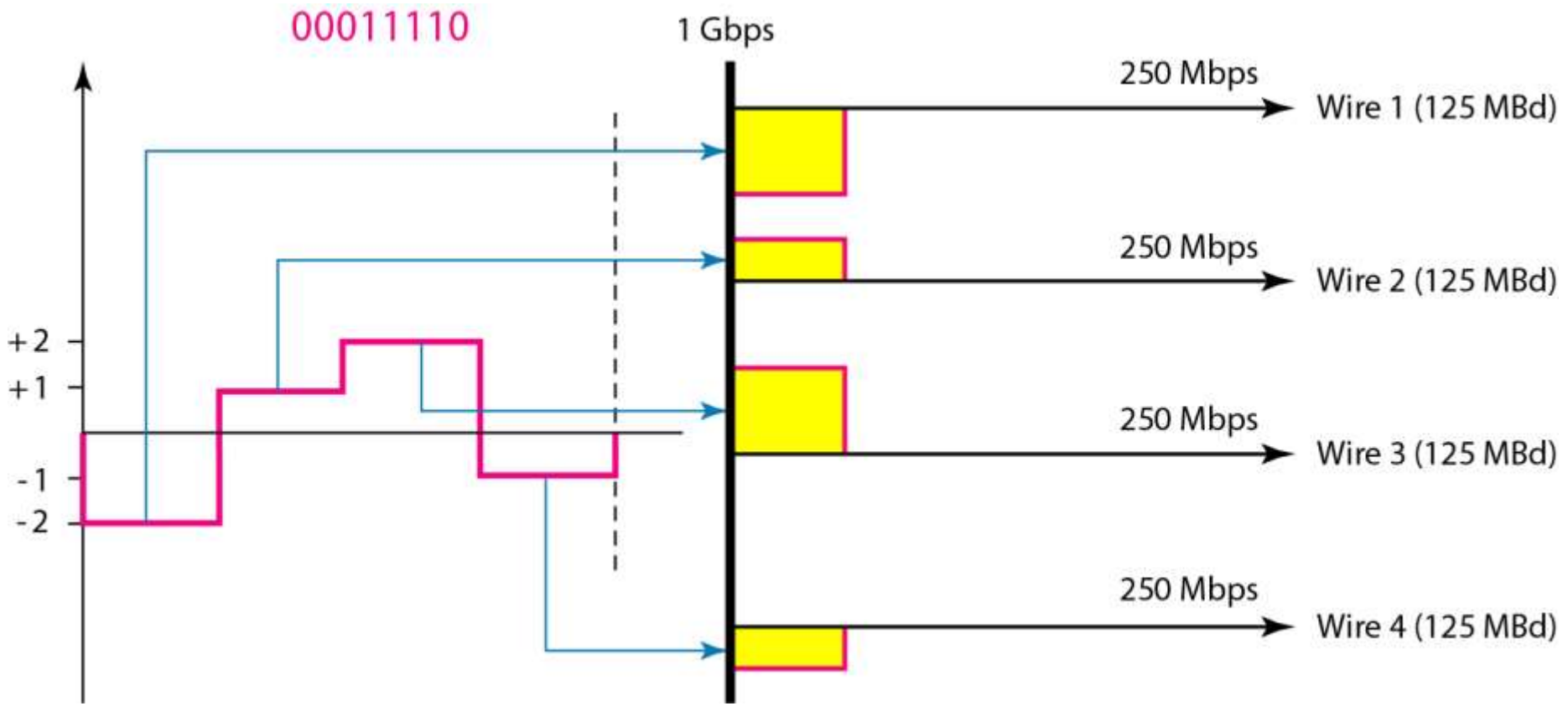
11 = 0001 0001
53 = 0101 0011
50 = 0101 0000
(Hexadecimal Numbers)

DC Balance
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Number of Data Pattern = 2^m = 256
Number of Signal Pattern = L^n = 729
Unused = 729 – 256 = 473 (Redundant and used for Synchronization, Error detection, DC balance)

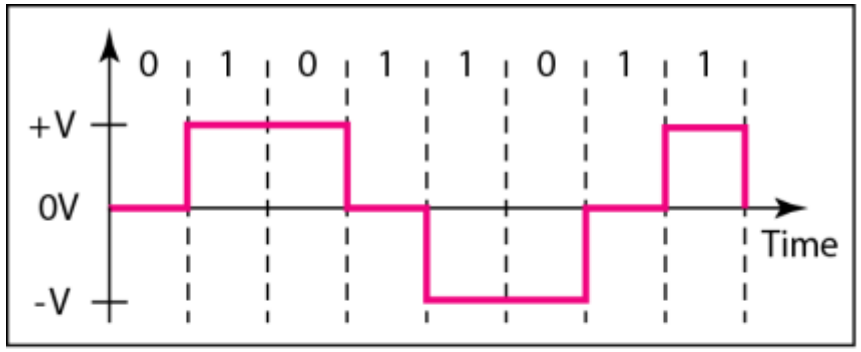
Figure 4.12 *Multilevel: 4D-PAM5 scheme (Pulse Amplitude Modu*



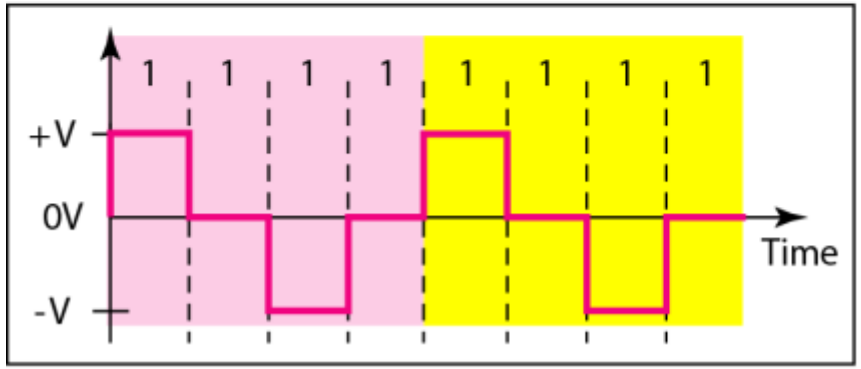
Note: 8B4Q, 0 Level works for Error detection

Figure 4.13 *Multitransition: MLT-3 scheme (Multiline Transi*

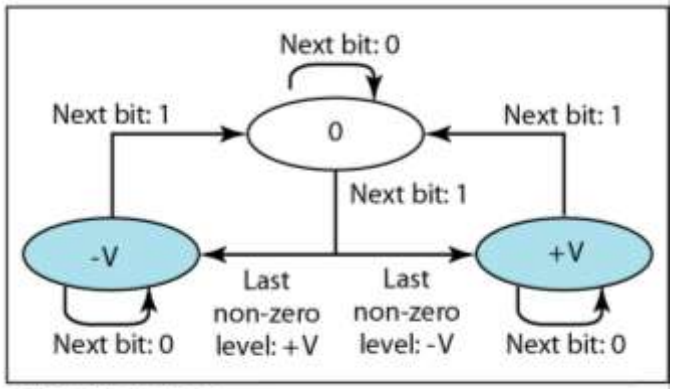
Let us assume that, the last level was at 0 voltage and the last non-zero puls



a. Typical case



b. Worse case



c. Transition states

Rules:

Next Bit 0 = No transition

Next Bit 1 = 0 (Current level not 0)

Next Bit 1 = Opposite of last non-zero (Current level 0)

Table 4.1 *Summary of line coding schemes*

Category	*Scheme*	*Bandwidth (average)*	*Characteristics*
Unipolar	NRZ	$B = N/2$	Costly, no self-synchronization if long 0s or 1s, DC
Unipolar	NRZ-L	$B = N/2$	No self-synchronization if long 0s or 1s, DC
	NRZ-I	$B = N/2$	No self-synchronization for long 0s, DC
	Biphase	$B = N$	Self-synchronization, no DC, high bandwidth
Bipolar	AMI	$B = N/2$	No self-synchronization for long 0s, DC
Multilevel	2B1Q	$B = N/4$	No self-synchronization for long same double bits
	8B6T	$B = 3N/4$	Self-synchronization, no DC
	4D-PAM5	$B = N/8$	Self-synchronization, no DC
Multiline	MLT-3	$B = N/3$	No self-synchronization for long 0s

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Instr Class B, CSE Dept

MIST

Lec 9 (Digital-to-digital Conversion)

Note

Block coding is normally referred to as *m*B/*n*B coding; it replaces each *m*-bit group with an *n*-bit group.

Figure 4.14 *Block coding concept*

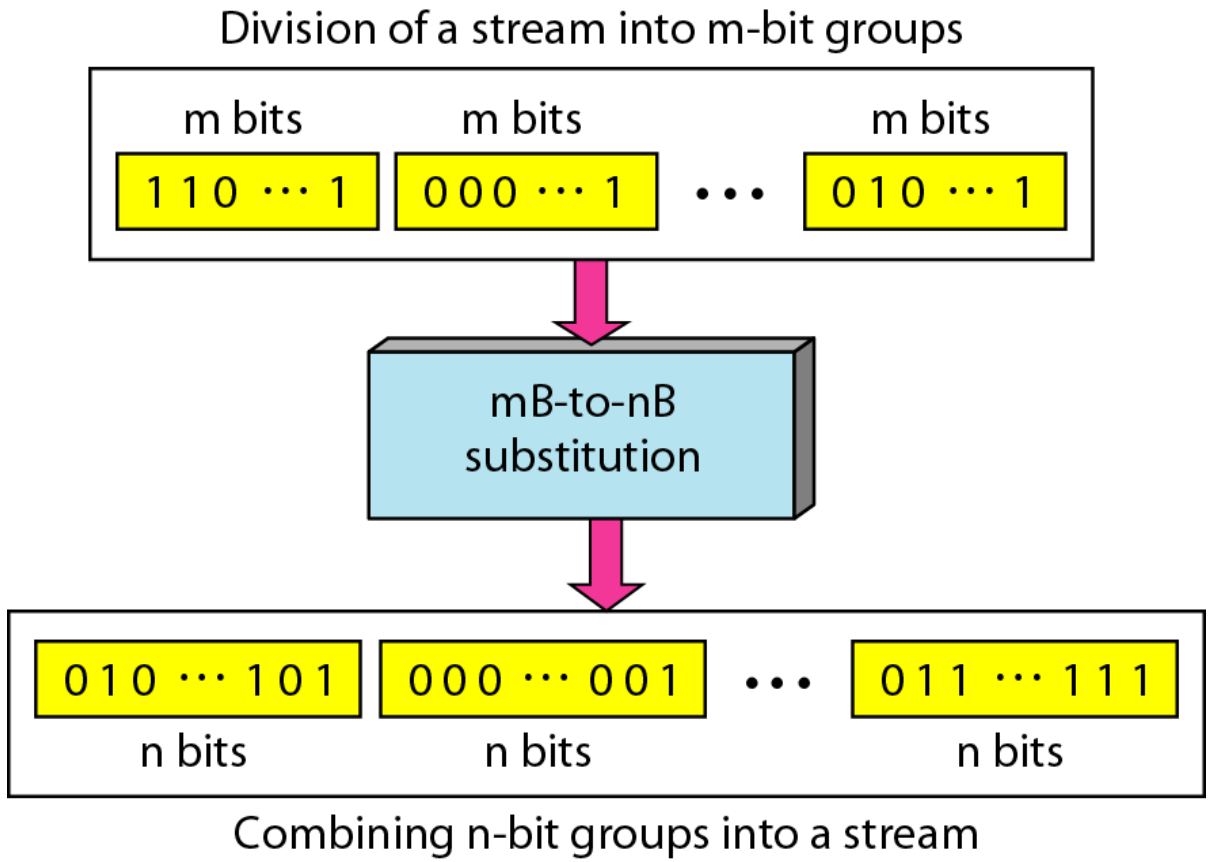


Figure 4.15 *Using block coding 4B/5B with NRZ-I line coding*

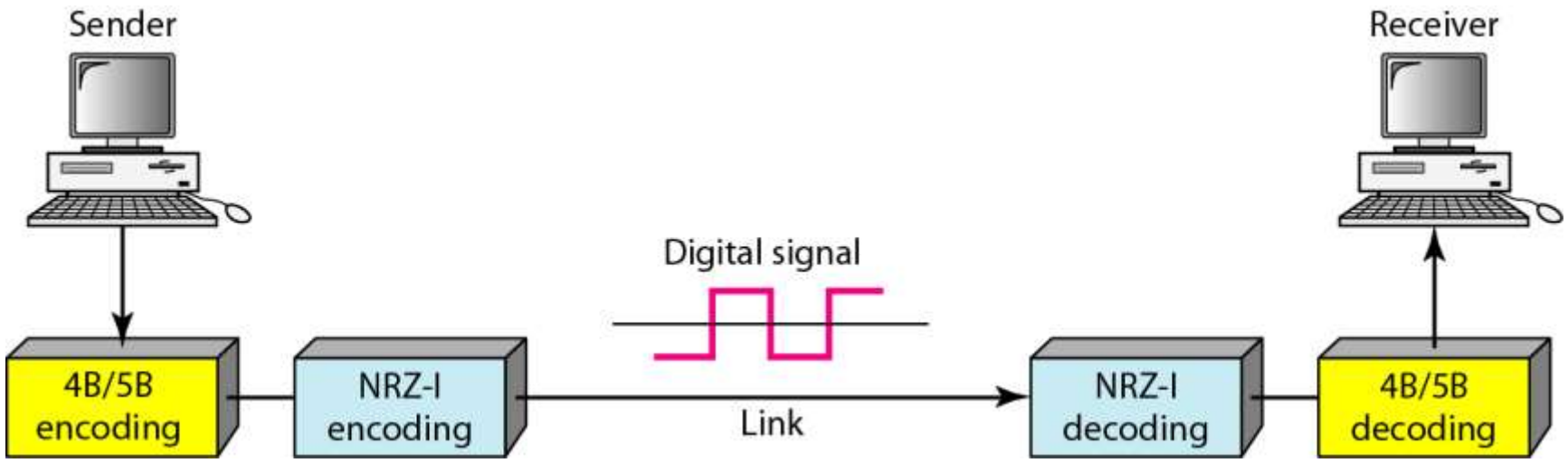
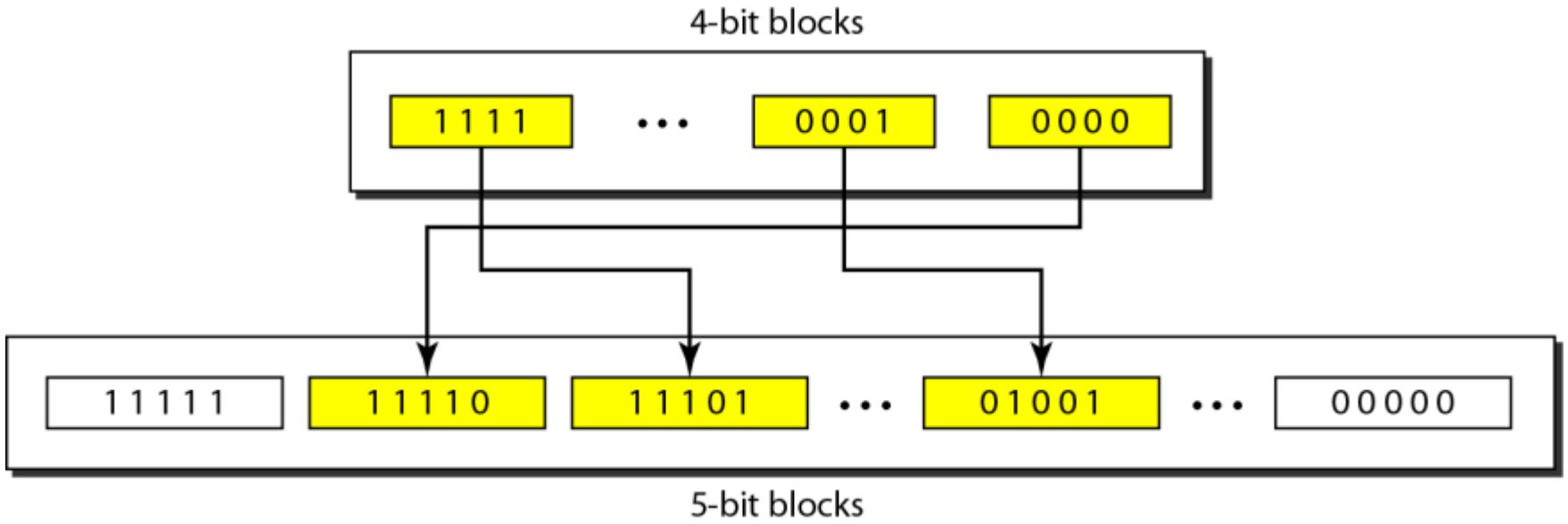


Table 4.2 4B/5B mapping codes

Data Sequence	Encoded Sequence	Control Sequence	Encoded Sequence
0000	11110	Q (Quiet)	00000
0001	01001	I (Idle)	11111
0010	10100	H (Halt)	00100
0011	10101	J (Start delimiter)	11000
0100	01010	K (Start delimiter)	10001
0101	01011	T (End delimiter)	01101
0110	01110	S (Set)	11001
0111	01111	R (Reset)	00111
1000	10010		
1001	10011		
1010	10110		
1011	10111		
1100	11010		
1101	11011		
1110	11100		
1111	11101		

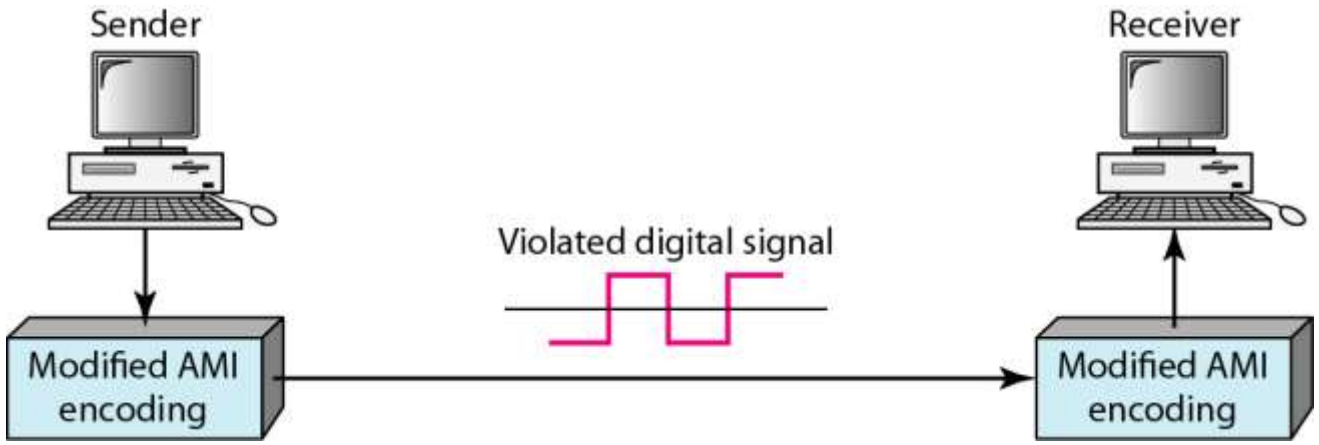
Figure 4.16 *Substitution in 4B/5B block coding*



B8ZS Scrambling

(Bipolar with 8 Zero Substitution)

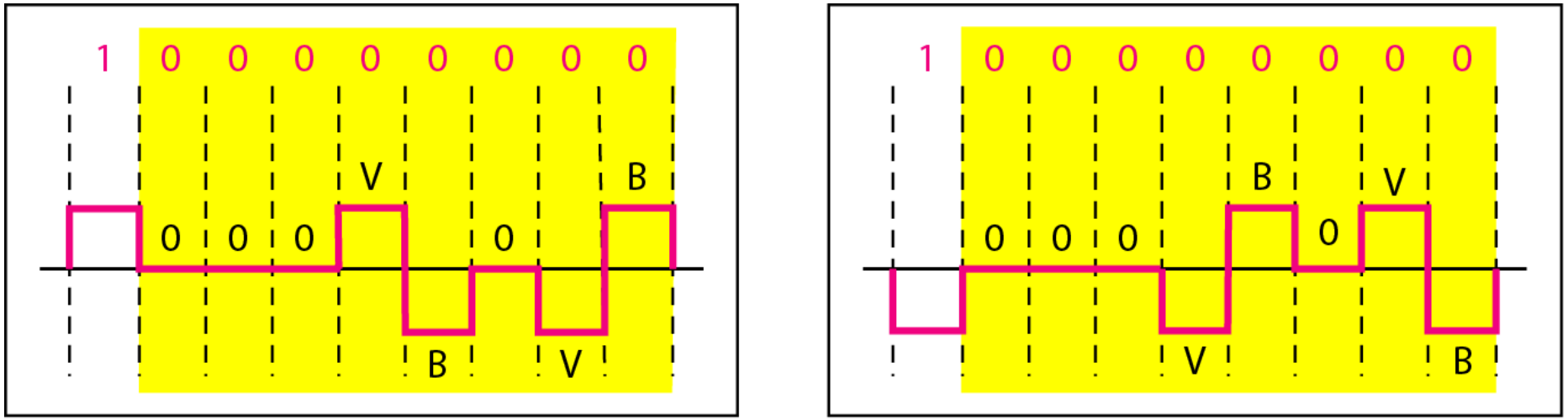
Figure 4.18 *AMI used with scrambling*



Note

B8ZS substitutes eight consecutive zeros with 000VB0VB.

Figure 4.19 *Two cases of B8ZS scrambling technique*



a. Previous level is positive.

b. Previous level is negative.

Rules:
Replace consecutive 8 Zero with 000VB0VB
V (Violation) = Same as last non-zero level
B (Bipolar) = Opposite to last non-zero level

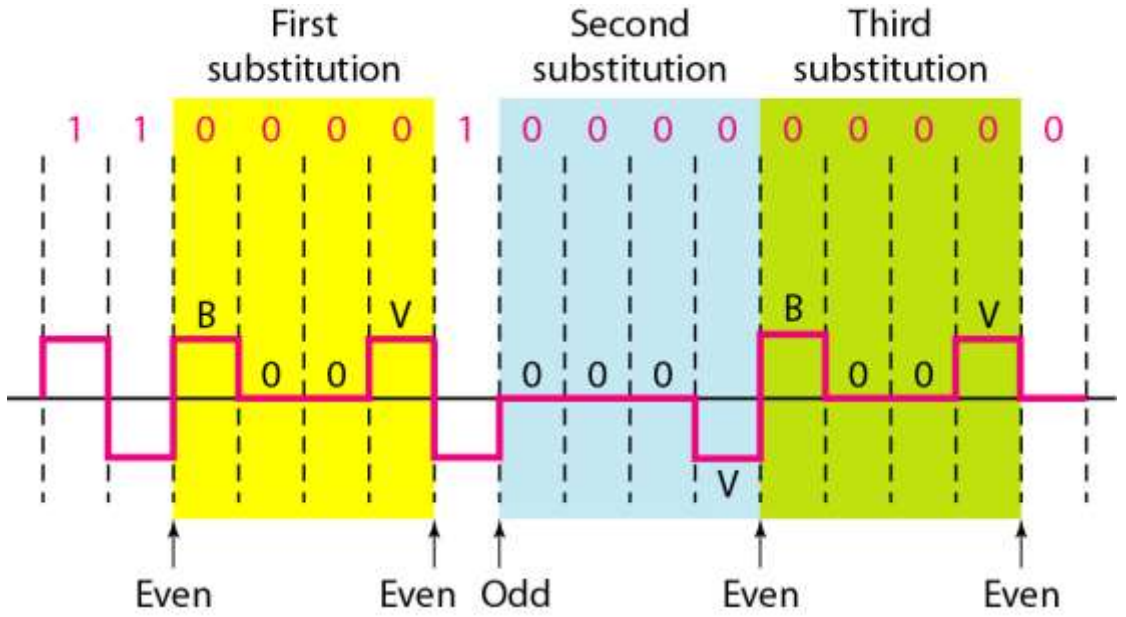
HDB3 *Scrambling*

(High Density Bipolar 3 Zero)

Note

HDB3 substitutes four consecutive zeros with 000V or B00V depending on the number of nonzero pulses after the last substitution.

Figure 4.20 *Different situations in HDB3 scrambling technique*



Rules:

Replace consecutive 4 Zero with 000V (ODD) or B00V (EVEN)

V (Violation) = Same as last non-zero level

B (Bipolar) = Opposite to last non-zero level

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Data Communication Level – 3 (Spring Term)

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Instr Class B, CSE Dept

MIST

Lec 10 (ASK, FSK, PSK & QAM)



Data Communications and Networking Fourth Edition

Forouzan

Chapter 5

Analog Transmission

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5-1 DIGITAL-TO-ANALOG CONVERSION

Digital-to-analog conversion is the process of changing one of the characteristics of an analog signal based on the information in digital data.

Topics discussed in this section:

Aspects of Digital-to-Analog Conversion
Amplitude Shift Keying
Frequency Shift Keying
Phase Shift Keying
Quadrature Amplitude Modulation

Figure 5.1 *Digital-to-analog conversion*

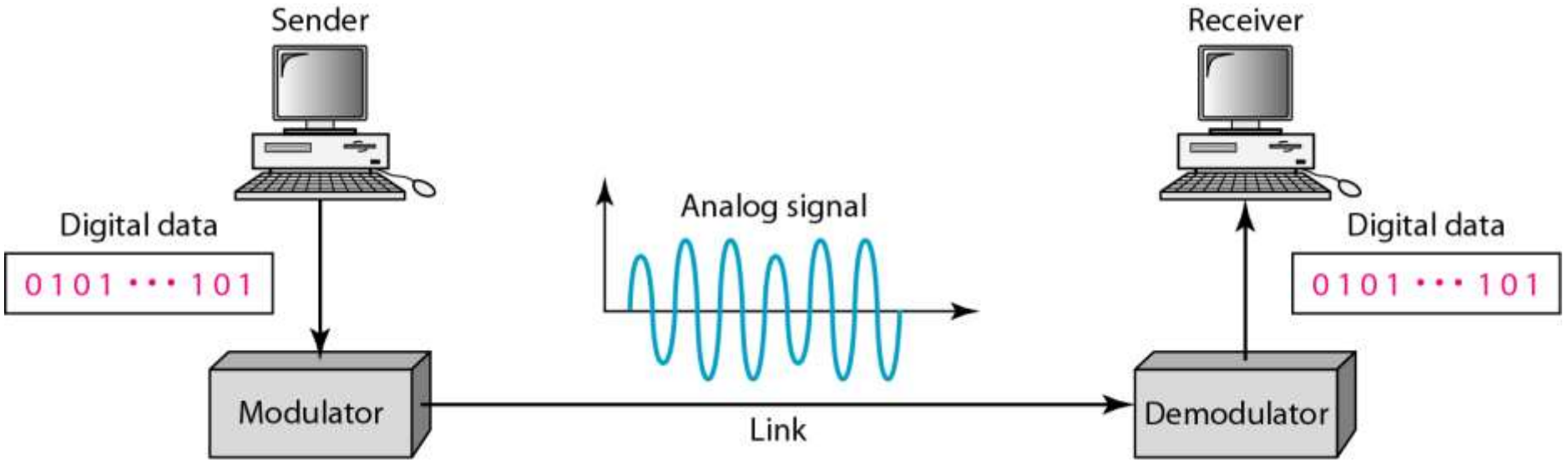


Figure *Amplitude, Frequency and Phase of a Sign Wave*

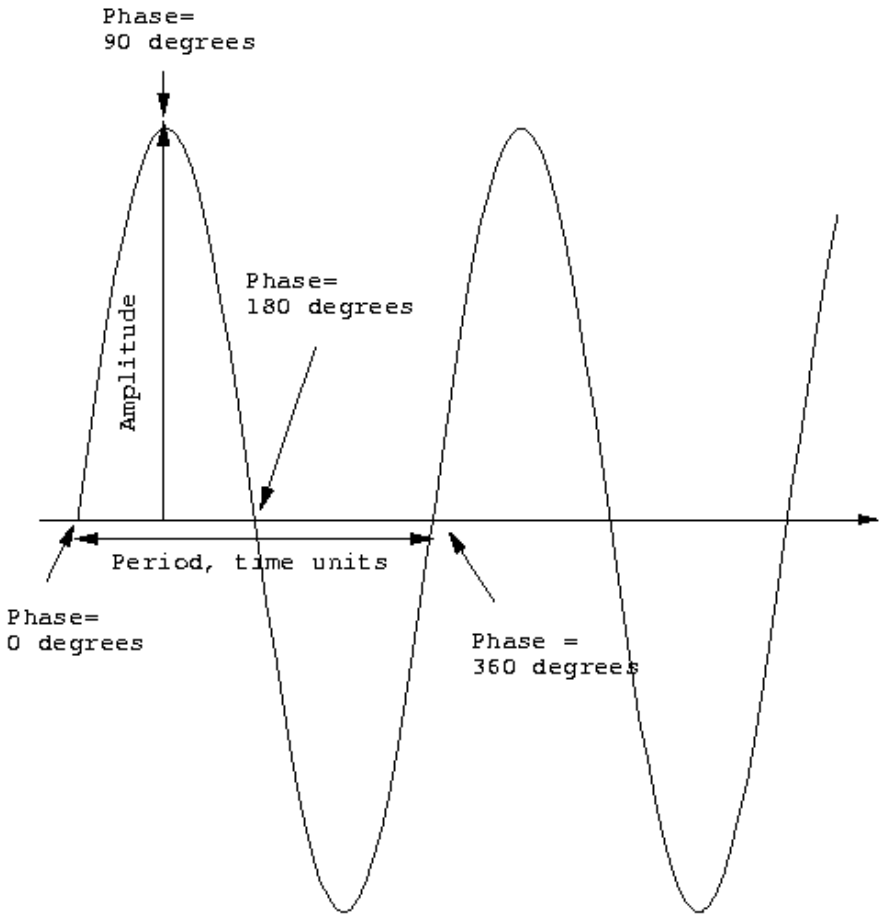


Figure 5.2 *Types of digital-to-analog conversion*

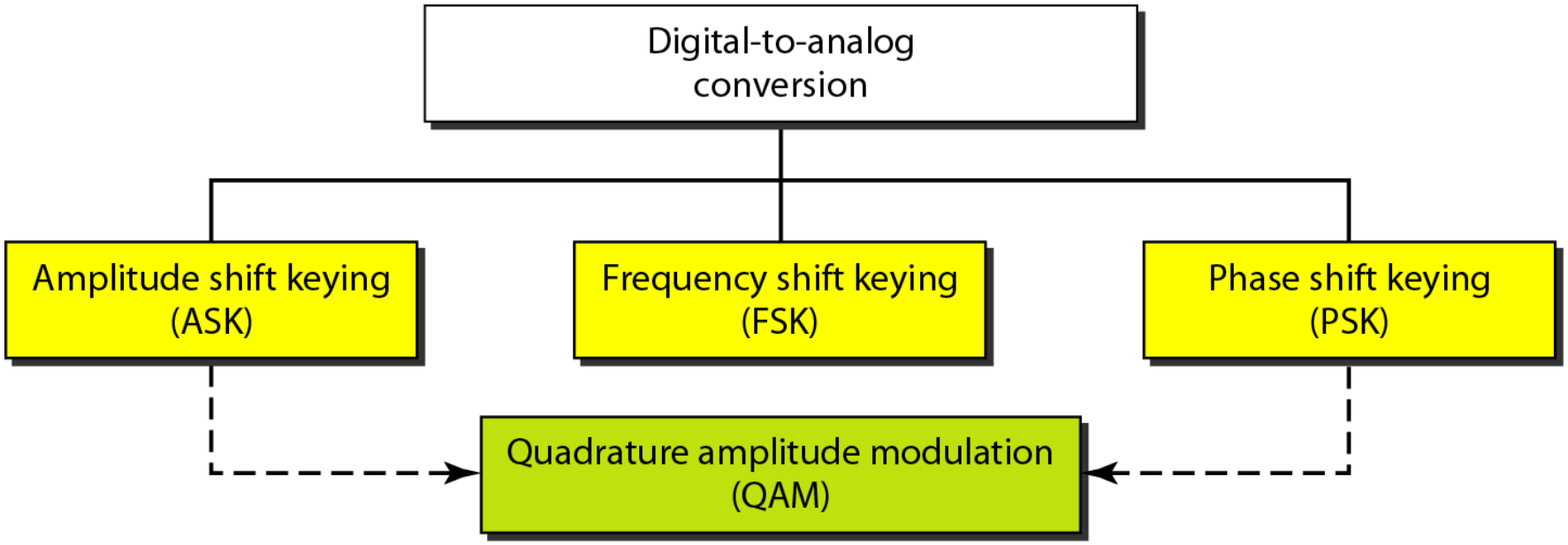
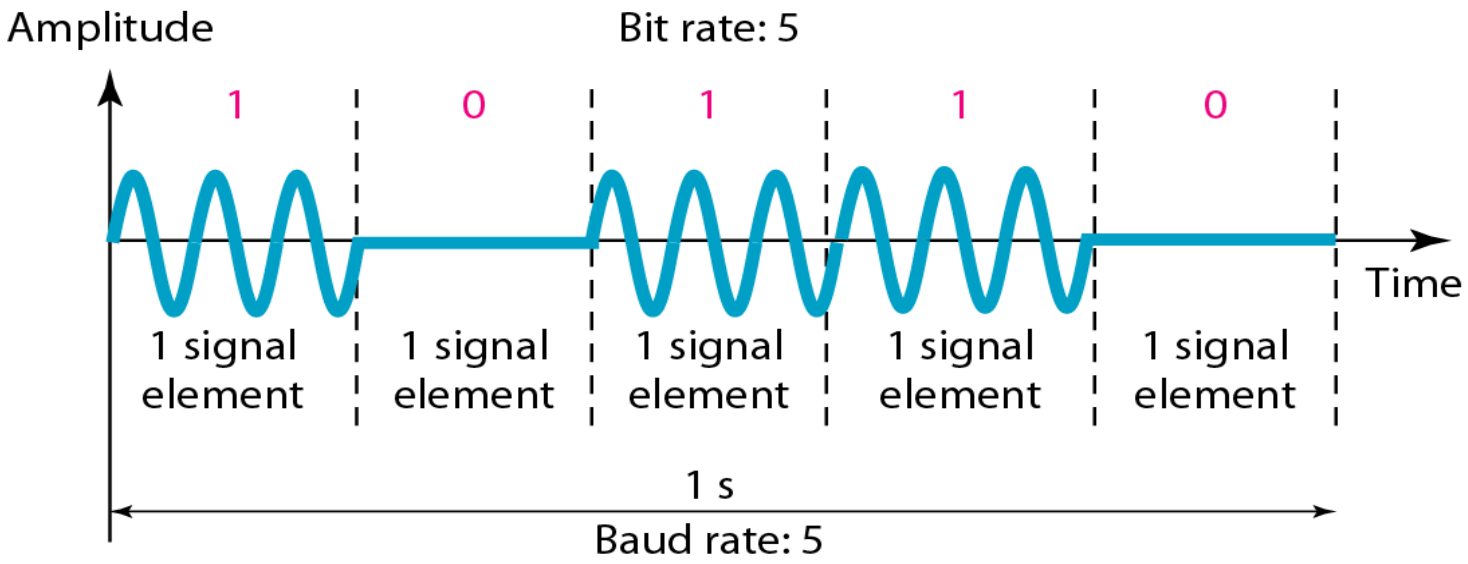


Figure *Some uses of analog-to-analog and digital-to-analog conversion*



Figure 5.3 *Binary amplitude shift keying (on-off keying)*



Also known as On Off Keying (OOK)

Figure 5.4 *Implementation of binary ASK*

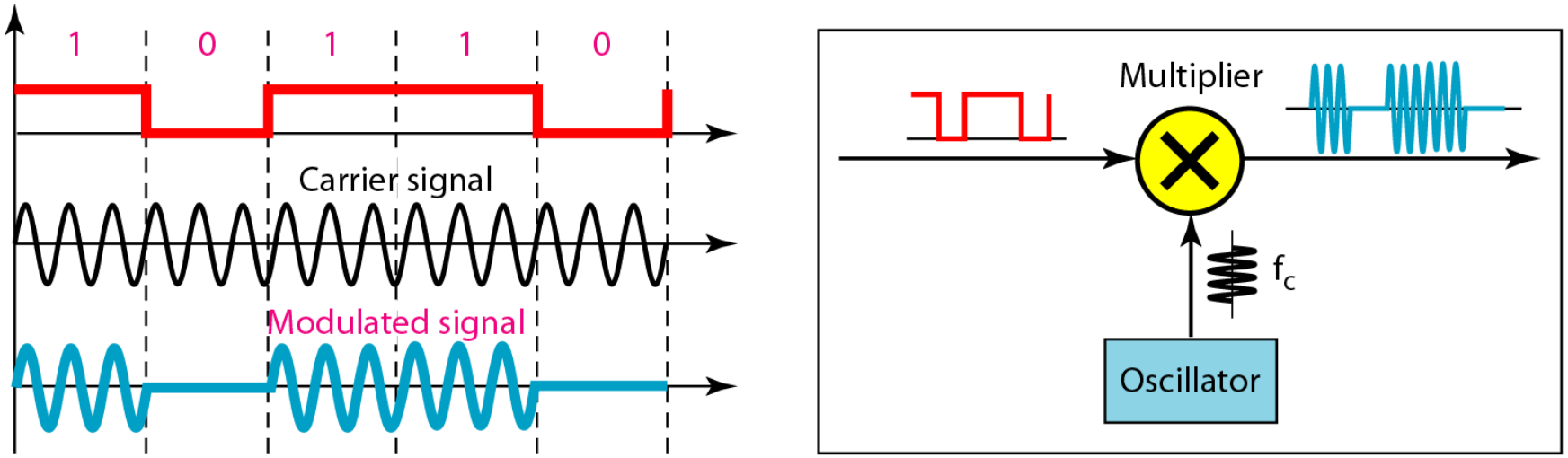


Figure 5.6 *Binary frequency shift keying*

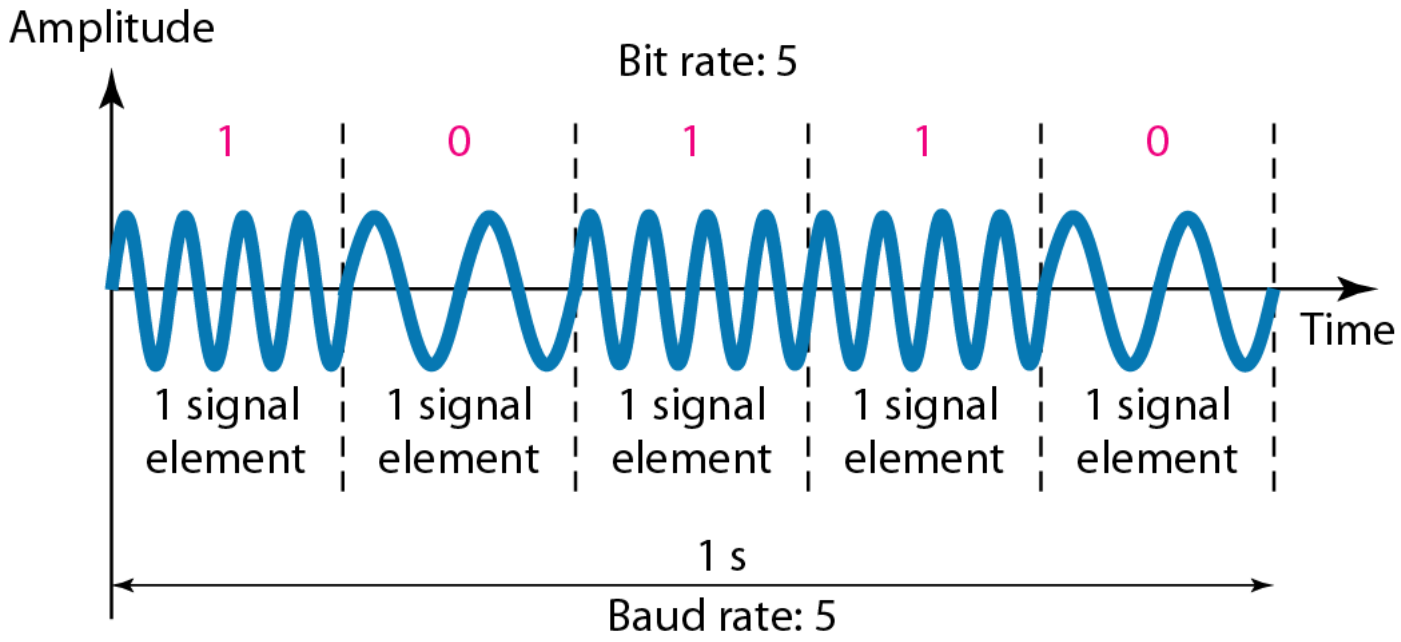


Figure 5.7 *Bandwidth of MFSK used in Example 5.6*

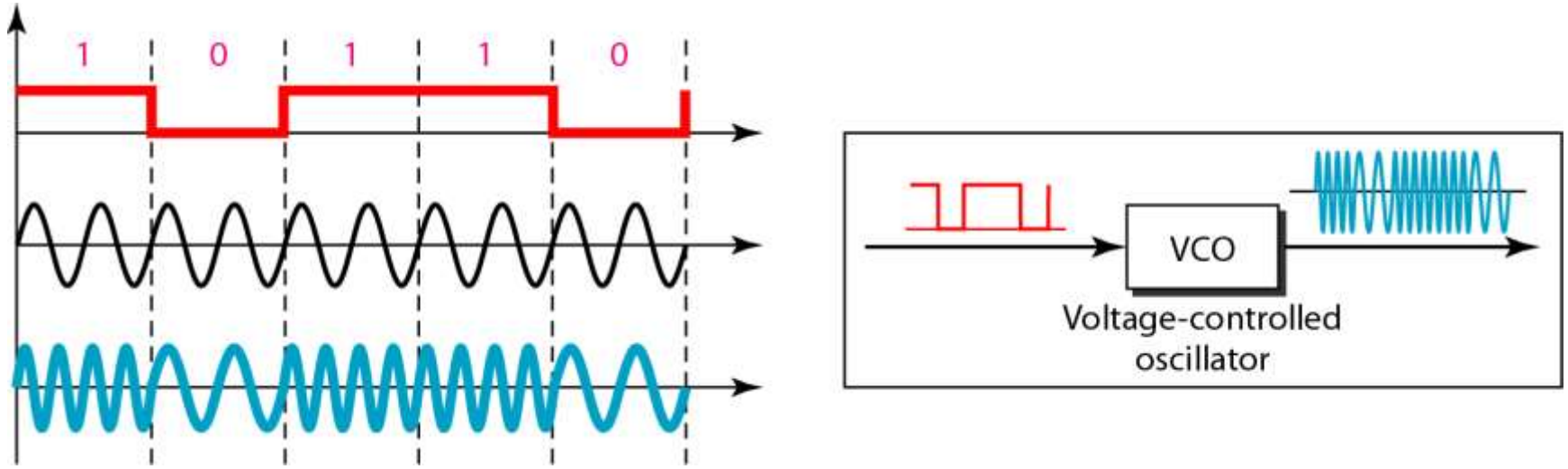
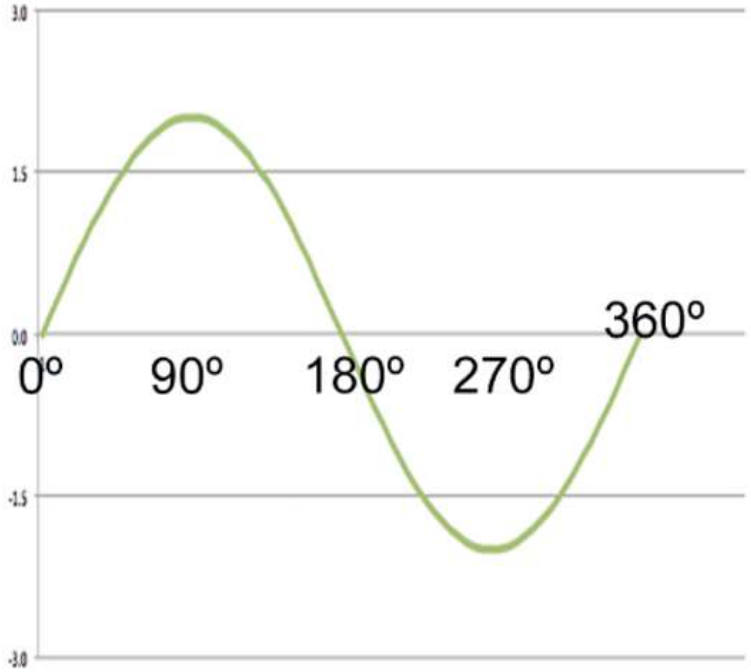
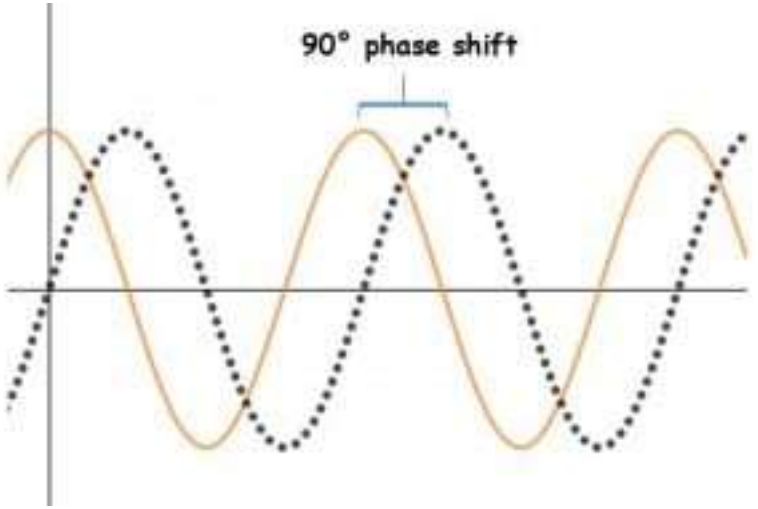
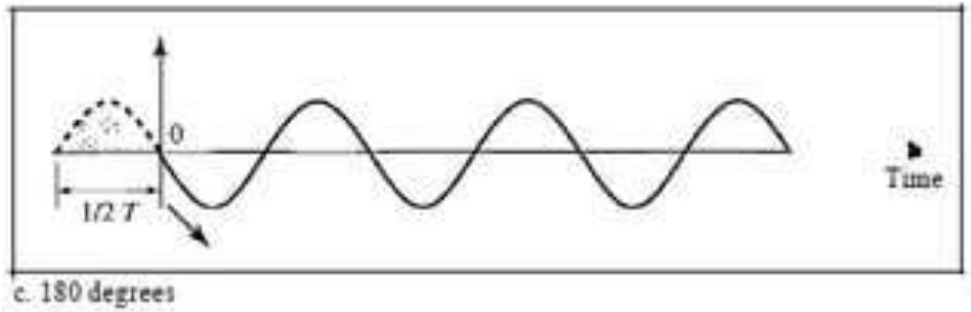


Figure *Phase Shift*







c. 180 degrees

Figure 5.9 *Binary phase shift keying*

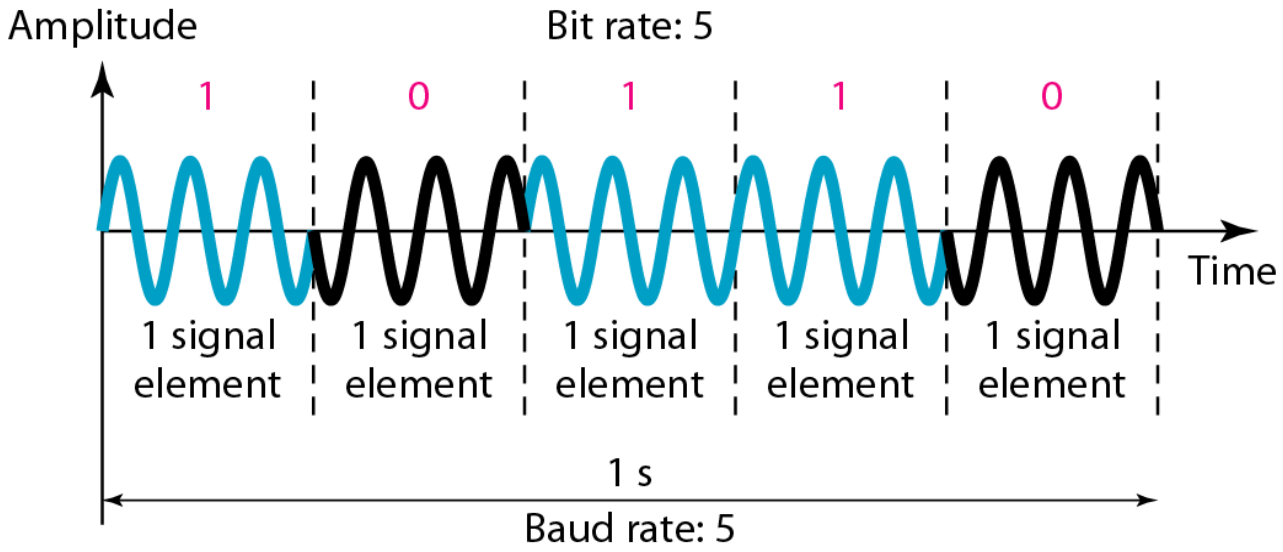


Figure 5.10 *Implementation of BPSK*

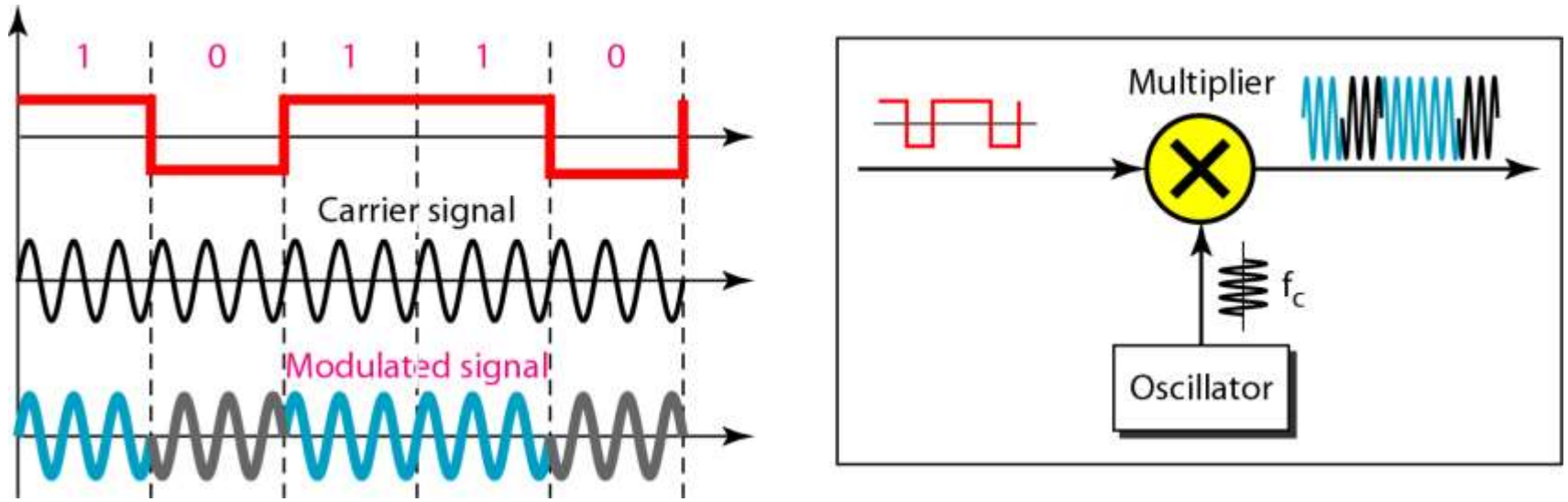


Figure 5.11 *QPSK and its implementation (Quadrature)*

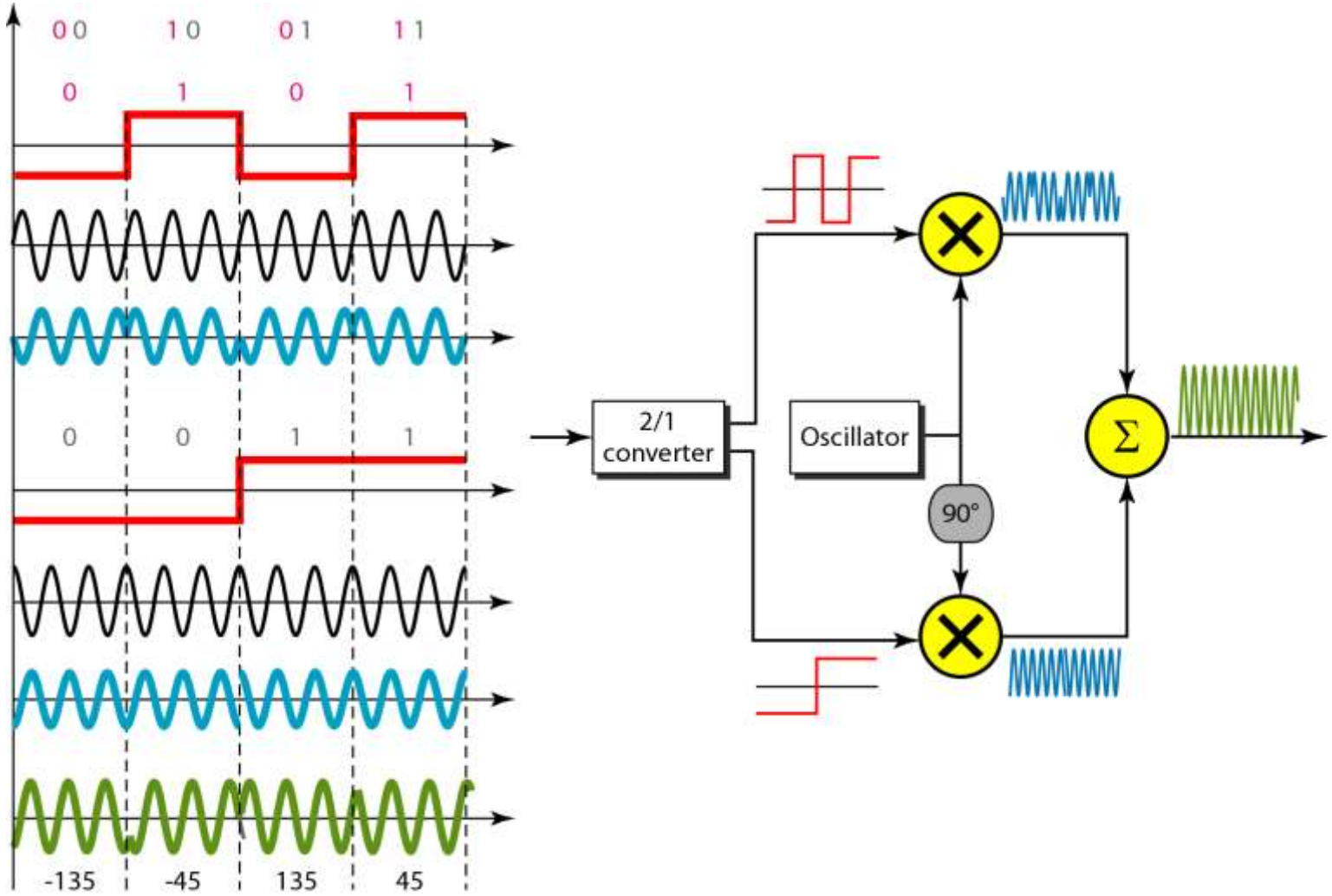


Figure 5.12 *Concept of a constellation diagram*

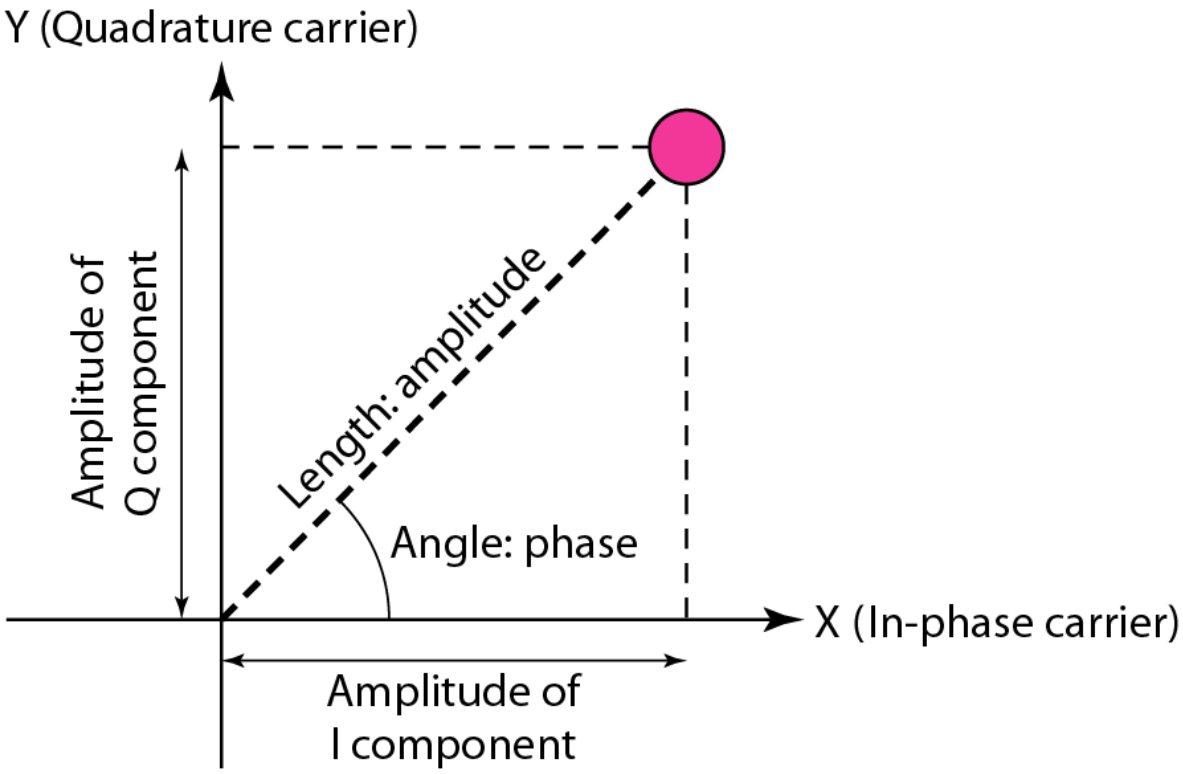
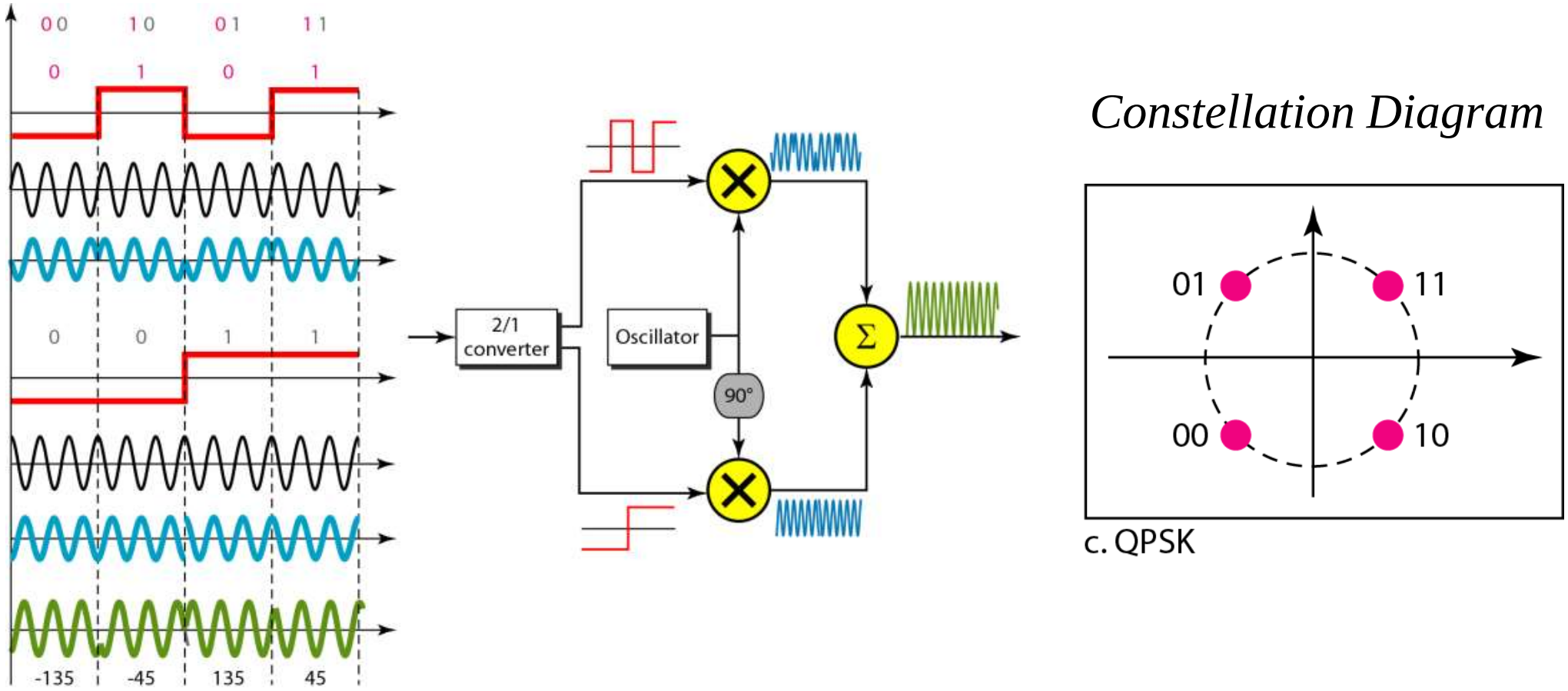


Figure 5.11 *QPSK and its implementation (Quadrature)*

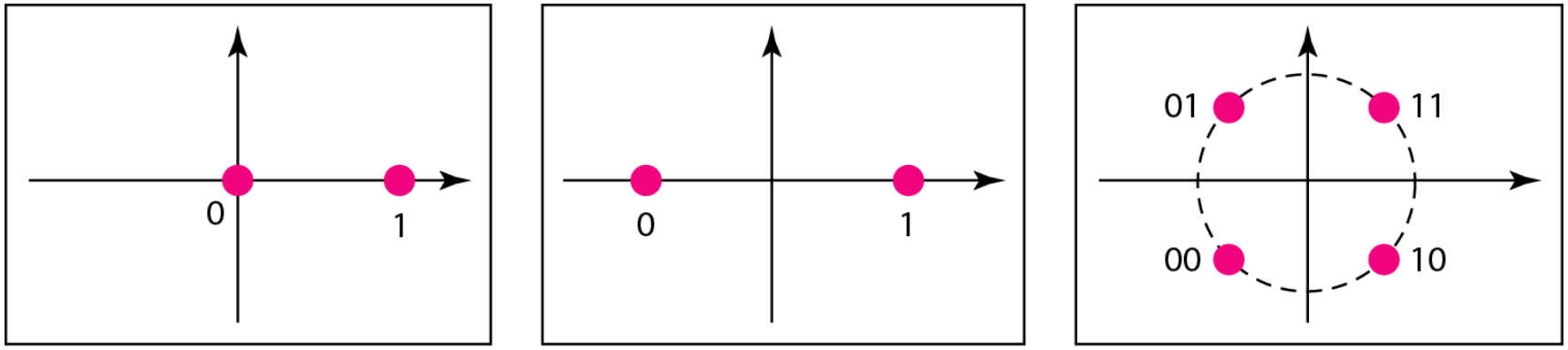


Show the constellation diagrams for an ASK (OOK), BPSK, and QPSK signals.

Solution

Figure 5.13 shows the three constellation diagrams.

Figure 5.13 *Three constellation diagrams*

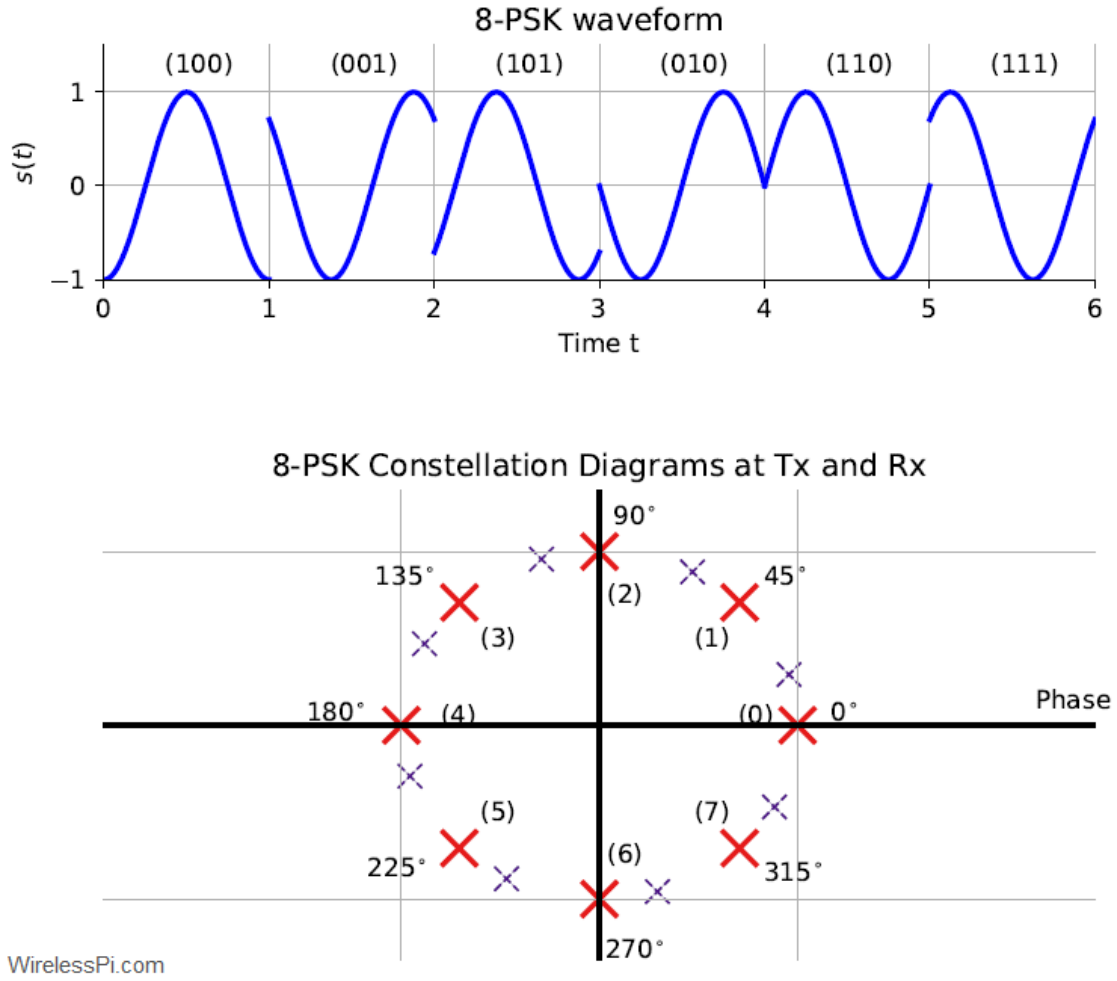


a. ASK (OOK)

b. BPSK

c. QPSK

Figure 5.14 *Constellation diagrams for some QAMs*



Note

Quadrature Amplitude Modulation (QAM) is a combination of ASK and PSK.

Figure 5.14 *Constellation diagrams for some QAMs*

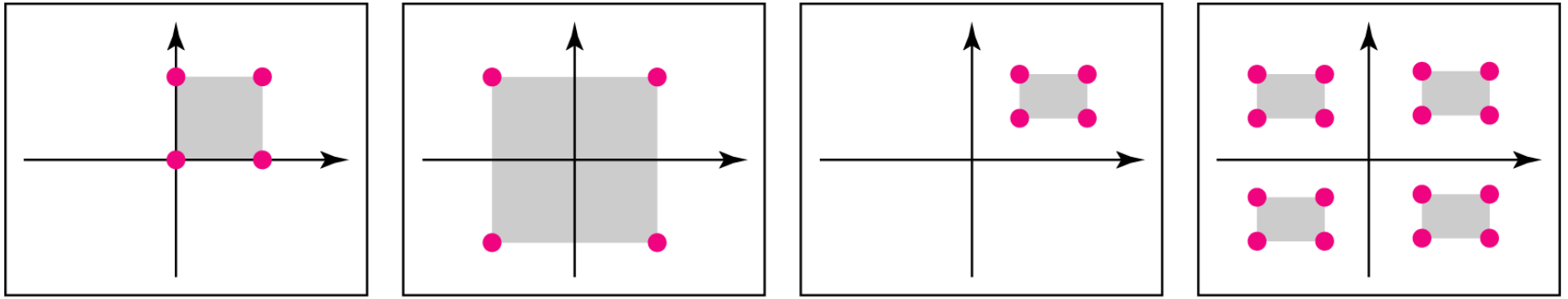
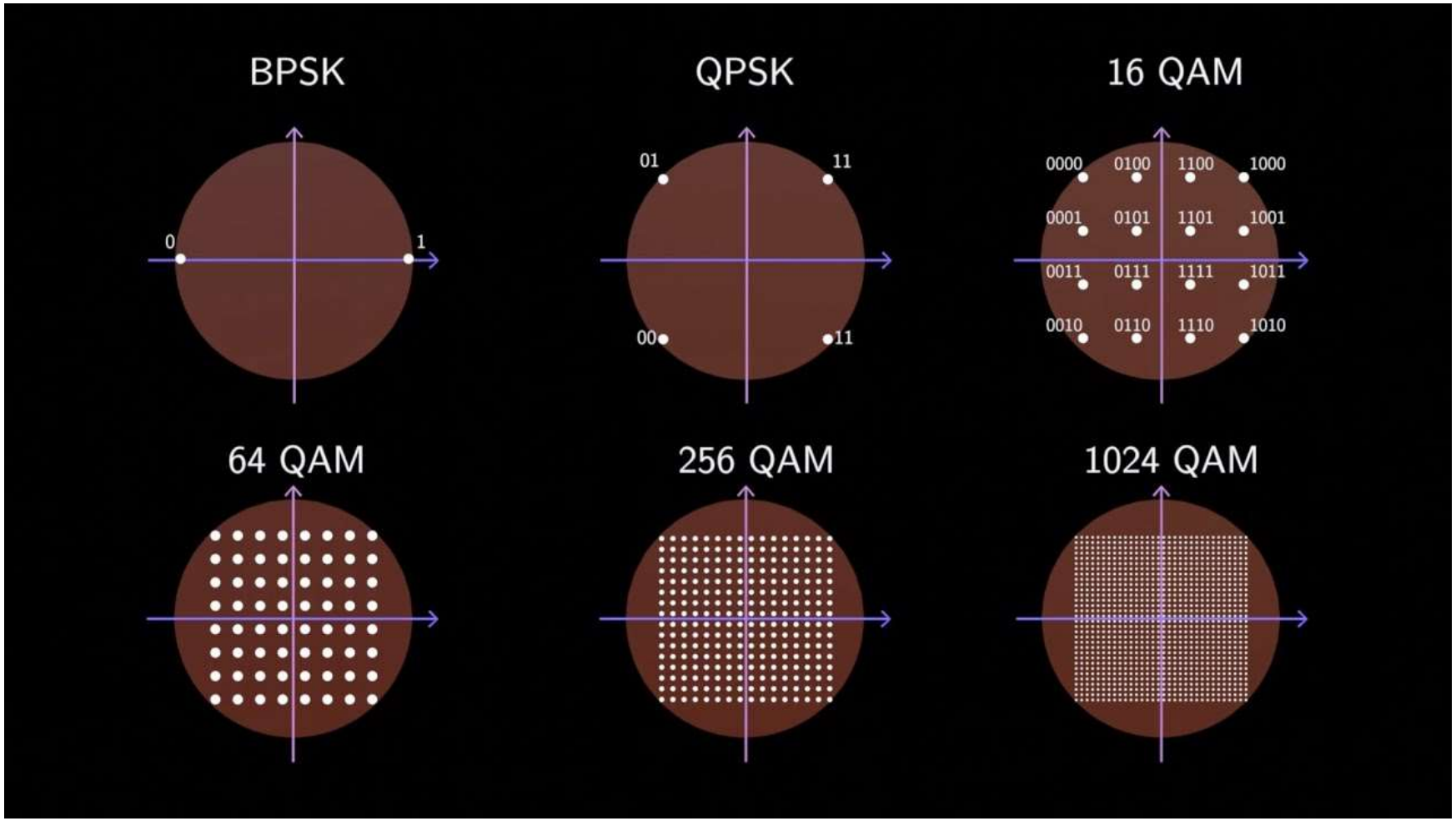


Figure 5.14 *Constellation diagrams for some QAMs*



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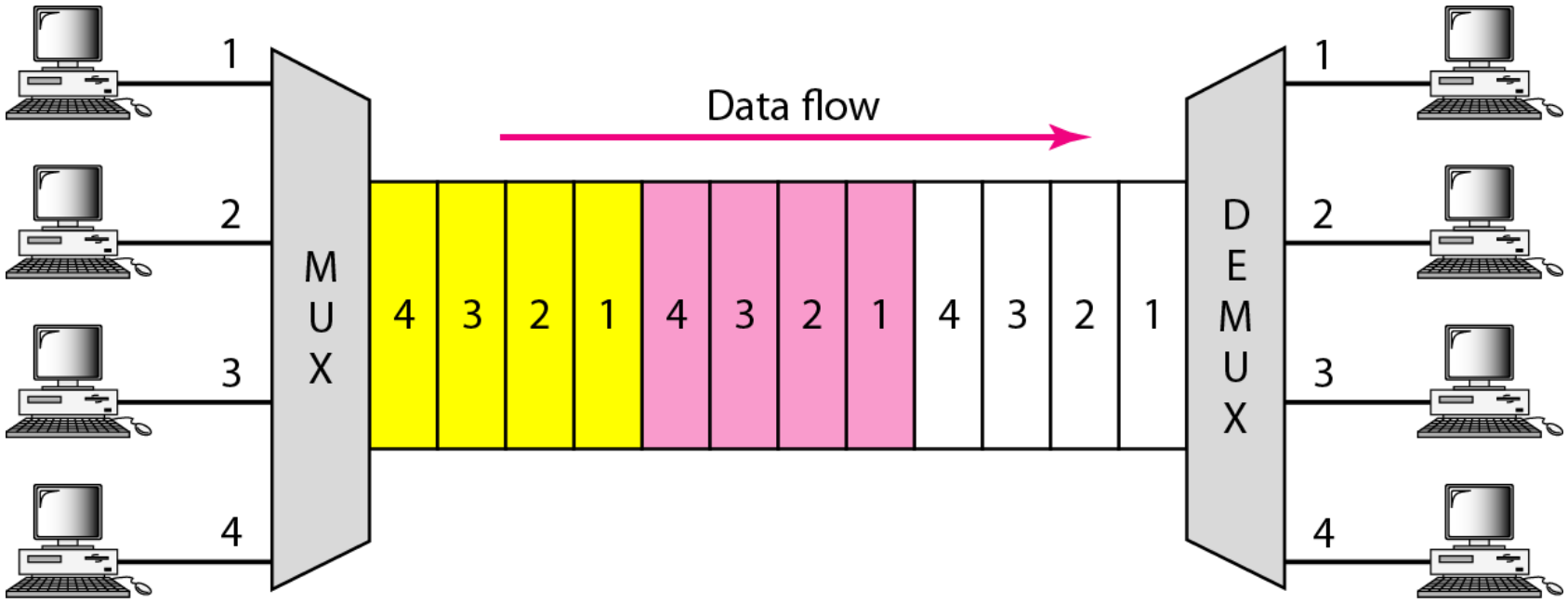
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Lec 12-14 (WDM, TDM, FHSS & DSSS)

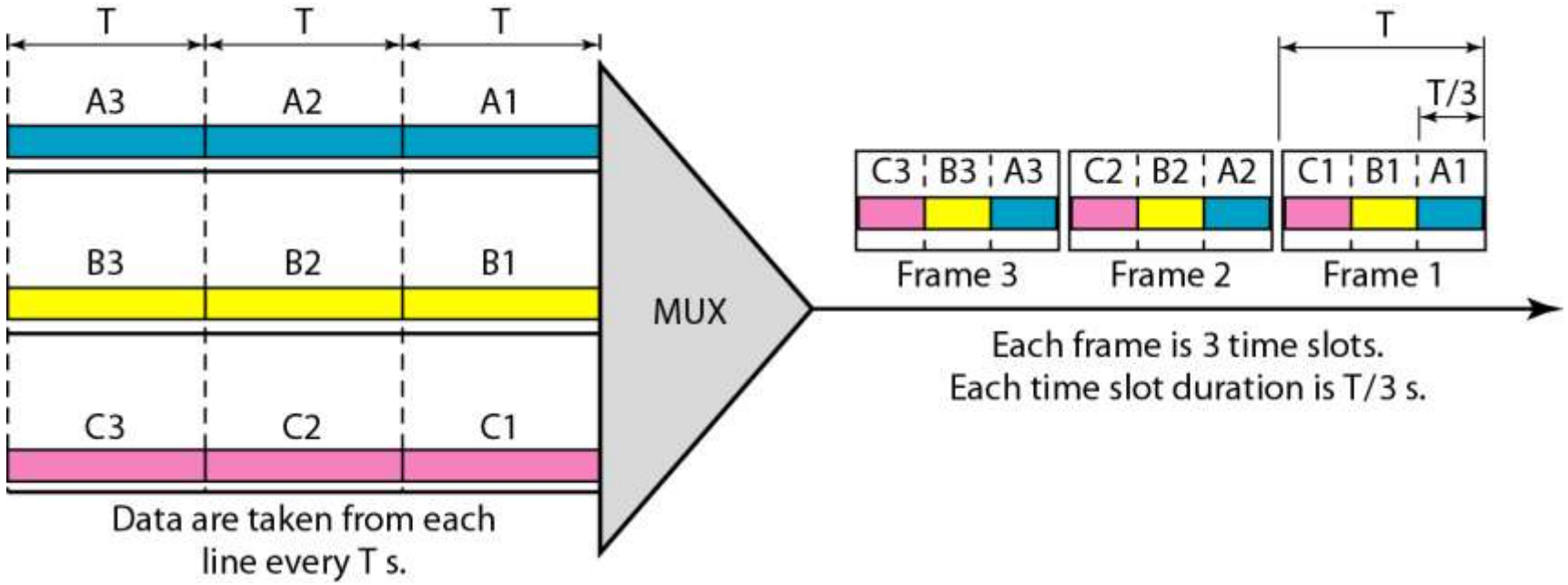
Figure 6.12 *TDM*



Note

TDM is a digital multiplexing technique for combining several low-rate channels into one high-rate one.

Figure 6.13 *Synchronous time-division multiplexing*



Example 6.5

In Figure 6.13, the data rate for each input connection is 3 kbps. If 1 bit at a time is multiplexed (a unit is 1 bit), what is the duration of (a) each input slot, (b) each output slot, and (c) each frame?

Solution

We can answer the questions as follows:

a. *The data rate of each input connection is 1 kbps. This means that the bit duration is 1/1000 s or 1 ms. The duration of the input time slot is 1 ms (same as bit duration).*

b. *The duration of each output time slot is one-third of the input time slot. This means that the duration of the output time slot is 1/3 ms.*

c. *Each frame carries three output time slots. So the duration of a frame is 3 × 1/3 ms, or 1 ms. The duration of a frame is the same as the duration of an input unit.*

Figure 6.14 *Example 6.6*

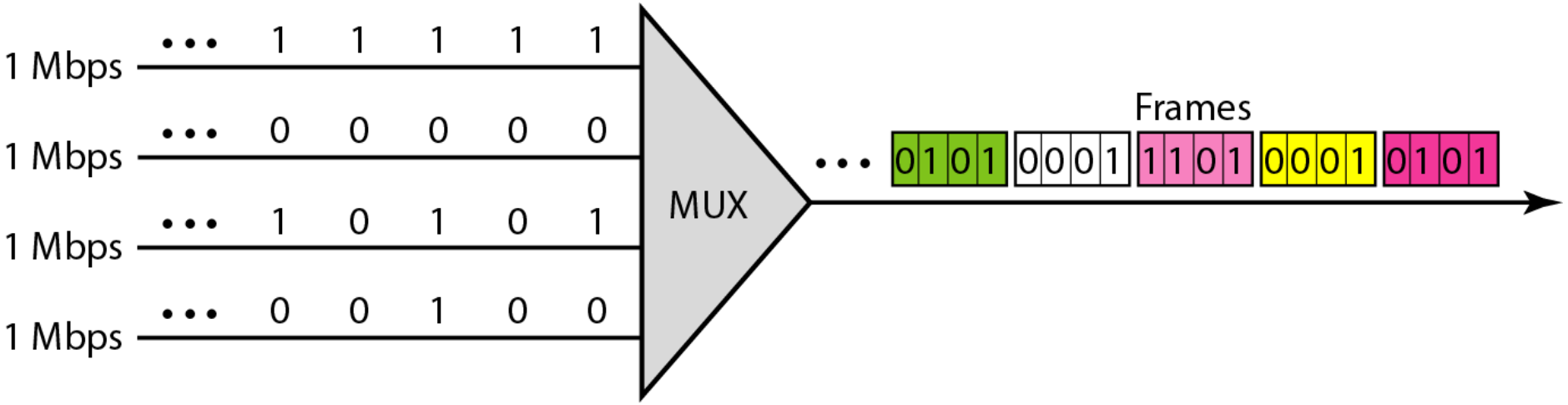


Figure 6.14 shows synchronous TDM with a data stream for each input and one data stream for the output. The unit of data is 1 bit. Find (a) the input bit duration, (b) the output bit duration, (c) the output bit rate, and (d) the output frame rate.

Solution

We can answer the questions as follows:

a. *The input bit duration is the inverse of the bit rate: 1/1 Mbps = 1 μs.*

b. *The output bit duration is one-fourth of the input bit duration, or ¼ μs.*

c. *The output bit rate is the inverse of the output bit duration or 1/(4μs) or 4 Mbps. This can also be deduced from the fact that the output rate is 4 times as fast as any input rate; so the output rate = 4 × 1 Mbps = 4 Mbps.*

d. *The frame rate is always the same as any input rate. So the frame rate is 1,000,000 frames per second. Because we are sending 4 bits in each frame, we can verify the result of the previous question by multiplying the frame rate by the number of bits per frame.*

Example 6.7

Four 1-kbps connections are multiplexed together. A unit is 1 bit. Find (a) the duration of 1 bit before multiplexing, (b) the transmission rate of the link, (c) the duration of a time slot, and (d) the duration of a frame.

Solution

We can answer the questions as follows:

a. The duration of 1 bit before multiplexing is 1 / 1 kbps, or 0.001 s (1 ms).

b. The rate of the link is 4 times the rate of a connection, or 4 kbps.

c. *The duration of each time slot is one-fourth of the duration of each bit before multiplexing, or 1/4 ms or 250 μs. Note that we can also calculate this from the data rate of the link, 4 kbps. The bit duration is the inverse of the data rate, or 1/4 kbps or 250 μs.*

d. *The duration of a frame is always the same as the duration of a unit before multiplexing, or 1 ms. We can also calculate this in another way. Each frame in this case has four time slots. So the duration of a frame is 4 times 250 μs, or 1 ms.*



Note

In synchronous TDM, the data rate of the link is *n* times faster, and the unit duration is *n* times shorter.

Figure 6.15 *Interleaving*

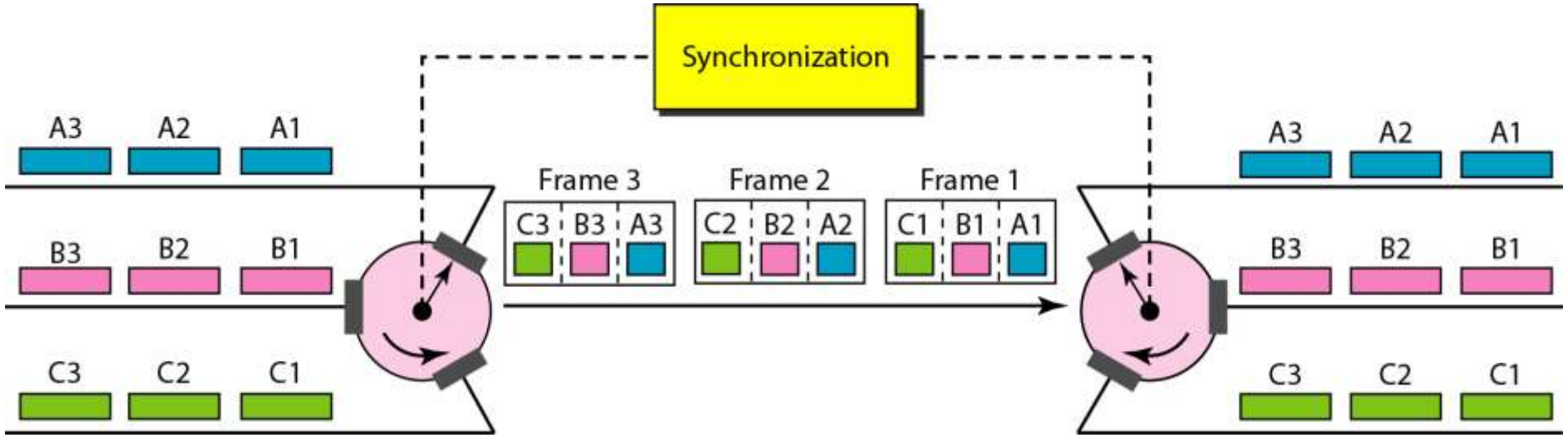
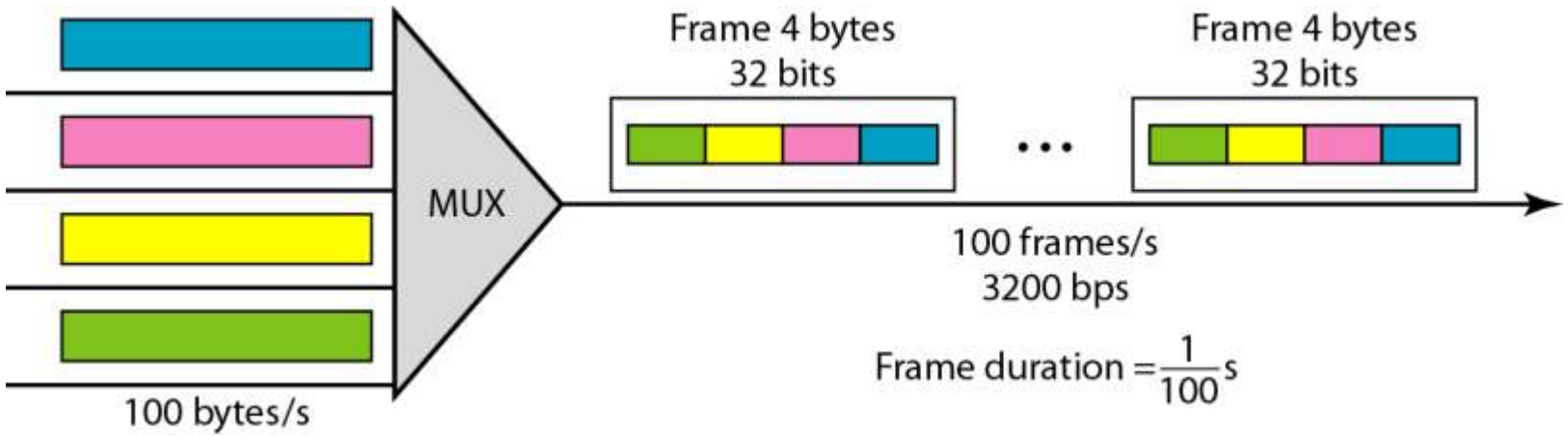


Figure 6.16 *Example 6.8*



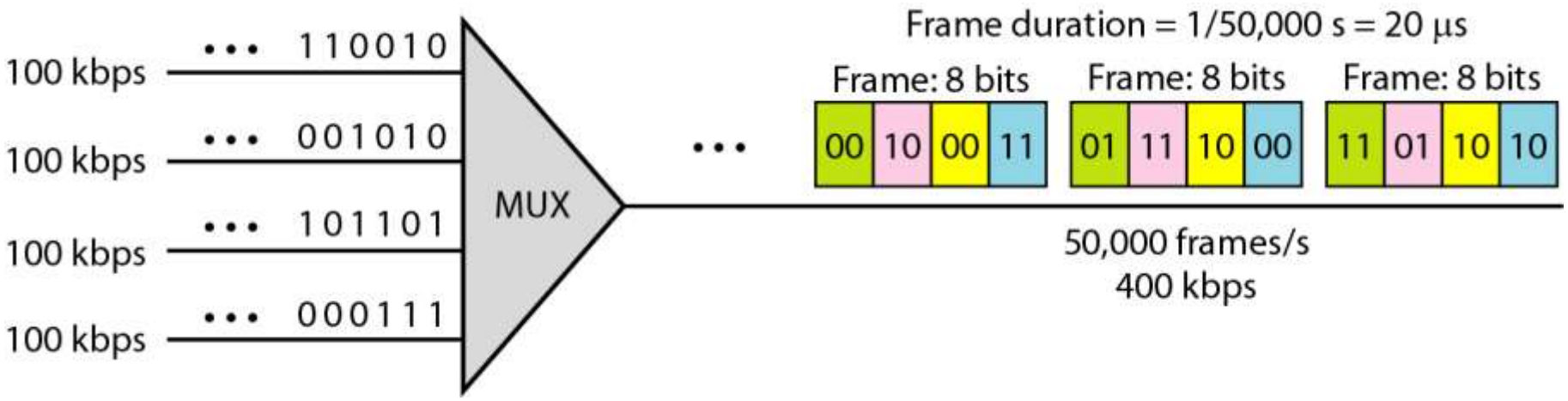
Example 6.8

Four channels are multiplexed using TDM. If each channel sends 100 bytes /s and we multiplex 1 byte per channel, show the frame traveling on the link, the size of the frame, the duration of a frame, the frame rate, and the bit rate for the link.

Solution

The multiplexer is shown in Figure 6.16. Each frame carries 1 byte from each channel; the size of each frame, therefore, is 4 bytes, or 32 bits. Because each channel is sending 100 bytes/s and a frame carries 1 byte from each channel, the frame rate must be 100 frames per second. The bit rate is 100 × 32, or 3200 bps.

Figure 6.17 *Example 6.9*



Example 6.9

A multiplexer combines four 100-kbps channels using a time slot of 2 bits. Show the output with four arbitrary inputs. What is the frame rate? What is the frame duration? What is the bit rate? What is the bit duration?

Solution

Figure 6.17 shows the output for four arbitrary inputs. The link carries 50,000 frames per second. The frame duration is therefore 1/50,000 s or 20 μs. The frame rate is 50,000 frames per second, and each frame carries 8 bits; the bit rate is 50,000 × 8 = 400,000 bits or 400 kbps. The bit duration is 1/400,000 s, or 2.5 μs.

Figure 6.18 *Empty slots*

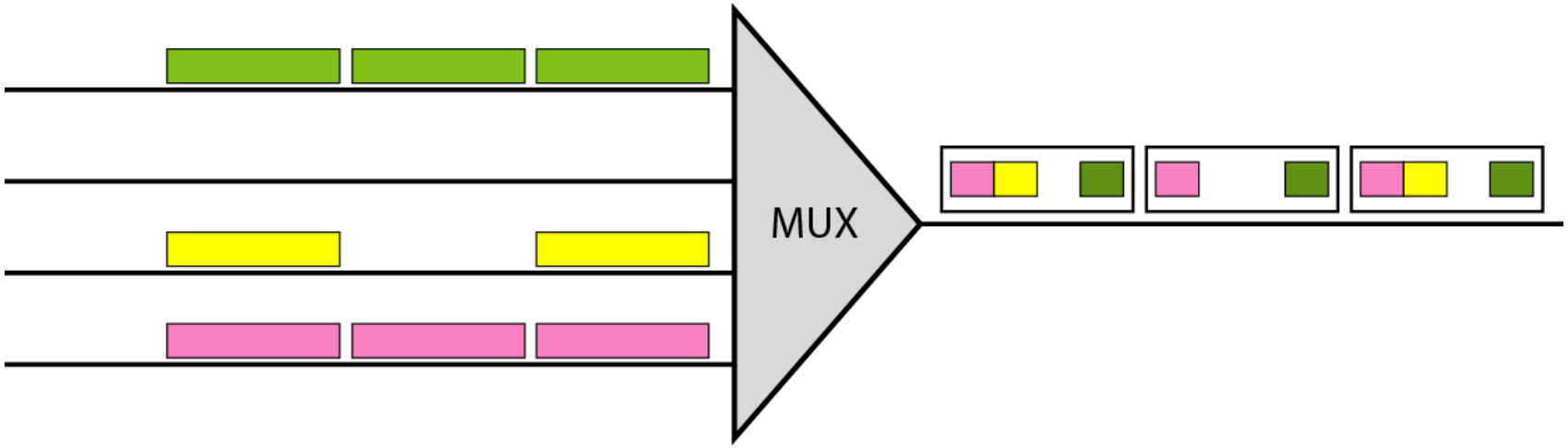


Figure 6.19 *Multilevel multiplexing*

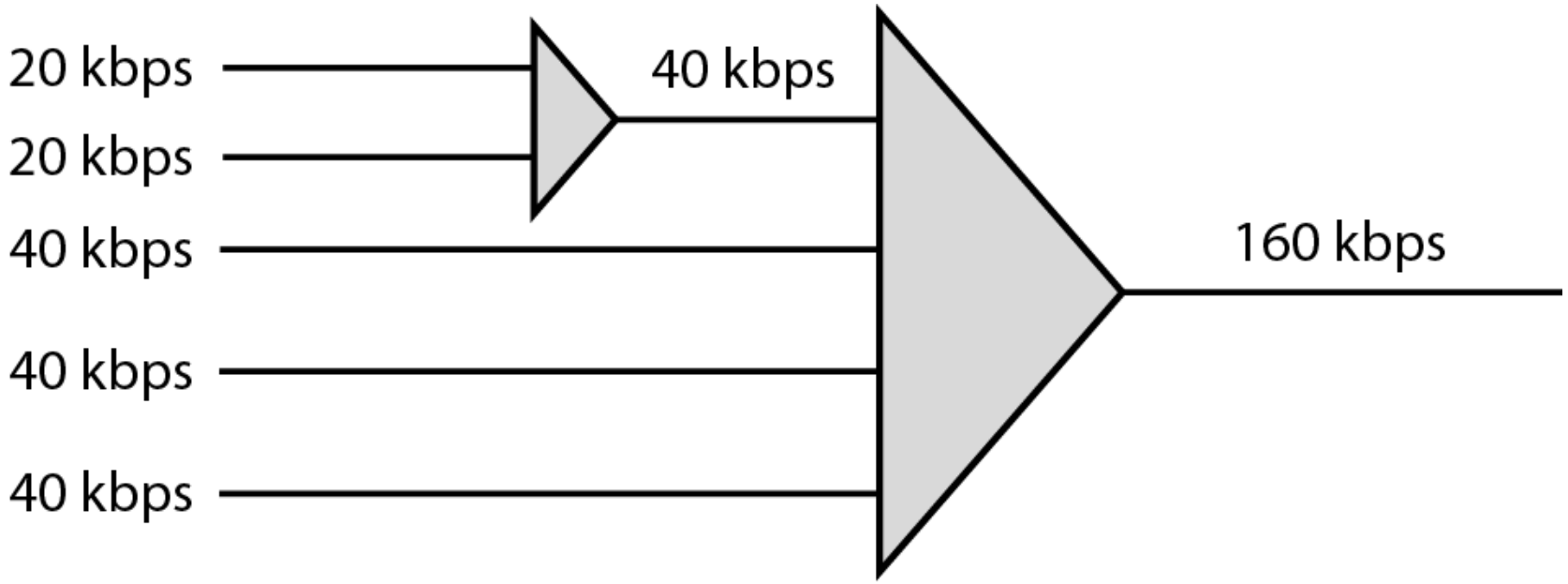


Figure 6.20 *Multiple-slot multiplexing*

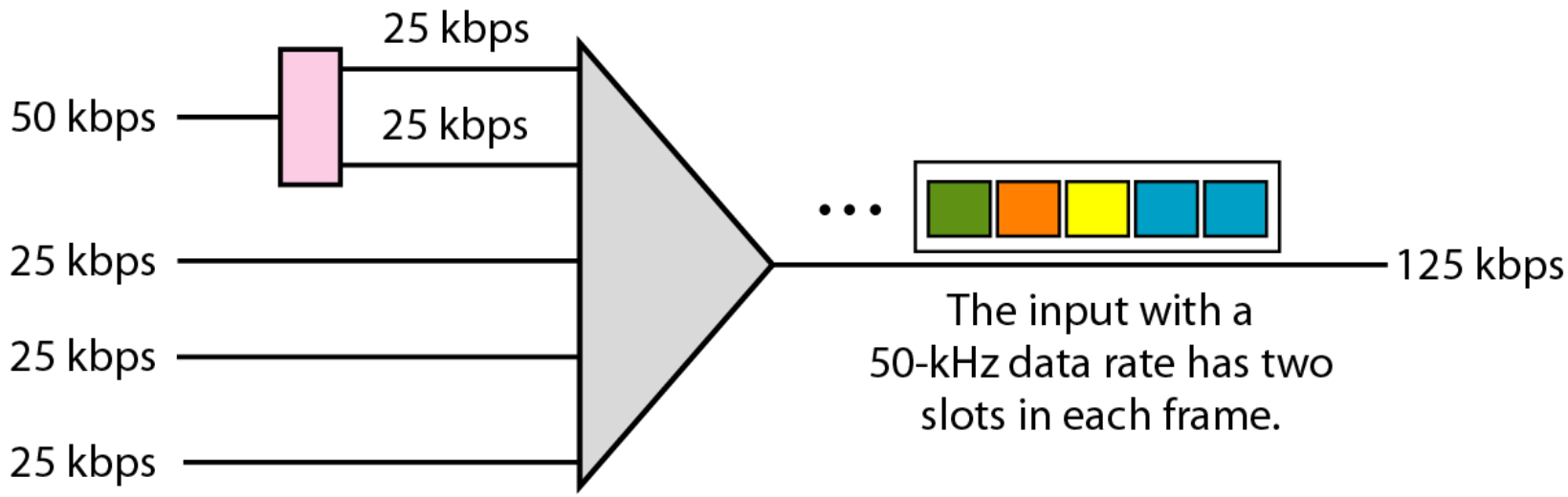


Figure 6.21 *Pulse stuffing*

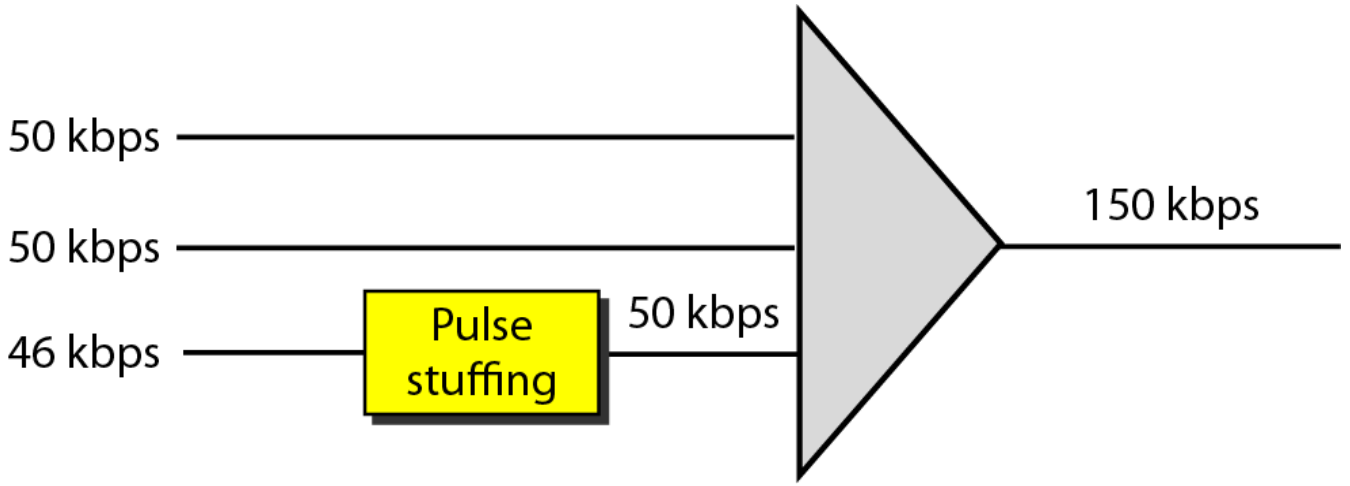
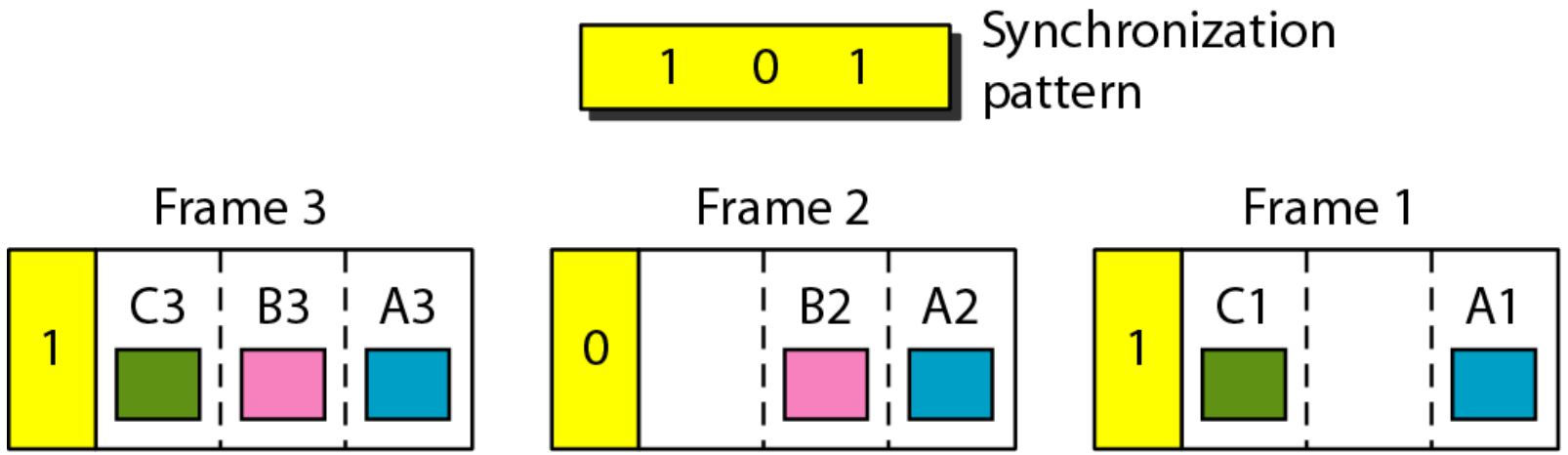


Figure 6.22 *Framing bits*



Example 6.10

We have four sources, each creating 250 characters per second. If the interleaved unit is a character and 1 synchronizing bit is added to each frame, find (a) the data rate of each source, (b) the duration of each character in each source, (c) the frame rate, (d) the duration of each frame, (e) the number of bits in each frame, and (f) the data rate of the link.

Solution

We can answer the questions as follows:

a. *The data rate of each source is 250 × 8 = 2000 bps = 2 kbps.*

b. *Each source sends 250 characters per second; therefore, the duration of a character is 1/250 s, or 4 ms.*

c. *Each frame has one character from each source, which means the link needs to send 250 frames per second to keep the transmission rate of each source.*

d. *The duration of each frame is 1/250 s, or 4 ms. Note that the duration of each frame is the same as the duration of each character coming from each source.*

e. *Each frame carries 4 characters and 1 extra synchronizing bit. This means that each frame is* $4 \times 8 + 1 = 33$ *bits.*

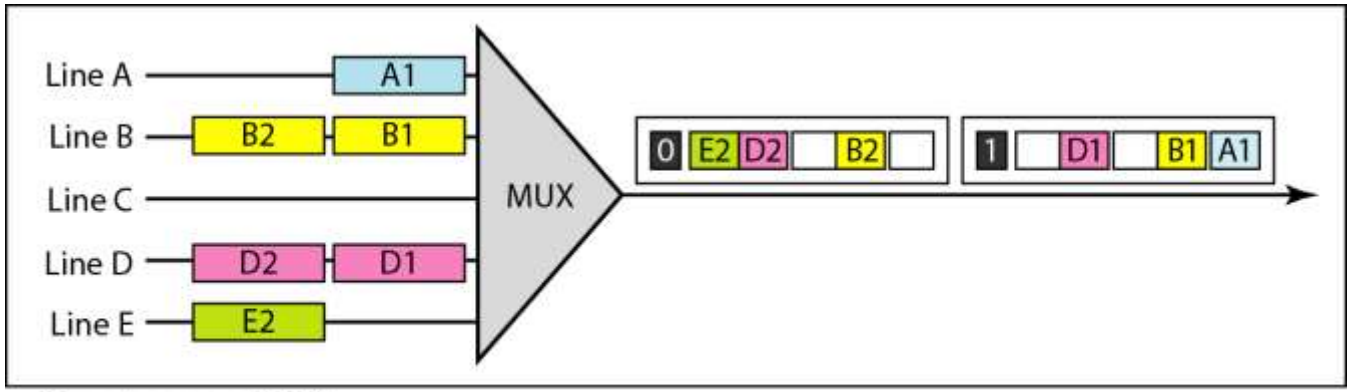
Example 6.11

Two channels, one with a bit rate of 100 kbps and another with a bit rate of 200 kbps, are to be multiplexed. How this can be achieved? What is the frame rate? What is the frame duration? What is the bit rate of the link?

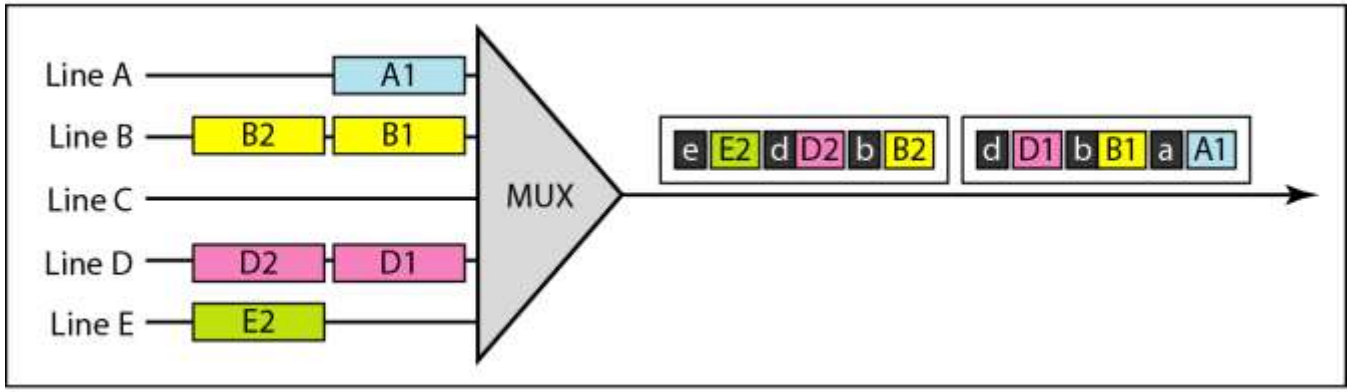
Solution

We can allocate one slot to the first channel and two slots to the second channel. Each frame carries 3 bits. The frame rate is 100,000 frames per second because it carries 1 bit from the first channel. The bit rate is 100,000 frames/s × 3 bits per frame, or 300 kbps.

Figure 6.26 *TDM slot comparison*



a. Synchronous TDM



b. Statistical TDM

6-1 SPREAD SPECTRUM

In spread spectrum (SS), we combine signals from different sources to fit into a larger bandwidth, but our goals are to prevent eavesdropping and jamming. To achieve these goals, spread spectrum techniques add redundancy.

Topics discussed in this section:

Frequency Hopping Spread Spectrum (FHSS)
Direct Sequence Spread Spectrum Synchronous (DSSS)

Figure 6.27 *Spread spectrum*

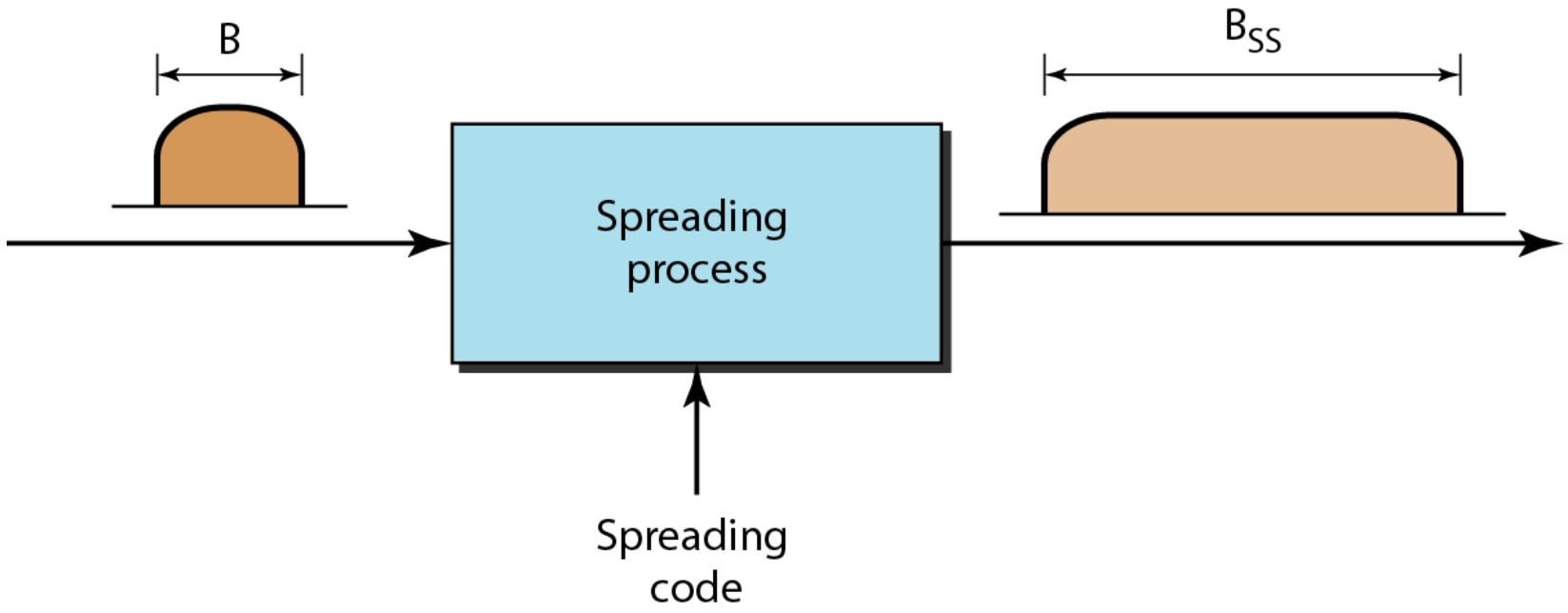


Figure 6.28 *Frequency hopping spread spectrum (FHSS)*

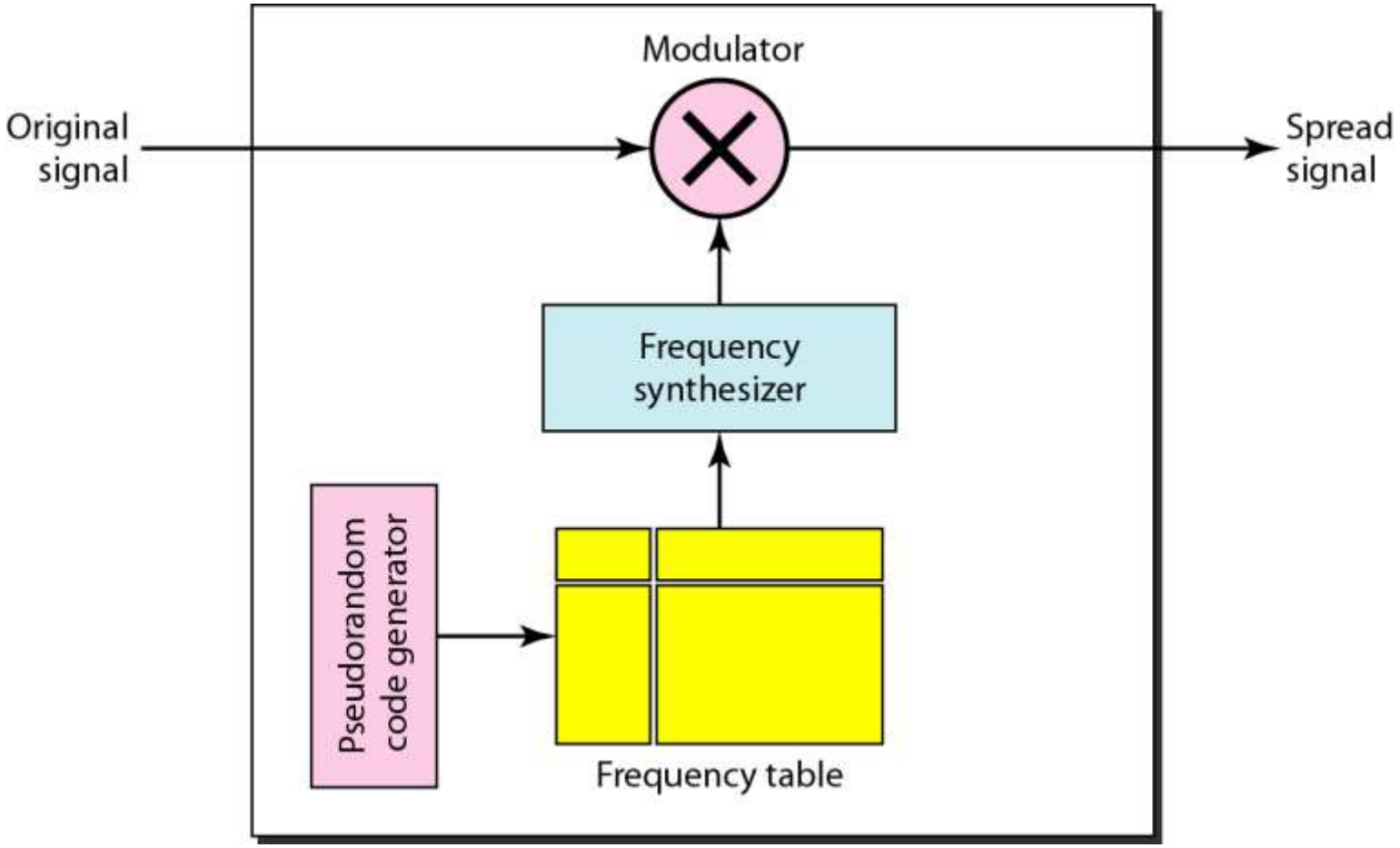


Figure 6.29 *Frequency selection in FHSS*

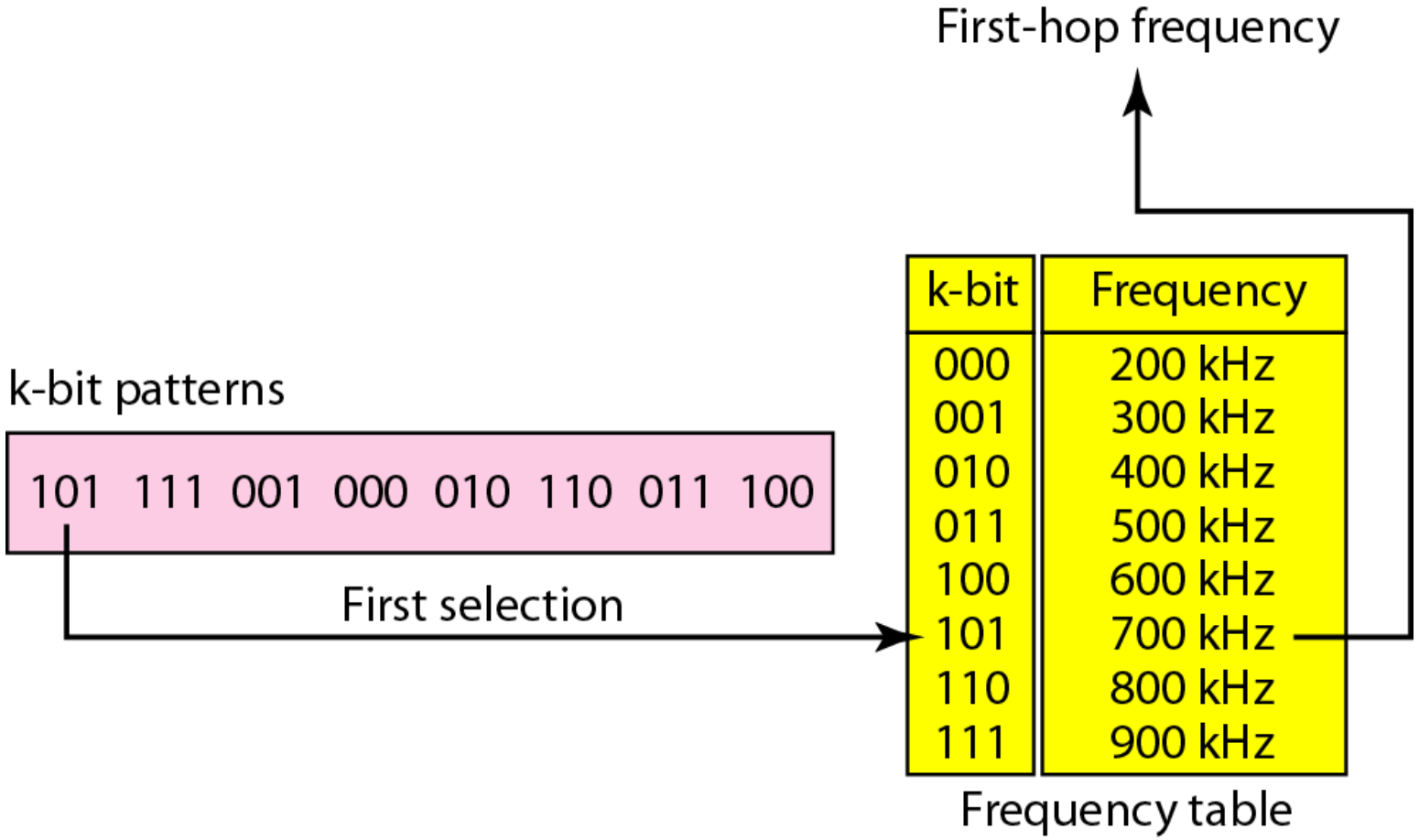


Figure 6.30 *FHSS cycles*

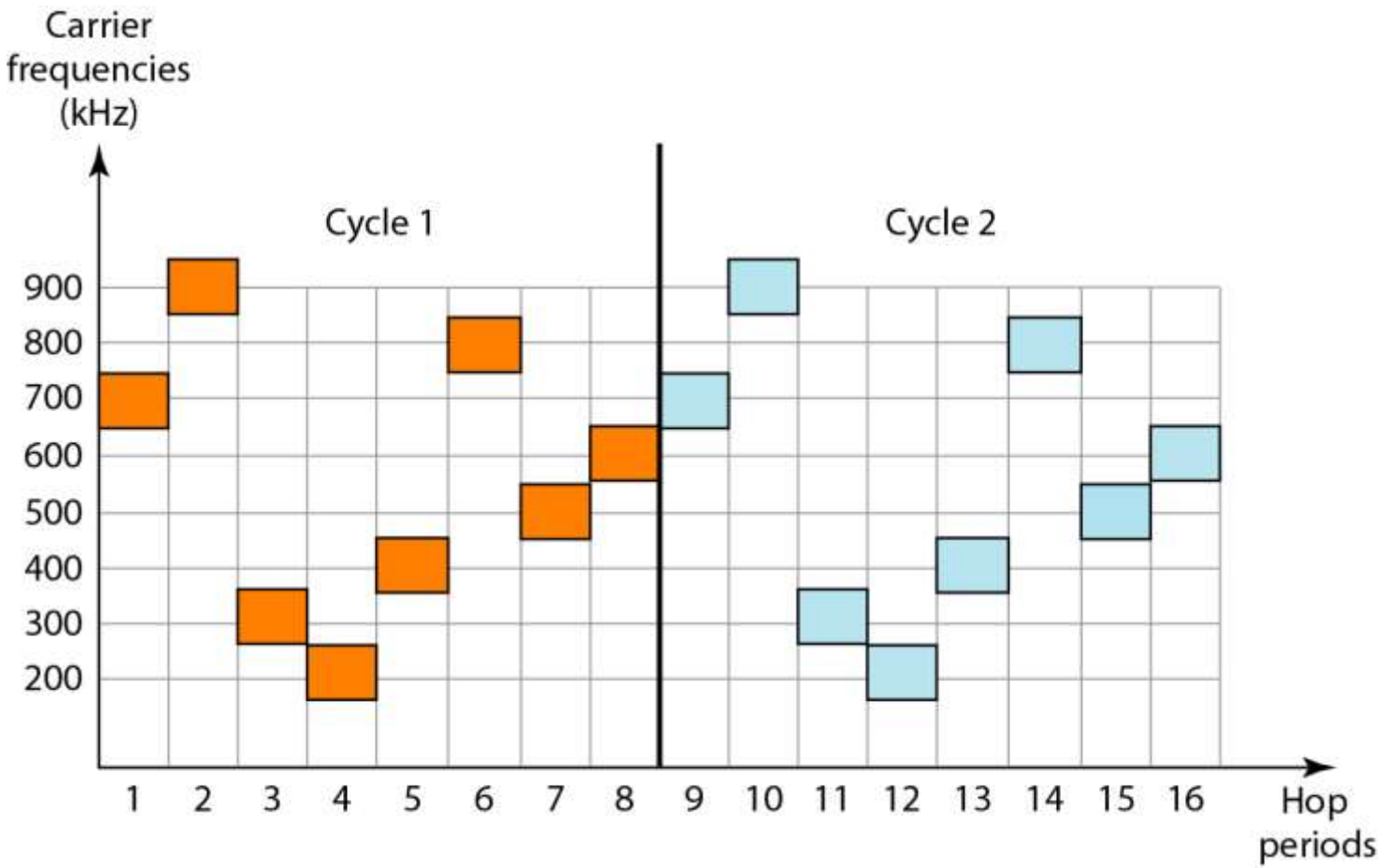


Figure 6.31 *Bandwidth sharing*

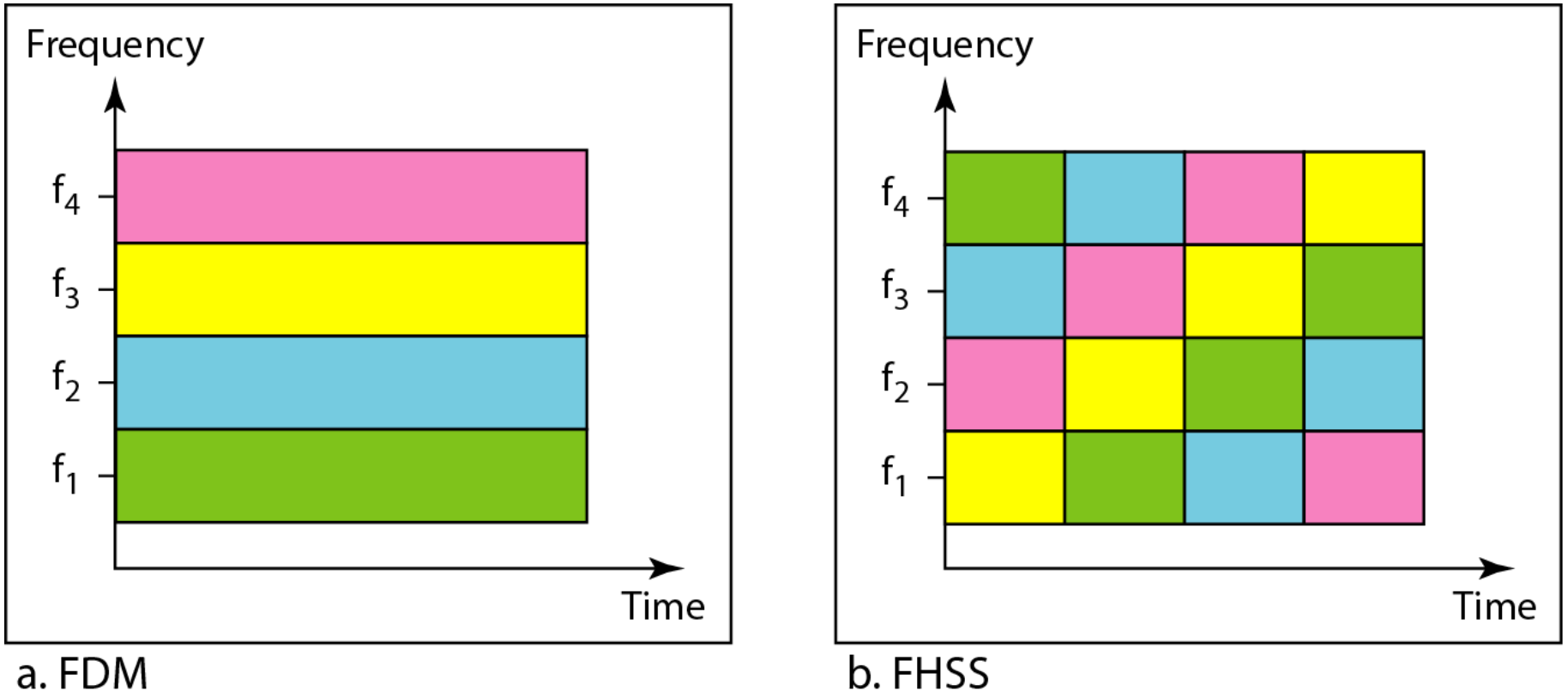


Figure 6.32 Direct Sequence Spread Spectrum Synchronous (DSSS)

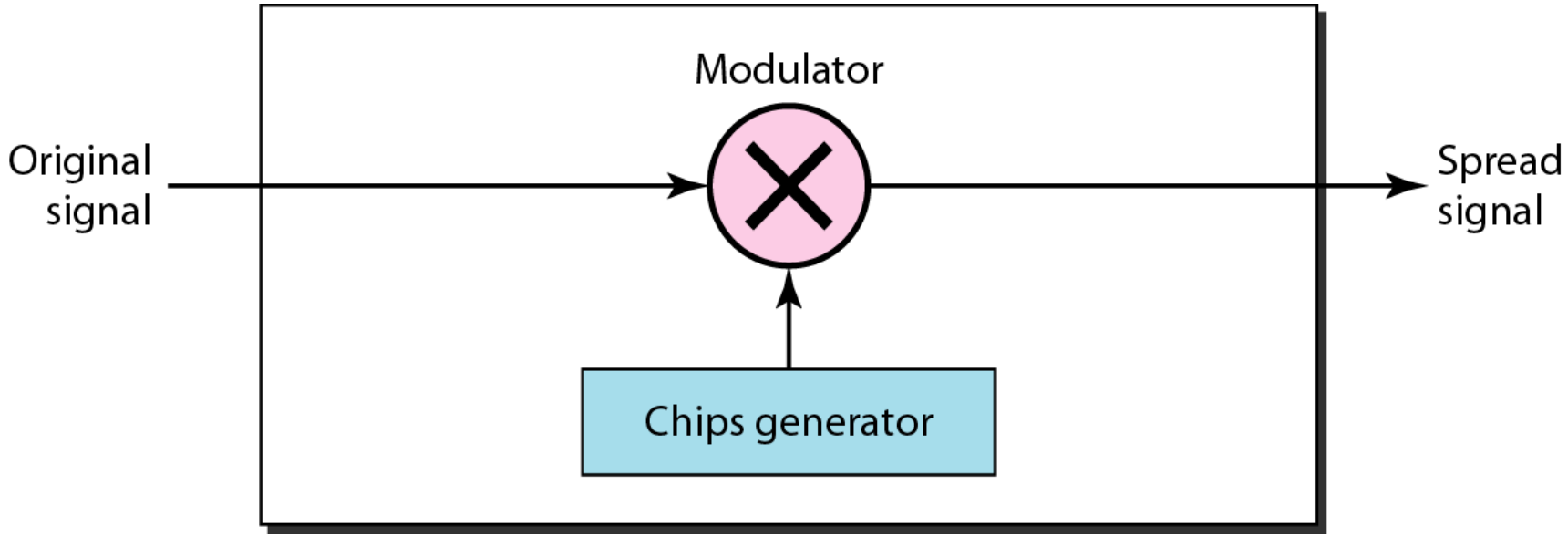
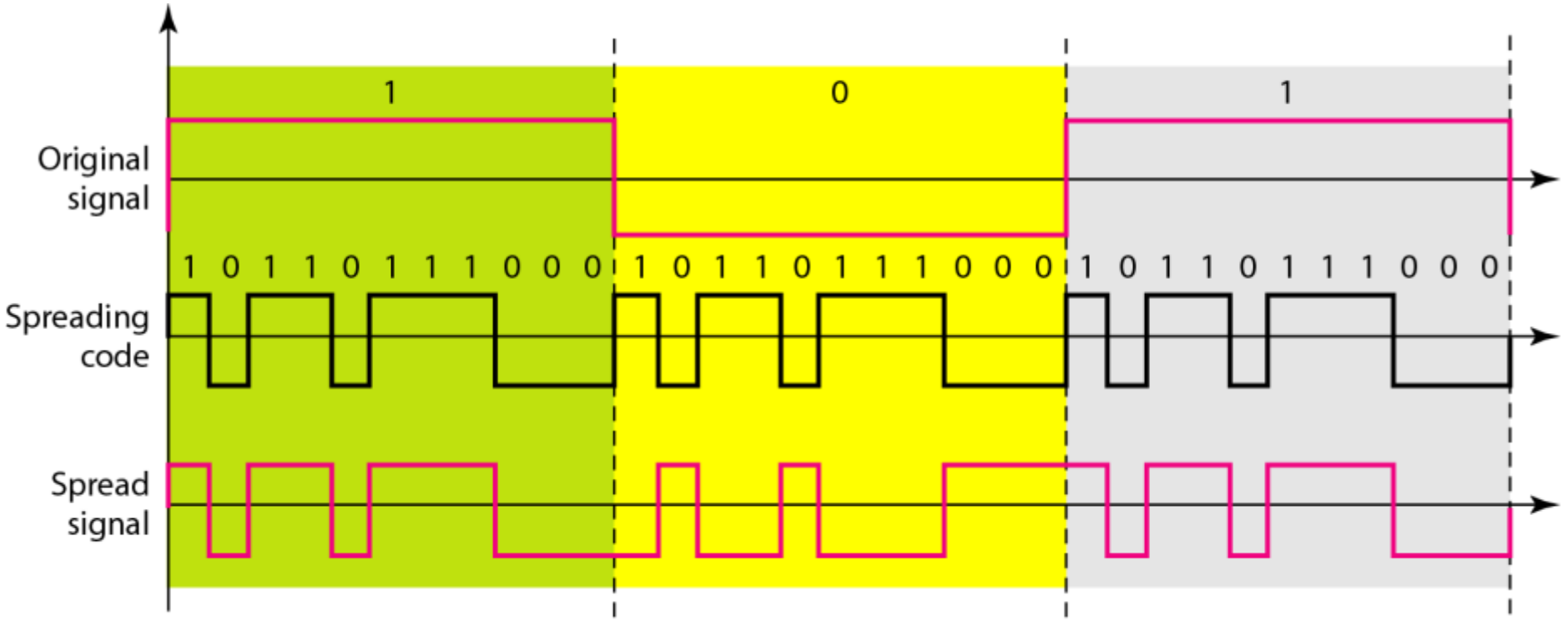


Figure 6.33 *DSSS example*



CSE 317

Data Communication Level – 3 (Spring Term)

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Instr Class B, CSE Dept

MIST

Lec 16 (Switch)

8-4 STRUCTURE OF A SWITCH

We use switches in circuit-switched and packet-switched networks. In this section, we discuss the structures of the switches used in each type of network.

Topics discussed in this section:

Structure of Circuit Switches
Structure of Packet Switches

Figure 8.17 *Crossbar switch with three inputs and four outputs*

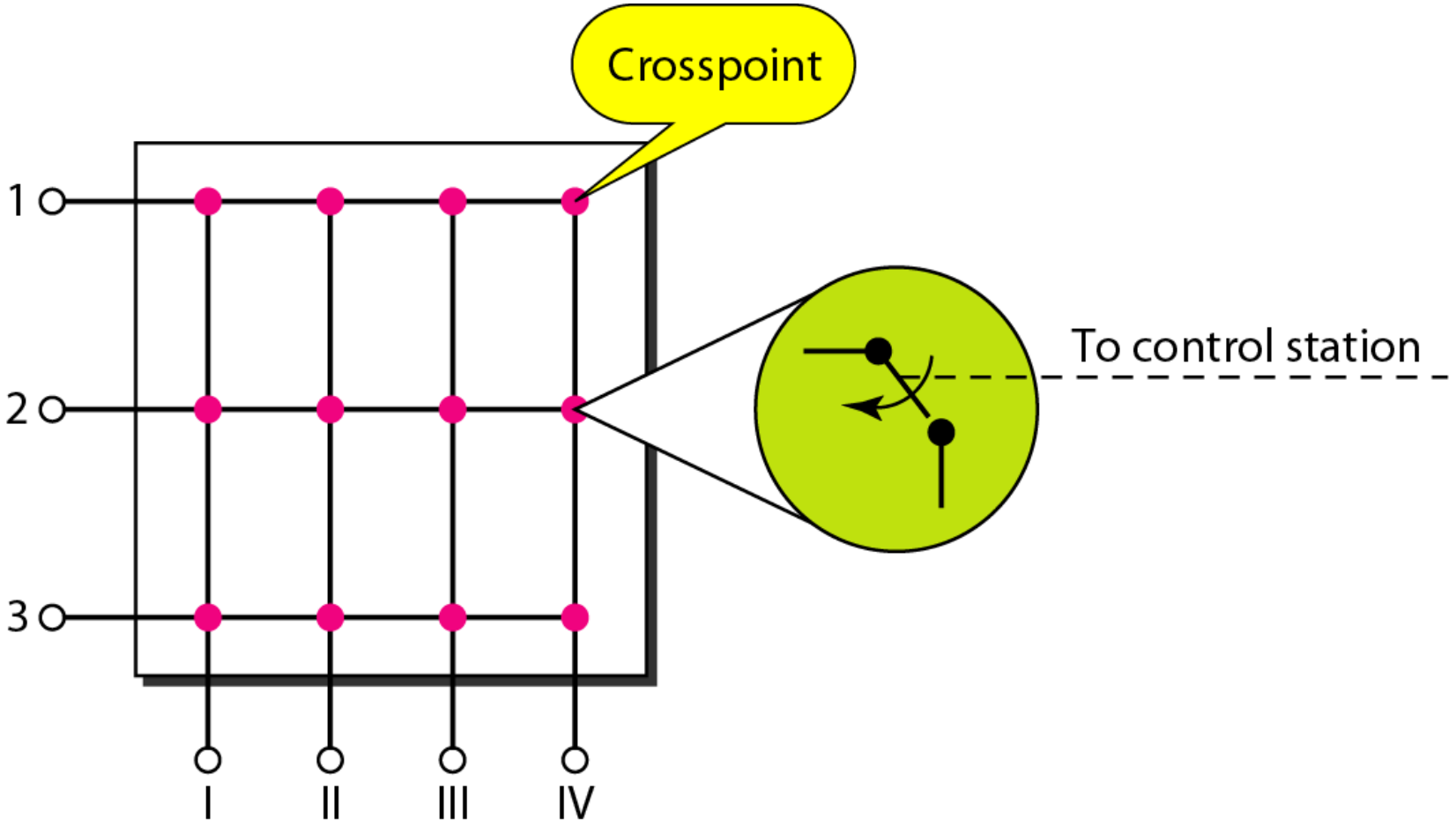
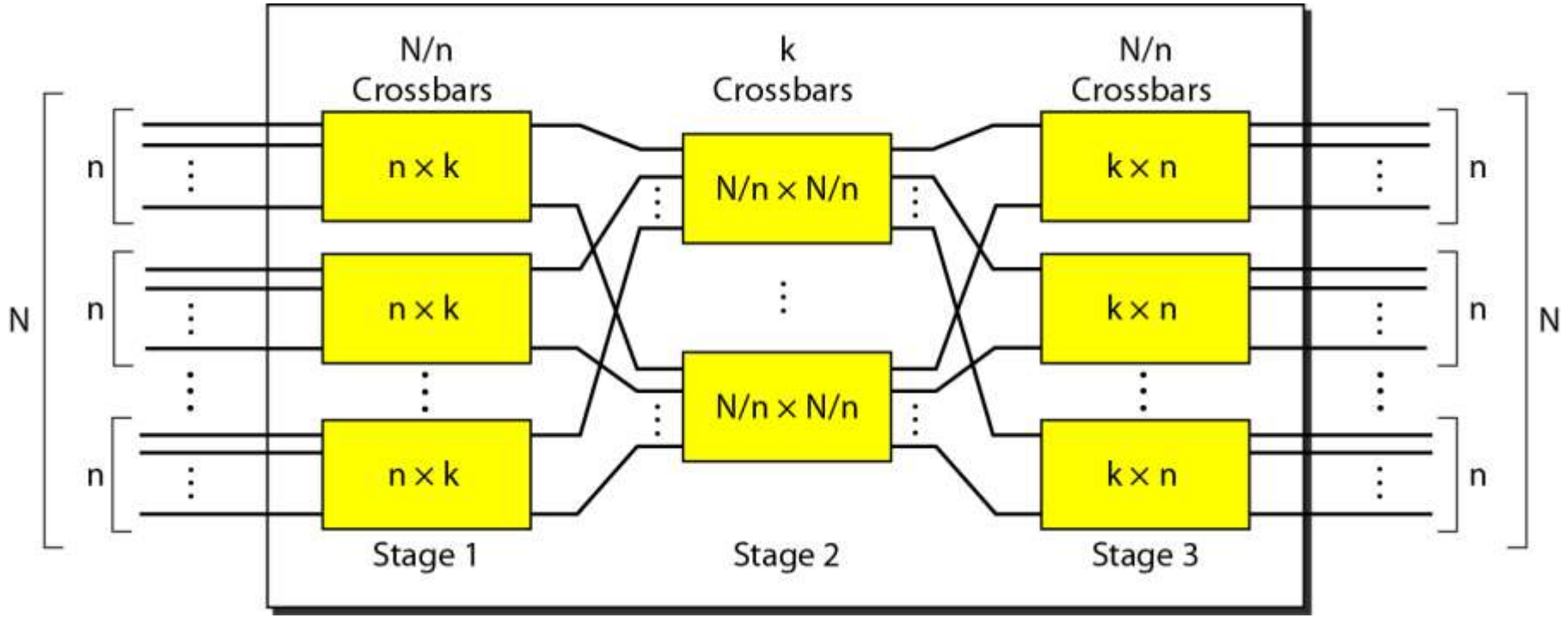


Figure 8.18 *Multistage switch*



Note

In a three-stage switch, the total number of crosspoints is $2kN + k(N/n)^2$ which is much smaller than the number of crosspoints in a single-stage switch (N^2).

Switching network	Crosspoints
Single-stage 64×64	4096
3-stage	1536

So the 3-stage network uses **1536/4096 ≈ 37.5%** as many crosspoints as the single-s

Example 8.3

Design a three-stage, 200 × 200 switch (N = 200) with k = 4 and n = 20.

Solution

In the first stage we have N/n or 10 crossbars, each of size 20 × 4. In the second stage, we have 4 crossbars, each of size 10 × 10. In the third stage, we have 10 crossbars, each of size 4 × 20. The total number of crosspoints is $2kN + k(N/n)^2$, or 2000 crosspoints. This is 5 percent of the number of crosspoints in a single-stage switch (200 × 200 = 40,000).

Note

According to the Clos criterion:

$n = (N/2)^{1/2}$

$k > 2n - 1$

Crosspoints $\geq 4N\,[(2N)^{1/2} - 1]$

Example 8.4

Redesign the previous three-stage, 200 × 200 switch, using the Clos criteria with a minimum number of crosspoints.

Solution

We let $n = (200/2)^{1/2}$, or $n = 10$. We calculate $k = 2n - 1 = 19$. In the first stage, we have 200/10, or 20, crossbars, each with 10 × 19 crosspoints. In the second stage, we have 19 crossbars, each with 10 × 10 crosspoints. In the third stage, we have 20 crossbars each with 19 × 10 crosspoints. The total number of crosspoints is 20(10 × 19) + 19(10 × 10) + 20(19 ×10) = 9500.

Figure 8.19 *Time-slot interchange*

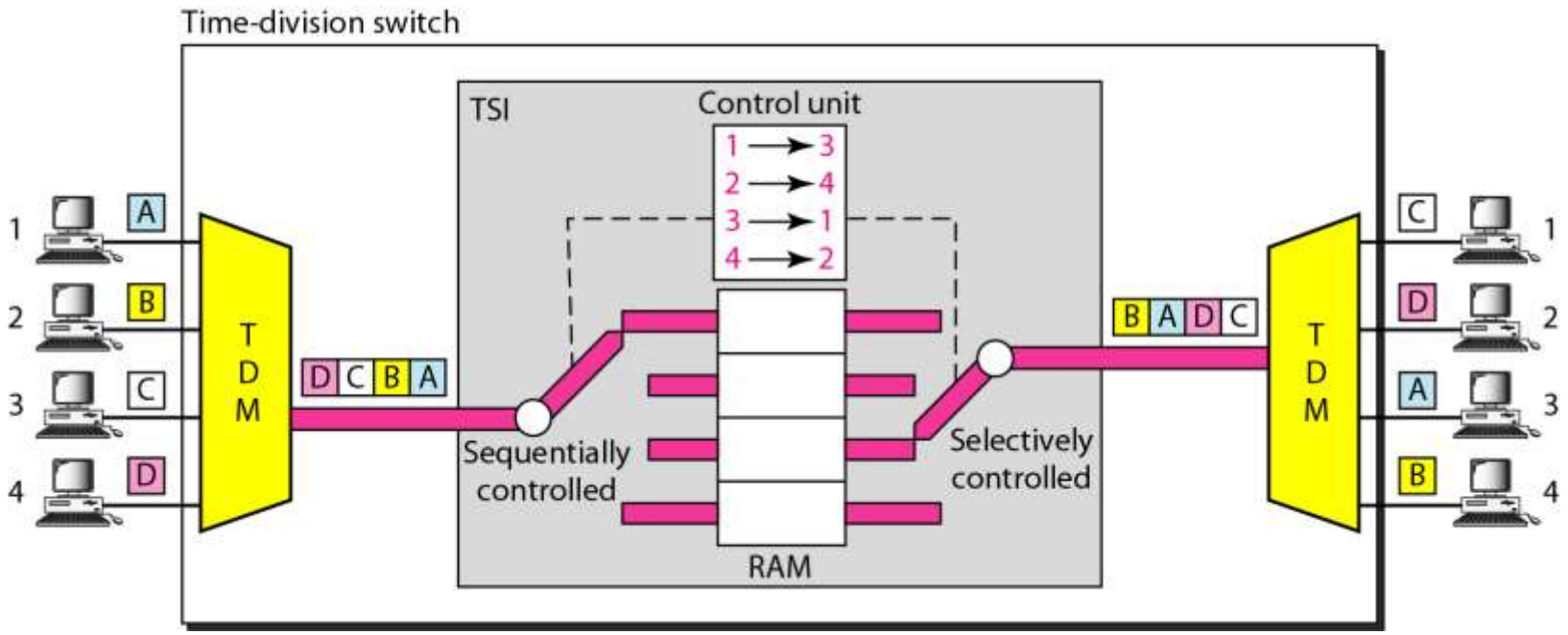


Figure 8.20 *Time-space-time switch*

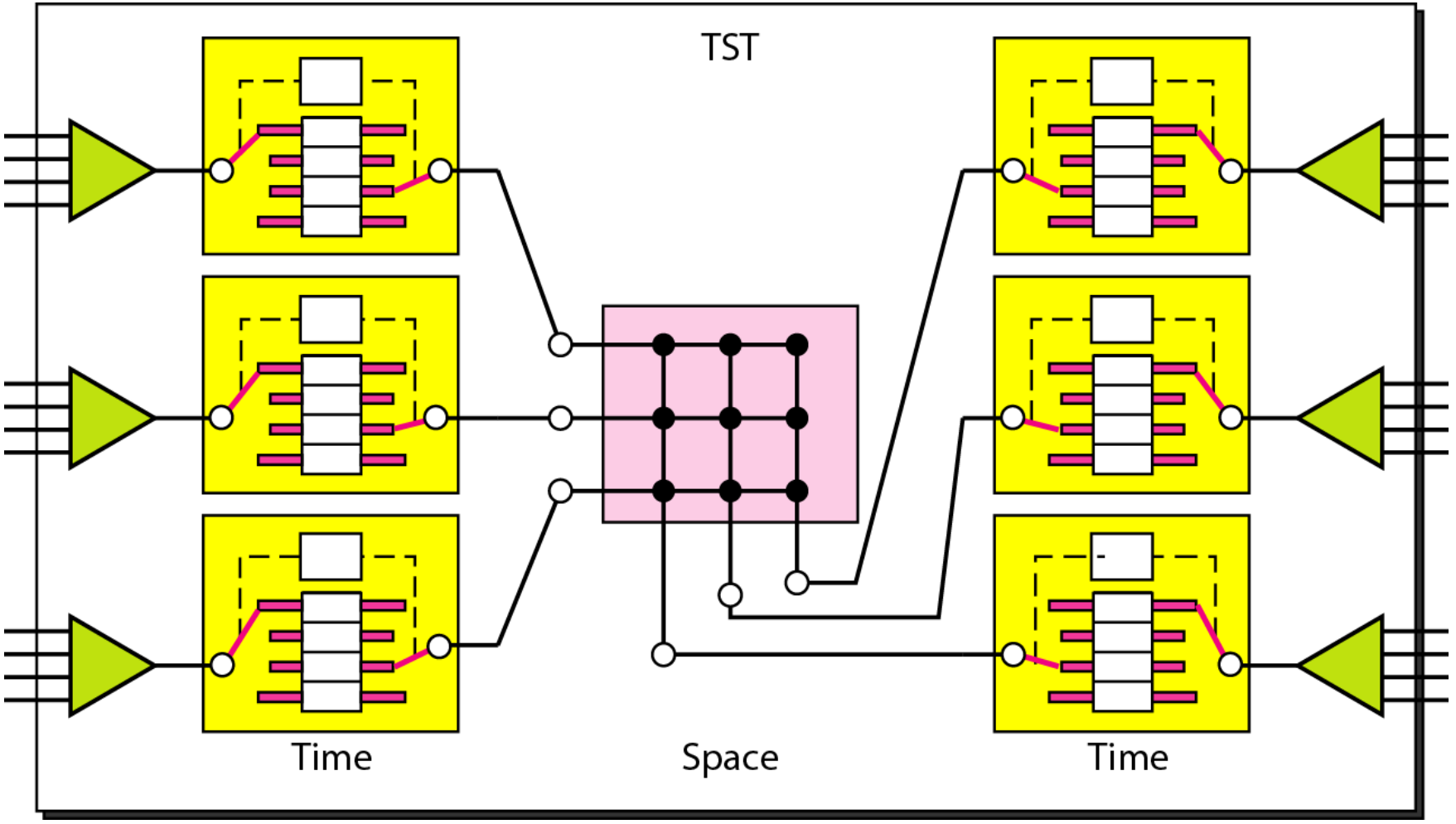


Figure 8.21 *Packet switch components*

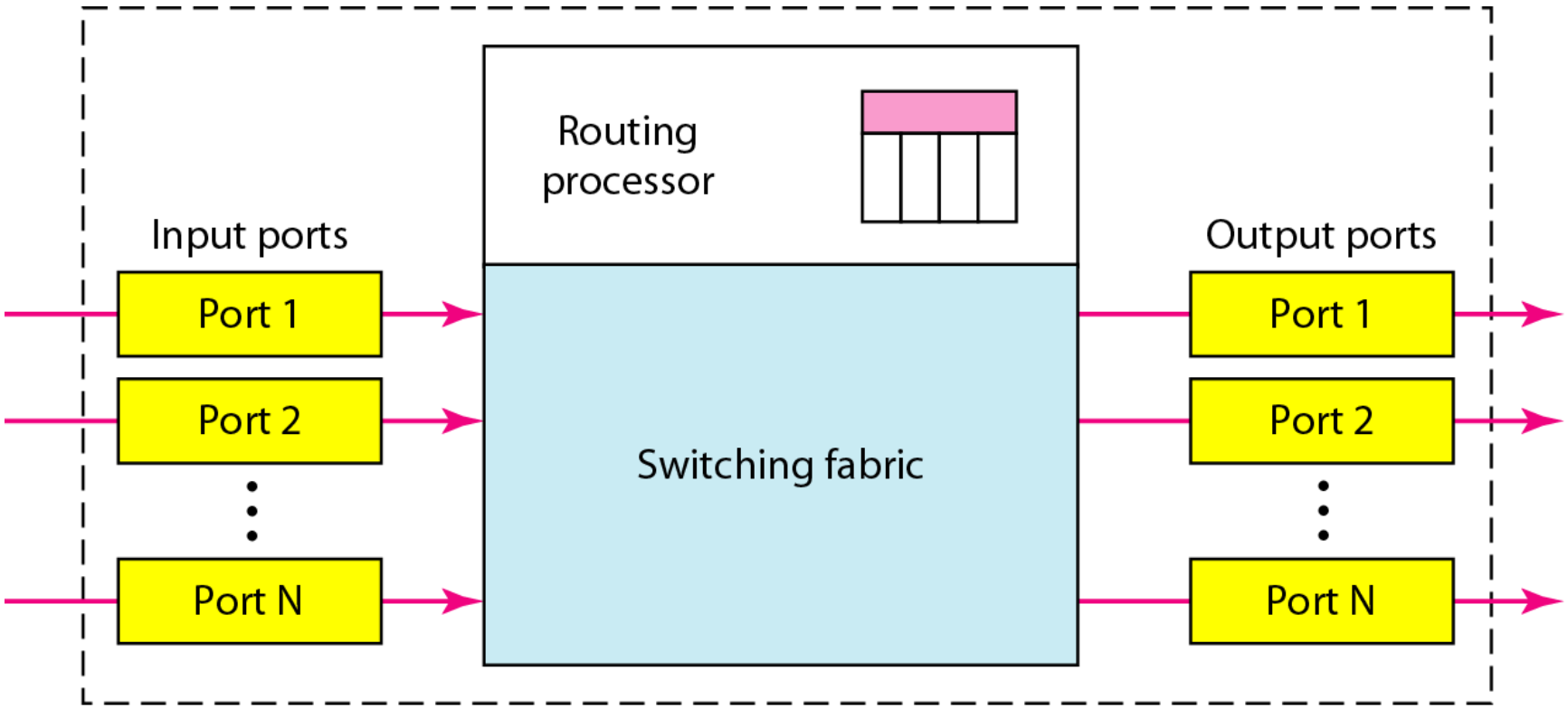


Figure 8.22 *Input port*

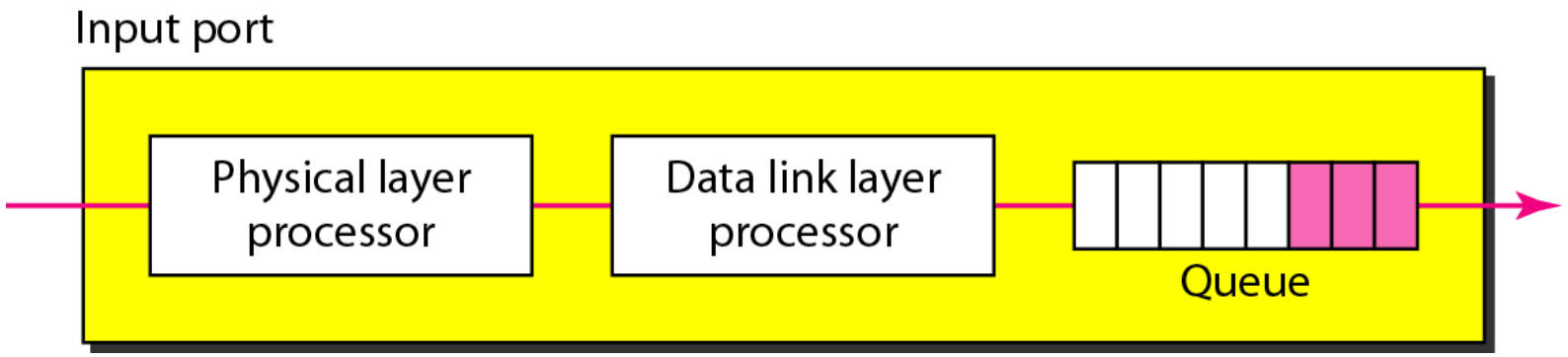


Figure 8.23 *Output port*

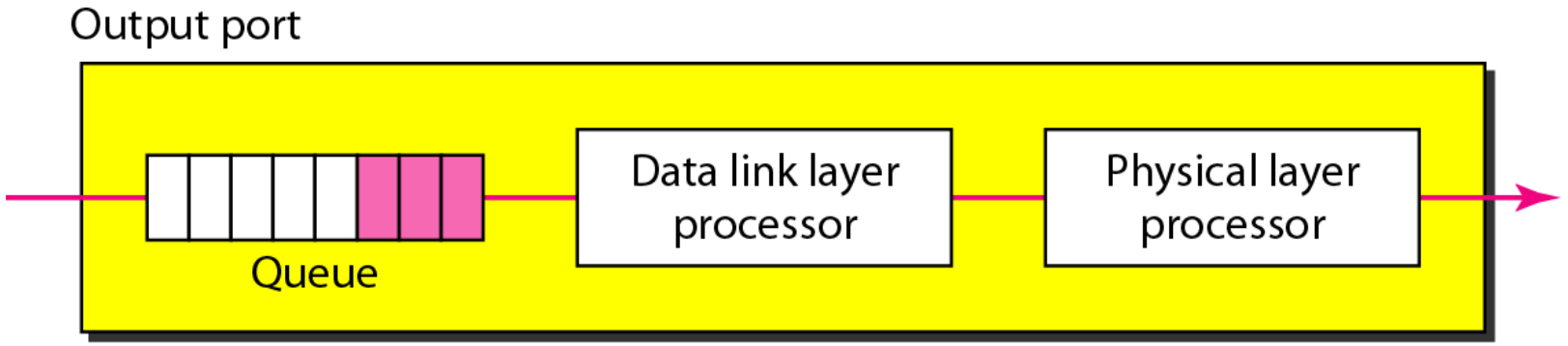


Figure 8.24 *A banyan switch*

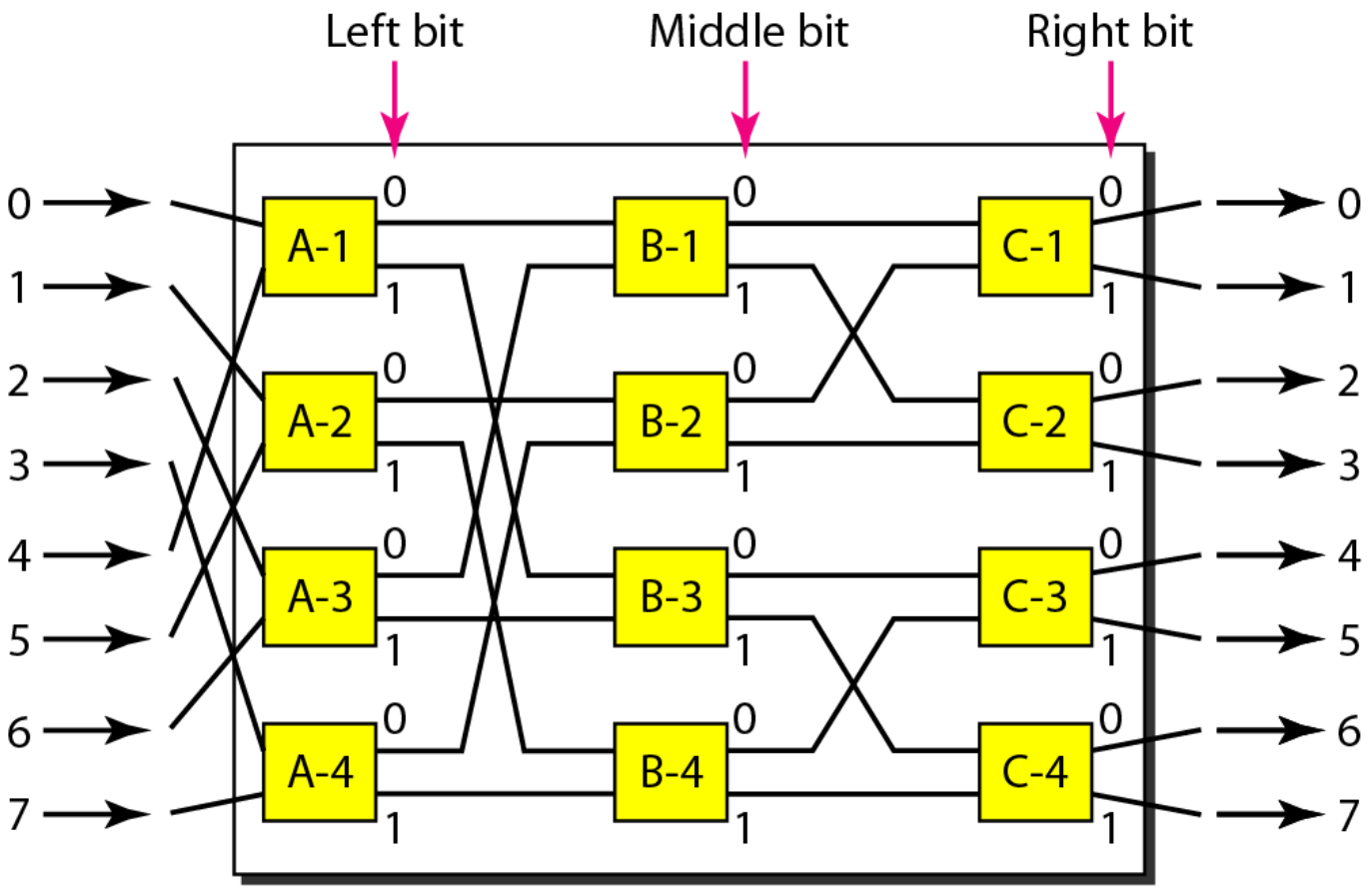
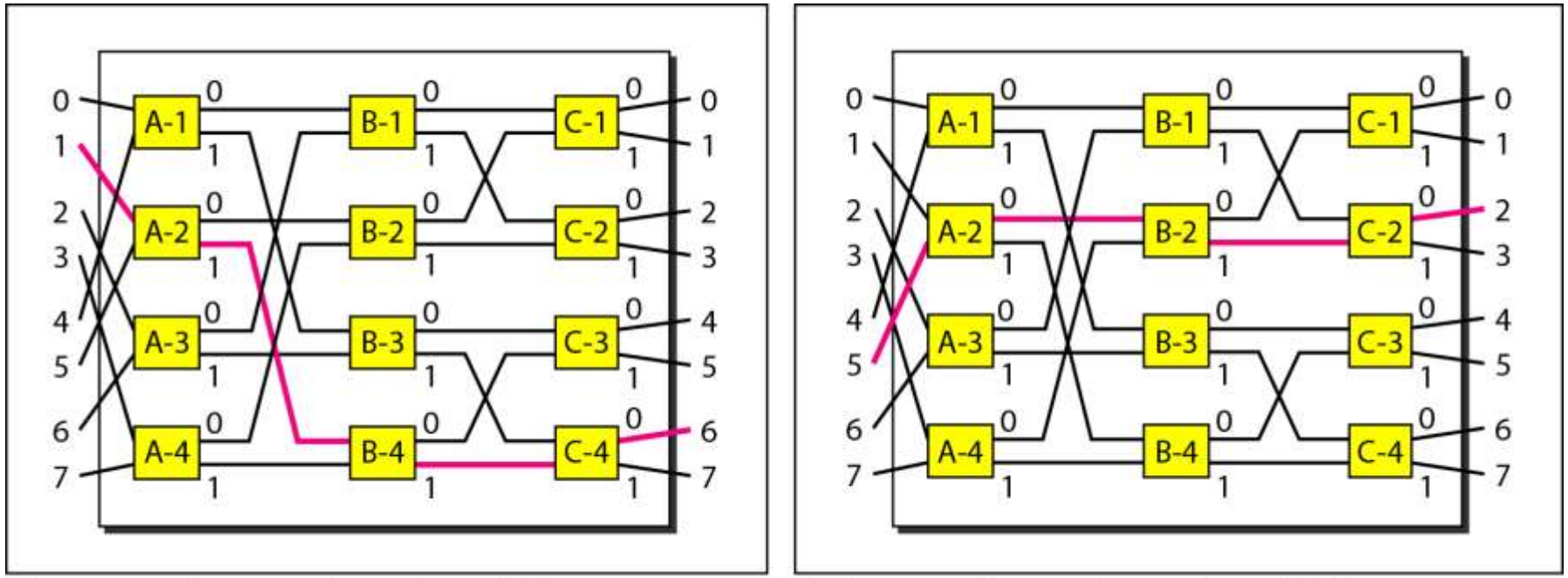


Figure 8.25 *Examples of routing in a banyan switch*



a. Input 1 sending a cell to output 6 (110)

b. Input 5 sending a cell to output 2 (010)

Figure 8.26 *Batcher-banyan switch*

